

DUBAI RAPID LINK CONSORTIUM

DUBAI METRO PROJECT OFFICE

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**Dubai Metro, UAE
Green Line – Al-Qusais Depot**

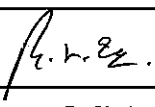
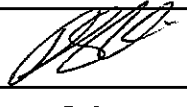
**Waste Stockyard
(Building 20)**

**Structural Design Calculations
(DCP3 Submission)**

Aug 2007

Client: JT Metro JV	Contract No. (if any):
Project Title: Dubai Metro, UAE	Project No.: JRD4561
Document No.: DM001-D-GAV-WWS-DR-DCC-545009	Controlled Copy No.:
Document Title: Green Line – Al-Qusais Depot Waste Stockyard (Building 20) Structural Design Calculations (DCP3 Submission)	
Covering Letter / Transmittal Ref. No.:	Date of Issue: 19 Aug 2007

Revision, Review and Approval/Authorization Records

				
A1	DCP3 Submission	D. Shah Aug 07	P. Scott Aug 07	J. Hill Aug 07
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1 Introduction

The Dubai Metro Project is under construction by the Dubai Rapid Link Consortium (DURL). The Civil Works are contracted to the Japan Turkey Metro Joint Venture (JTMJV), and the Rail Systems Works are contracted to Mitsubishi Heavy Industries (MHI). The rolling stock will be provided by Kinki Sharyo.

The waste stock yard building is 15m long, 5m wide, and 5m high, and is specially designed to store waste materials awaiting off site disposal. Materials to be stored will vary from paper and plastics, to oils and metal associated with the maintenance operations.

This submission for Waste Stockyard (Building 20) in Al-Qusais Depot is made by the Designers for the JTMJV, and as such concentrates on the scope of the Civil Works Contract, which includes the Architecture, Structure and M&E Design.

2 Building Form and Structural System

The waste stock yard building is 15m long, 5m wide, and 5m high, for storage of waste materials like paper and plastics, to oils and metal associated with the maintenance operations. This building is identical as that of Red Line Rashidiya Depot, Building 20, Waste Stockyard.

A steel portal frame with shell type roof is proposed in the harmony of other identical buildings. The building is to be clad with insulated standing seam aluminium sheeting, fixed to cold formed purlins which span onto structural frame.

A 300 mm thick Ground level slab monolithic with Reinforced Concrete Beam is proposed to house the facilities. The RC Pedestal raised from RC pad footing will support structural steel columns with holding down bolts fixed in position at formation level.

3 Peer Review of Red Line building and comparison

Essentially this building is identical as that of Red Line Rashidiya Main Depot, Building 20 Waste Stockyard. So the form and structural system has been replicated from the Rashidiya building.

The structural report DM001-E-MAW-CDP-DR-DCC-371948-A3 (Ref. Appendix 'A') for the Rashidiya building 20-Waste Stockyard has been peer reviewed for loadings, analysis, strength & deflection checks for all the members and foundation design. The drawings have also been peer reviewed for the design intent translation on to the details, which have then been replicated onto the Green Line Depot building-20-Waste Stockyard as listed in sect.4.

The basis of this report is to utilise Red Line Main Depot Waste Stockyard (Bldg-20) to Green Line Al-Qusais Depot Waste Stockyard (Bldg-20) with sufficient review to allow submission of Green Line DCP3 Phase.

4 Drawings

To complement the purpose of this submission, the following Structural drawings are submitted along with this report.

	<u>Drawing Title</u>	<u>Drawing No.</u>
Structural Drawings		
1	Al-Qusais-Depot - General Notes And Typical Details Sheet 1 of 3	DM001-D-GDP-CDP-DD-DCC-551990
2	Al-Qusais-Depot - General Notes And Typical Details Sheet 2 of 3	DM001-D-GDP-CDP-DD-DCC-551991
3	Al-Qusais-Depot - General Notes And Typical Details Sheet 3 of 3	DM001-D-GDP-CDP-DD-DCC-551992
4	Al-Qusais-Depot, Building-20 – Waste Stockyard - Foundation Layout Plan	DM001-D-GAV-WWS-DD-DCC-551700
5	Al-Qusais-Depot, Building-20 – Waste Stockyard - Ground Slab Plan	DM001-D-GAV-WWS-DD-DCC-551701
6	Al-Qusais-Depot, Building-20 – Waste Stockyard - Roof Layout Plan	DM001-D-GAV-WWS-DD-DCC-551702
7	Al-Qusais-Depot, Building-20 – Waste Stockyard - Section A-A	DM001-D-GAV-WWS-DD-DCC-551710
8	Al-Qusais-Depot, Building-20 – Waste Stockyard - Section B-B & C-C	DM001-D-GAV-WWS-DD-DCC-551711
9	Al-Qusais-Depot, Building-20 – Waste Stockyard – Footing/Beams/Slab/Pit Reinforcement Details	DM001-D-GAV-WWS-DD-DCC-551712
10	Al-Qusais-Depot, Building-20 – Waste Stockyard – Roof Setting Out – Plan & Sections	DM001-D-GAV-WWS-DD-DCC-551713

Appendix A
Red Line Rashidiya Depot
Waste Stockyard Building - 20
Design Calculations (Enhanced Design)

**Dubai Metro, UAE
Rashidiya Depot**

**Structural Design Calculation
(Enhanced Design)
Building 20 – Waste Stockyard**

May 2007

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Document No.: DM001-E-MAW-CDP-DR-DCC-371948	Controlled Copy No.:
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Covering Letter / Transmittal Ref. No.: To be advised	Date of Issue: 16 May 2007

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A1	Internal Review	CK Lam Sep 06	P.Shepherd-Smith Sep 06	B. Maney Sep 06
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1 TYPICAL PORTAL FRAME TRANSVERSE ANALYSIS

Document No.: DM001-E-MAW-CDP-DR-DCC-371948 Rev A3	Date: 16 May 2007
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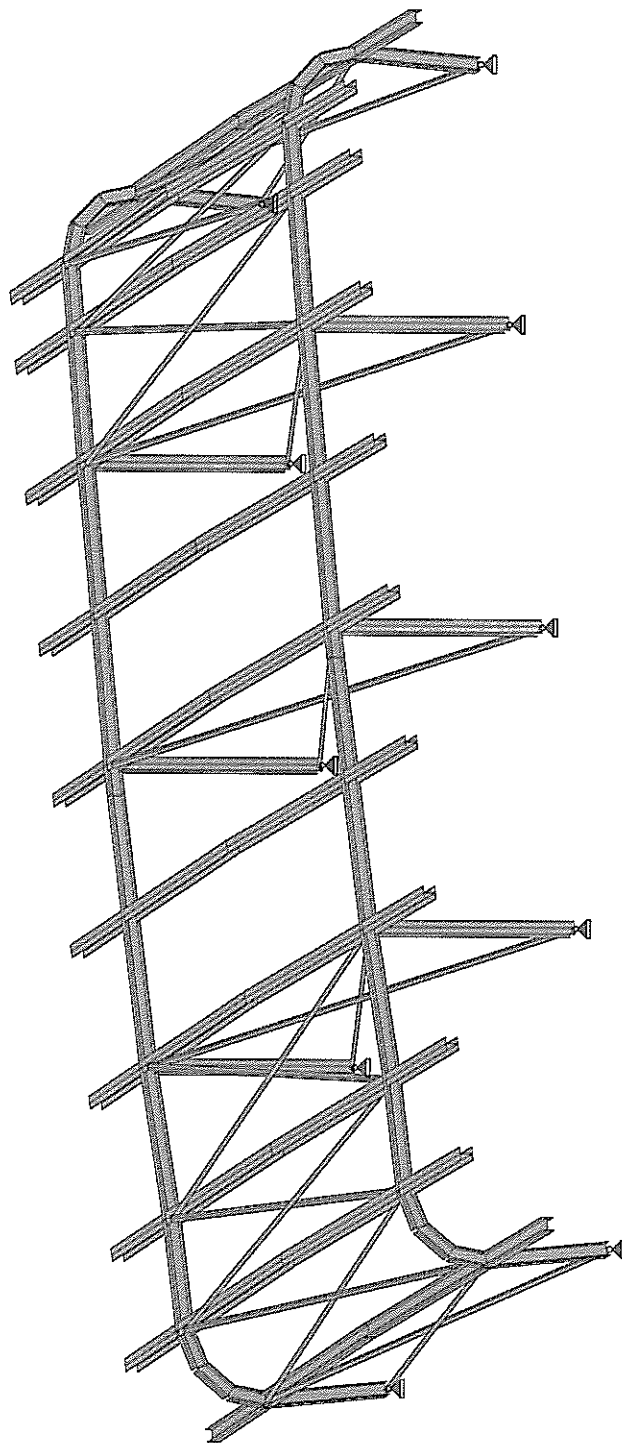
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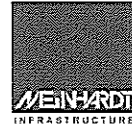
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WASTE STOCK YARD - RASHIDIYA

Load 1



Site:	Dubai Metro Project - Rashidiya Waste Stock Yard	Date:	4-May-07	Job No.:	68010.01
	Loadings	Designed by:	Rudy	Sheet No.:	

Roof Loadings :**DL** $W_{DL}:$

Sheeting	0.1 kPa		
Purlins	0.1 kPa		
Insulation	0.3 kPa		
M&E Finishes	0.3 kPa		
	<u>0.8 kPa</u>	x 1 =	0.8 kN/m

LL

0.6 kPa	x 1 =	0.6 kN/m
---------	-------	----------

$P_{LL}:$ A/C equipment S/W
 = 5 kN

(point load at mid-span of bay 2, 4)

Notional horizontal load, Ph

(Load Case of 1.4DL + 1.6LL)

Tributary load area for the roof

$$= (10+2*2.6)*(22.418+2*0.7)$$

$$= 362 \text{ m}^2$$

Total Dead Load
 290 kN

Total Live Load
 222 kN
 (Including A/C equipment)

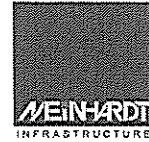
Tributary load area for the wall

$$= 4.4*2*(10+22.418)$$

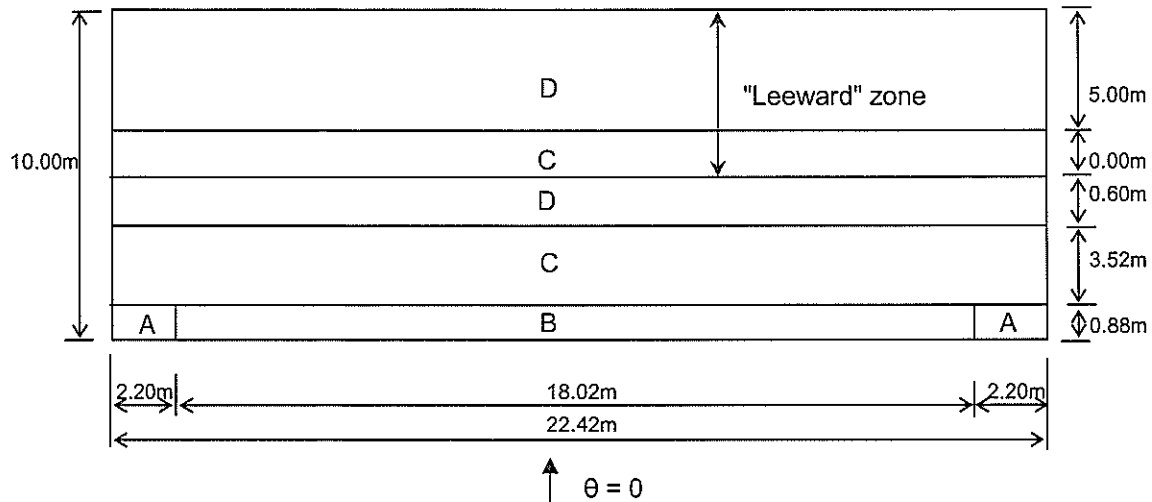
$$= 285 \text{ m}^2$$

Total Dead Load
 143 kN
 (excluding the M&E Finishes)

Total Live Load



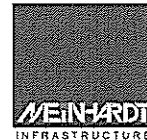
Site: Dubai Metro Project - Rashidiya	Date: 4-May-07	Job No.: 68010.01
Waste Stock Yard		
Wind Load Computation	Designed by: Rudy	Sheet No.:

Average Cpe for roof (Wind direction normal to eave)
PLAN


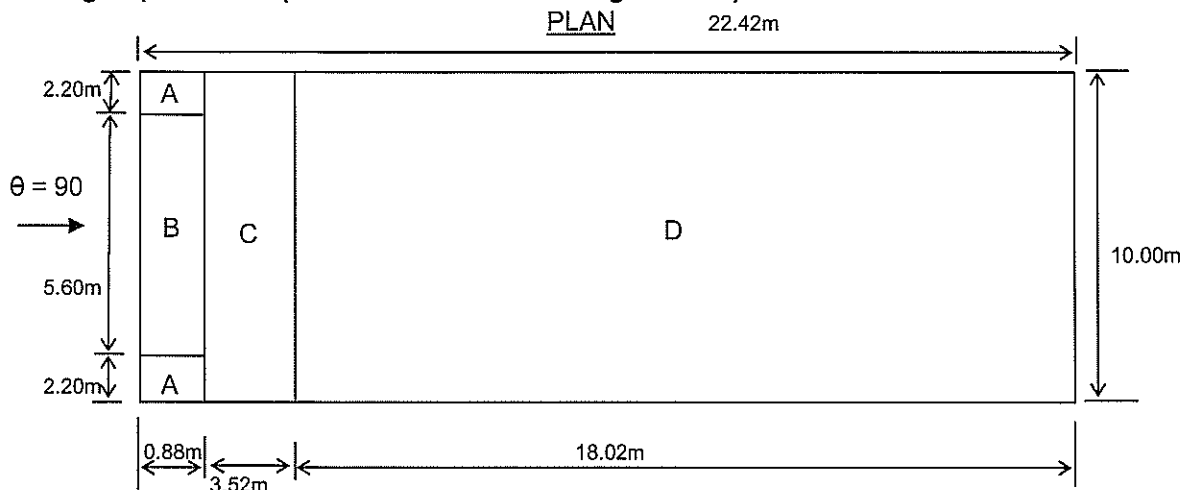
Design as flat roof (with sharp eaves)

$$\begin{aligned}
 B &= 22.418 \text{ m} \\
 H &= 4.4 \text{ m} \\
 b &= \text{smaller of } (B, 2H) \\
 &= 2 \times 4.4 \\
 &= 8.8 \text{ m}
 \end{aligned}$$

Zone	Windward		Leeward		
	Area (m ²)	Cpe	Area	Cpe1	Cpe2
A	3.9	-2			
B	15.9	-1.4			
C	78.9	-0.7	0.0	-0.7	-0.7
D	13.5	-0.2	112.1	-0.2	0.2
Total	112.1		112.1		
Average		-0.78		-0.20	0.20



Site:	Dubai Metro Project - Rashidiya	Date:	4-May-07	Job No.:	68010.01
	Waste Stock Yard				
	Wind Load Computation	Designed by:	Rudy	Sheet No.:	

Average Cpe for roof (Wind direction normal to gable end)


Design as flat roof (with sharp eaves)

$$\begin{aligned}
 B &= 10 \text{ m} \\
 H &= 4.4 \text{ m} \\
 b &= \text{smaller of } (B, 2H) \\
 &= 2 \times 4.4 \\
 &= 8.8 \text{ m}
 \end{aligned}$$

For maximum uplift, assume zone C coefficient =>

$$C_{pe} = -0.7$$

For maximum downward force, assume zone D coefficient =>

$$C_{pe} = 0.2$$

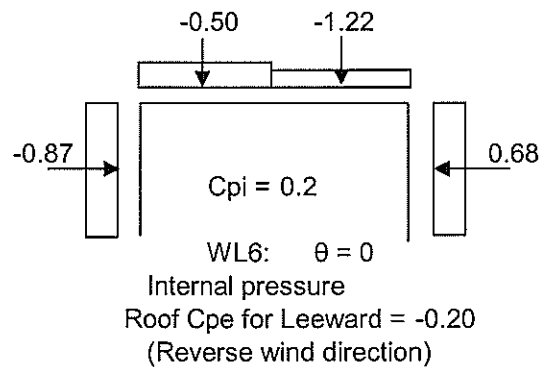
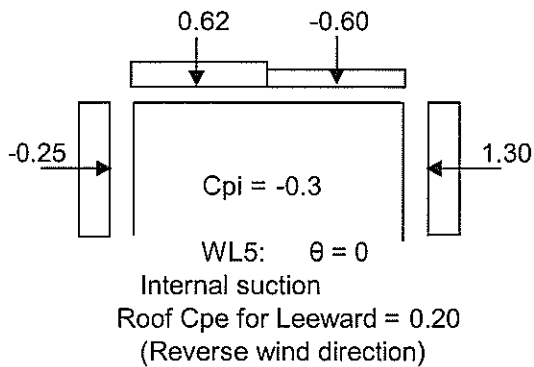
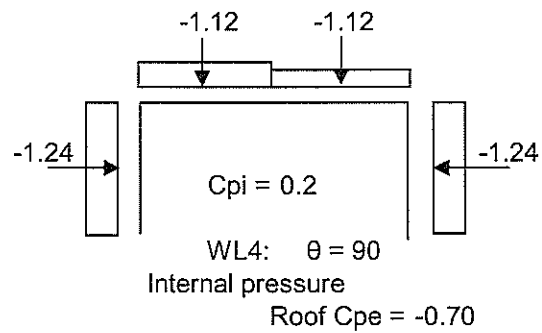
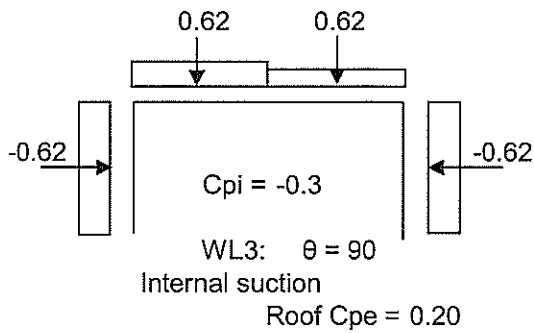
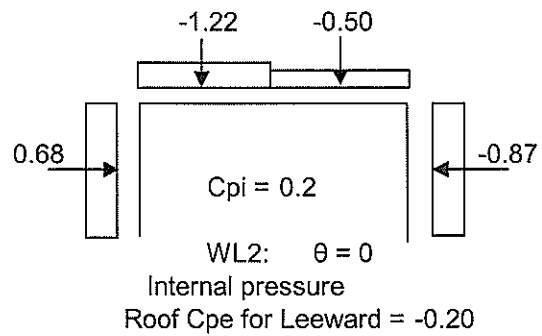
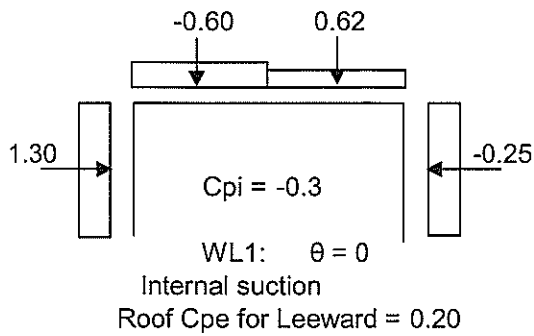
Wind Load Computation

$$\begin{aligned}
 D &= 10.00 \text{ m} \\
 H &= 4.4 \text{ m} \\
 D/H &= 2.27
 \end{aligned}$$

Elements	Effective Wind Speed V_e m/s	Dynamic Wind Pressure q kN/m ²	External Pressure Coefficient, C_{pe}			
			$\theta = 0$ (normal to eave)		$\theta = 90$ (normal to gable end)	
			Windward	Leeward		
Roof	45	1.24	-0.78	-0.20 0.20	-0.7 (Zone C)	0.2 (Zone D)
Wall	45	1.24	0.74	-0.5 -0.5	-0.8 (Zone B)	-0.8 (Zone B)

Site:	Dubai Metro Project - Rashidiya Waste Stock Yard	Date:	4-May-07	Job No.:	68010.01
	Wind Load Computation	Designed by:	Rudy	Sheet No.:	

Wind Load = 1 x q x (Cpe - Cpi) kN/m (For unit width)



Site: Dubai Metro Project - Rashidiya
Waste Stock Yard
Earthquake Loading

Date: 4-May-07

Job No.: 68010.01

Designed by: Rudy

Sheet No.:

Estimation of Earthquake Loading

Total Dead Load on the Building Column W = Given below (including part of Live Load)
Height of the Column h_n = Given below

Design Information:

- 1) Seismic Zone = 2A (from Design Brief)
2) Occupancy Category = 1 (Tanle- 16-K, Conservatively adopt 1)
3) Lateral Force procedure = **Static Procedure** (Section 1629.8 seismic zone 2A, structures not more than 5 stories)

Earthquake Loads

- 4) **Base Shear** $V_{base-S} = (C_v \times I \times W) / (R \times T)$ (Section 1630.2.1)
Seismic Coefficient $C_v = 0.32$ (Table 16-R assuming soil profile type S_D)
Importance Factor $I = 1.00$ (Table 16-K)
Numeric. Coef: Strength & Ductility $R = 4.50$ (Table 16-N for OMRT steel only)
Elastic Fundamental Period of Vibration $T = C_t (h_n)^{3/4}$ Sec.
Numerical Coefficient $C_t = 0.0853$ (Steel frame t.b.c)

- 5) **Limiting Values of Base shear** $V_{base-S-Max} = (2.5 \times C_a \times I \times W) / R = 66.41$
 $V_{base-S-Min} = 0.11 \times C_a \times I \times W = 13.15$

Design Earthquake Loads

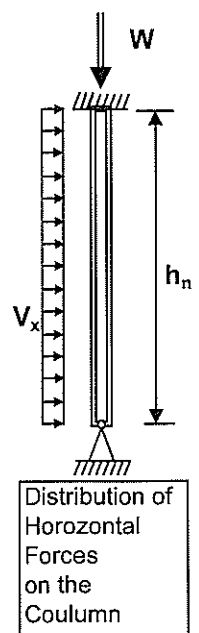
(Section 1630.6)

- $E_x = \rho \times \text{Min. } (E_h \text{ or } V_{base-S-Max}) + E_v$
 $E_{x-max} = \Omega \times \text{Min. } (E_h \text{ or } V_{base-S-Max})$
 $V_{base-S} = E_h =$ Earthquake load in the element due to the base shear, V , as set out above (for non-structural components refer to section 1632)
Vert. component of EQ ground motion $E_v = 0.5 C_a I D$ (Table 16-K)
Seismic Coefficient $C_a = 0.22$ (Table 16-Q, for soil type S_D)
Seismic Force Amplification Factor $\Omega = 2.80$ (Table 16-N)
Dead Load on the element $W = D =$ Given below
 $\rho = 2 - 6.1 / (r_{max} \sqrt{A_B}) = 1.25$ (conservative) ($1.0 \leq \rho \leq 1.5$)
 $r_{max} =$ The maximum element-storey shear ratio
 $A_B =$ Ground floor area of the structure

Distribution of Horizontal Forces:Distribution of Horizontal Force on the Column $V_x = E_x / h_n$

Col	Gr. Load, W, kN	Ht h_n , m	Time, T sec.	V_{base-S} kN	V_{base-S} kN	E_v kN	E_x kN	E_{x-max} kN	V_x kN/m	V_{x-max} kN/m
C1	543	4.40	0.26	149.1	66.41	59.7714	142.8	186.0	32.5	42.3

Load per column = $4.4 \times 32.5 / 10$
= 14.3 kN





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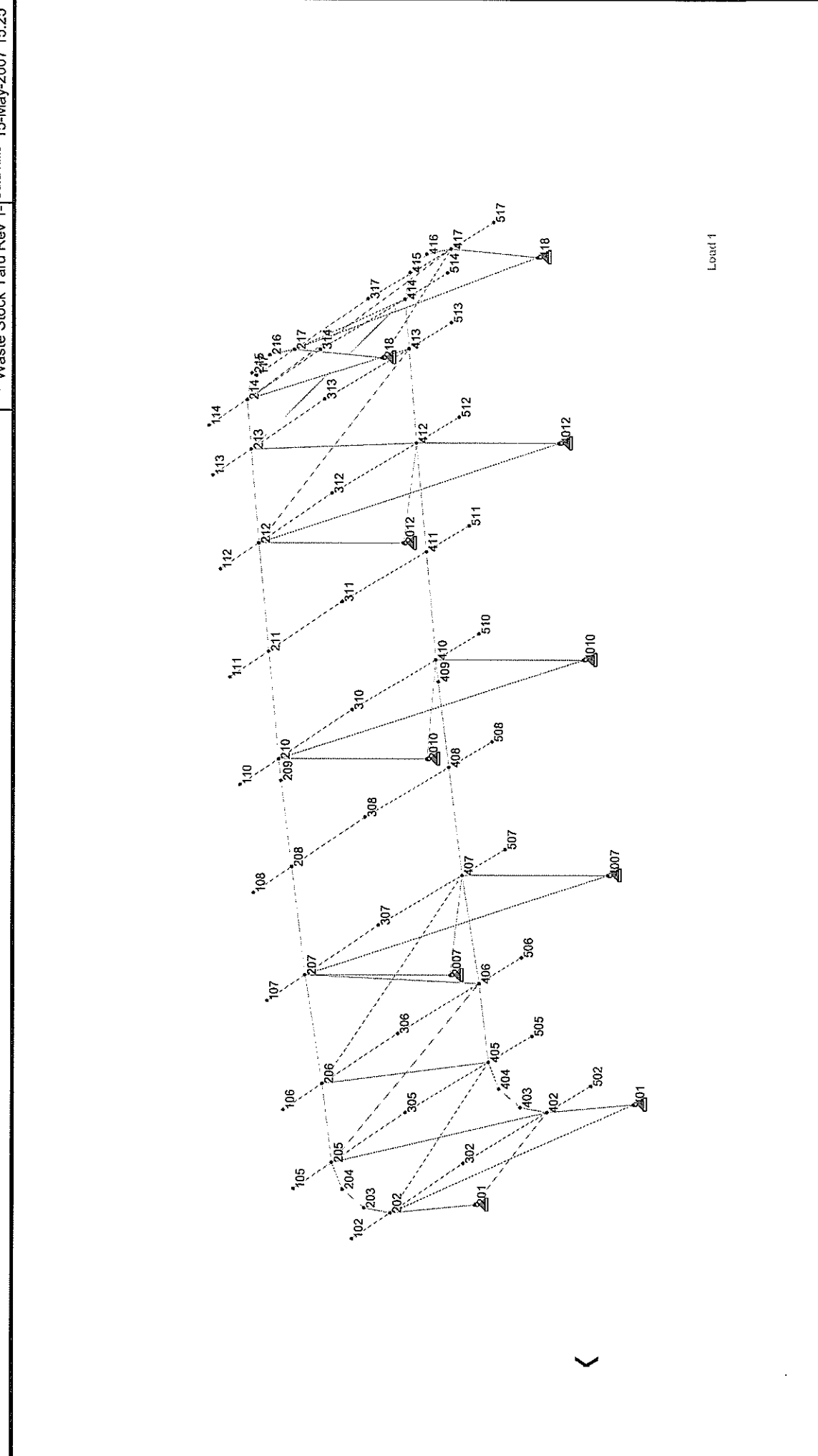
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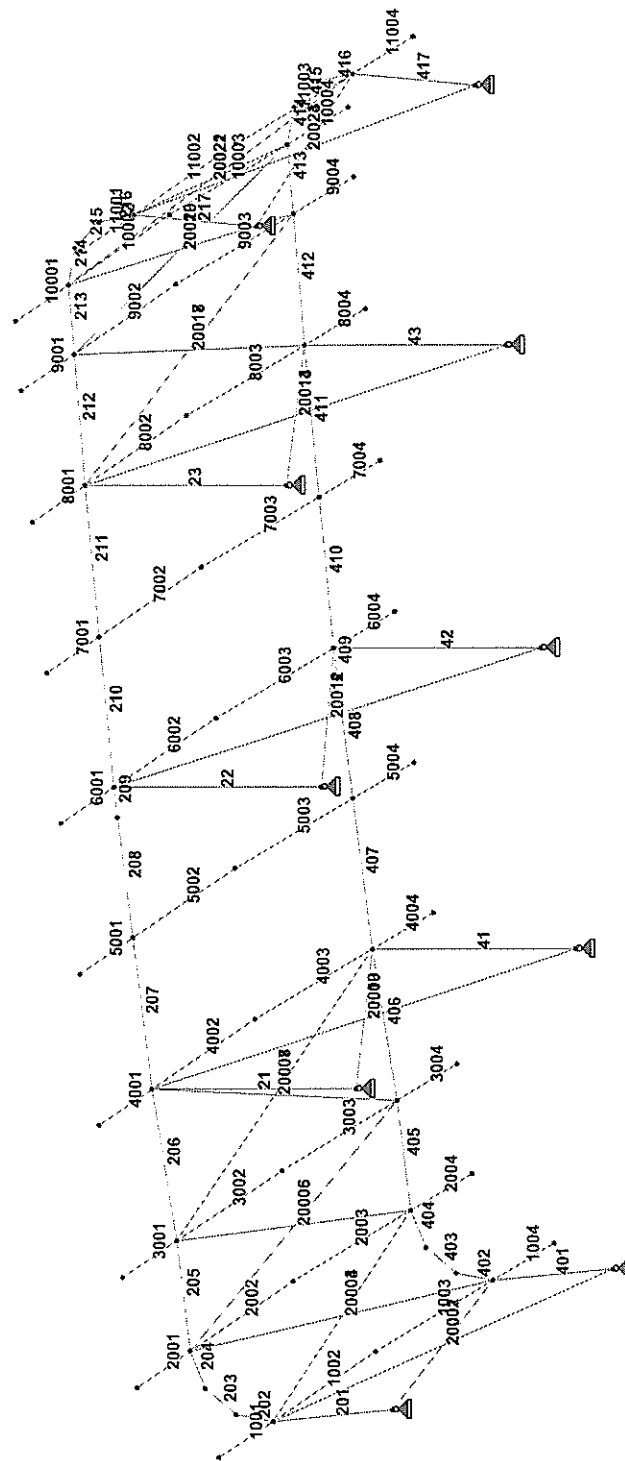
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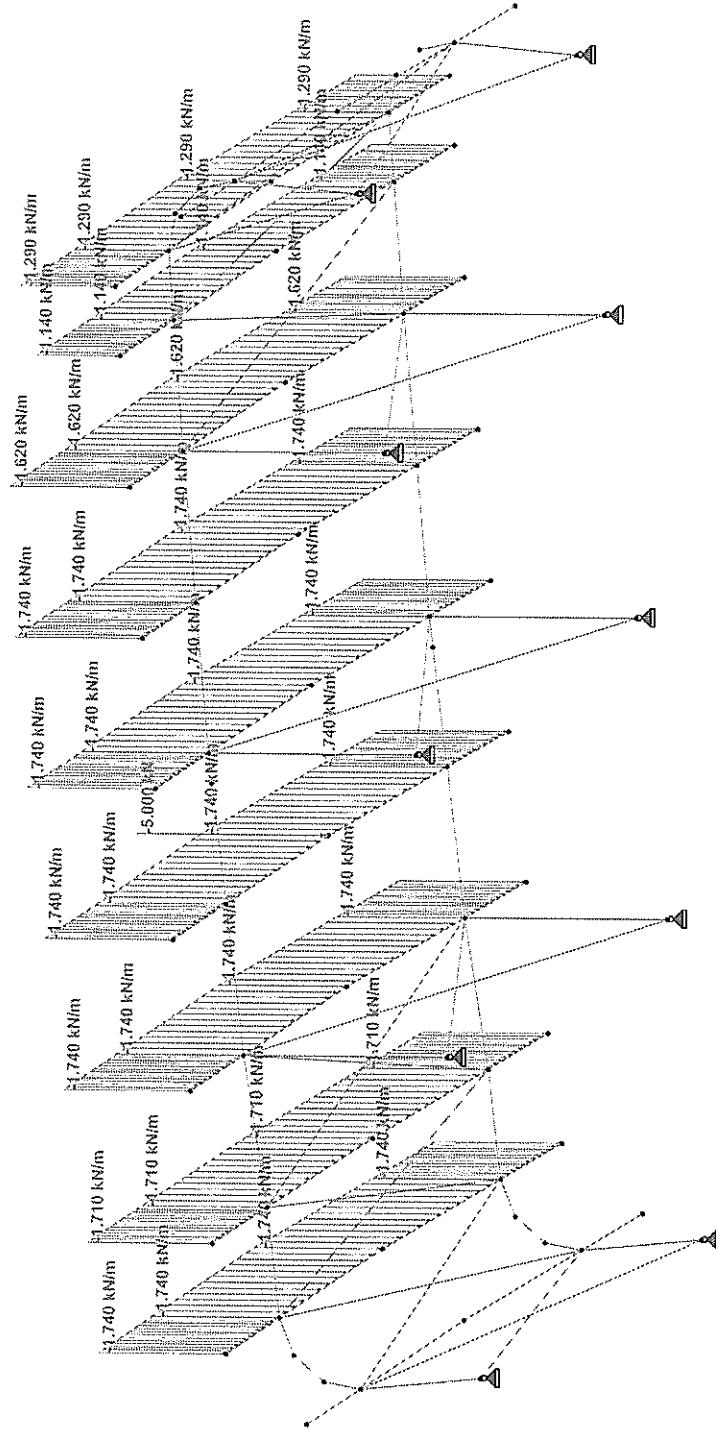


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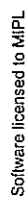
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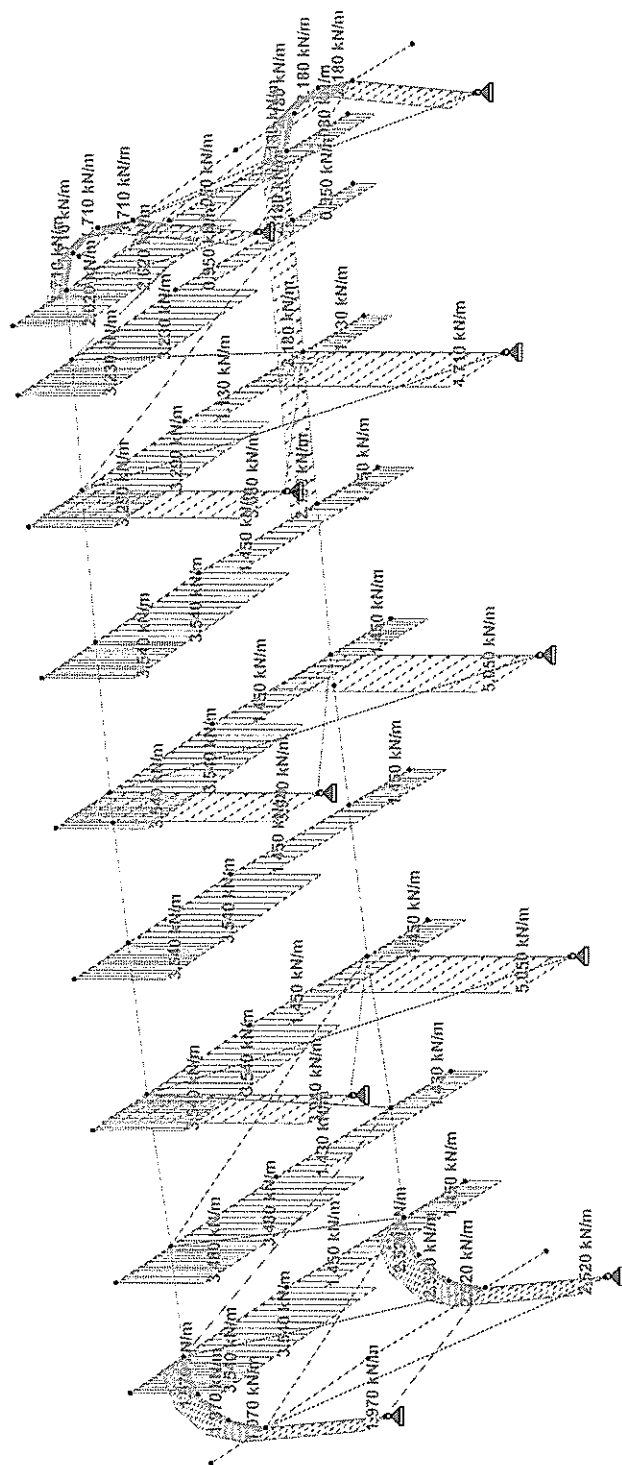
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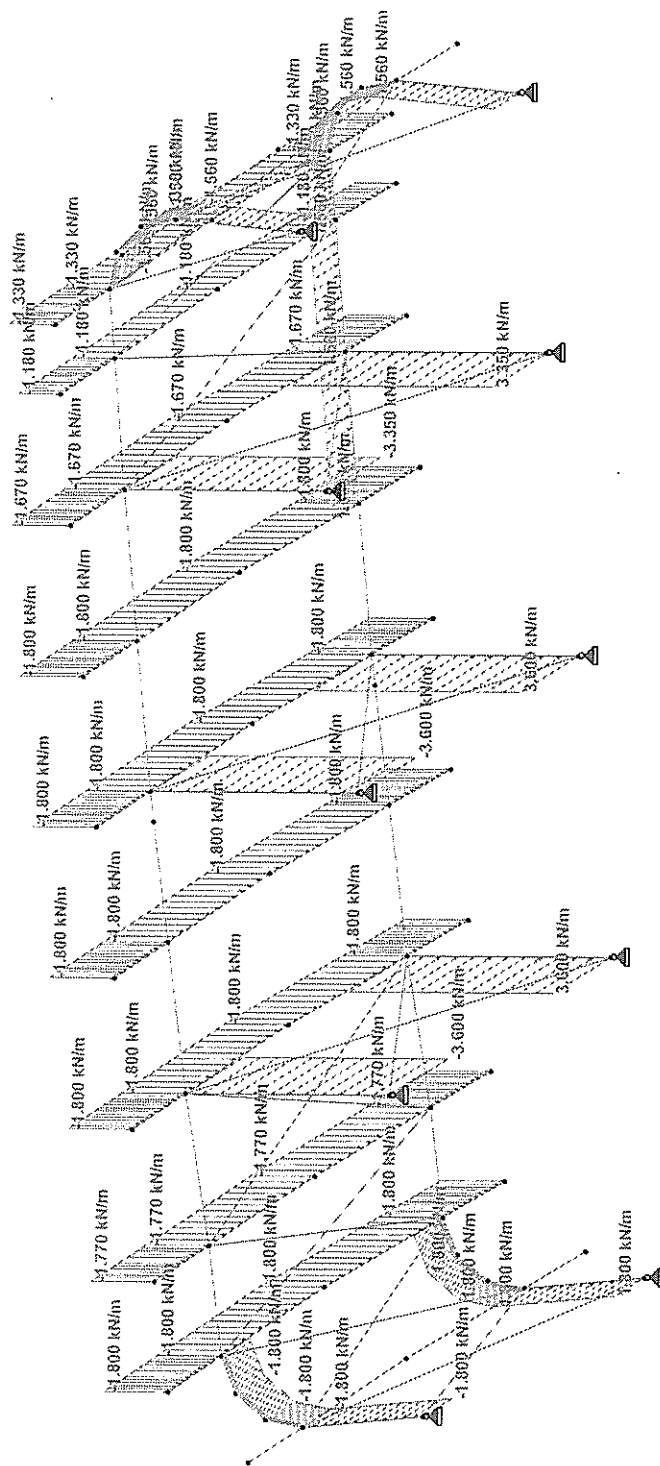
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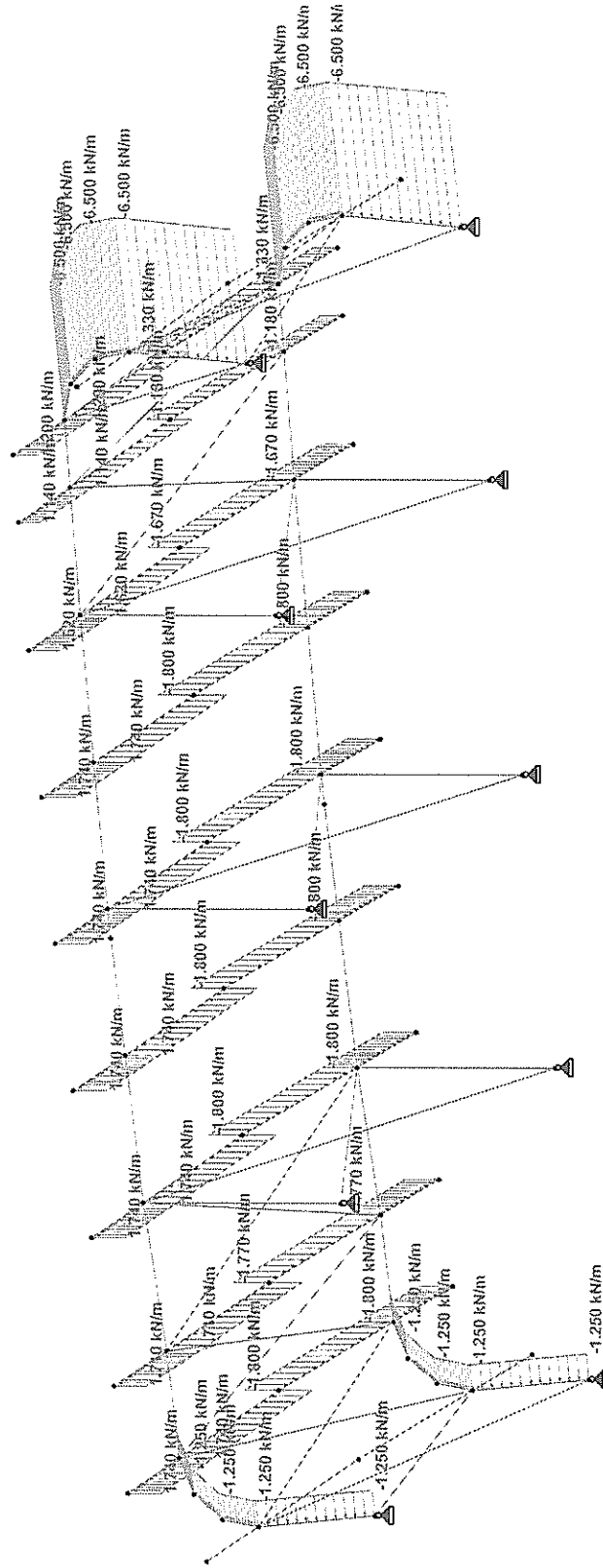
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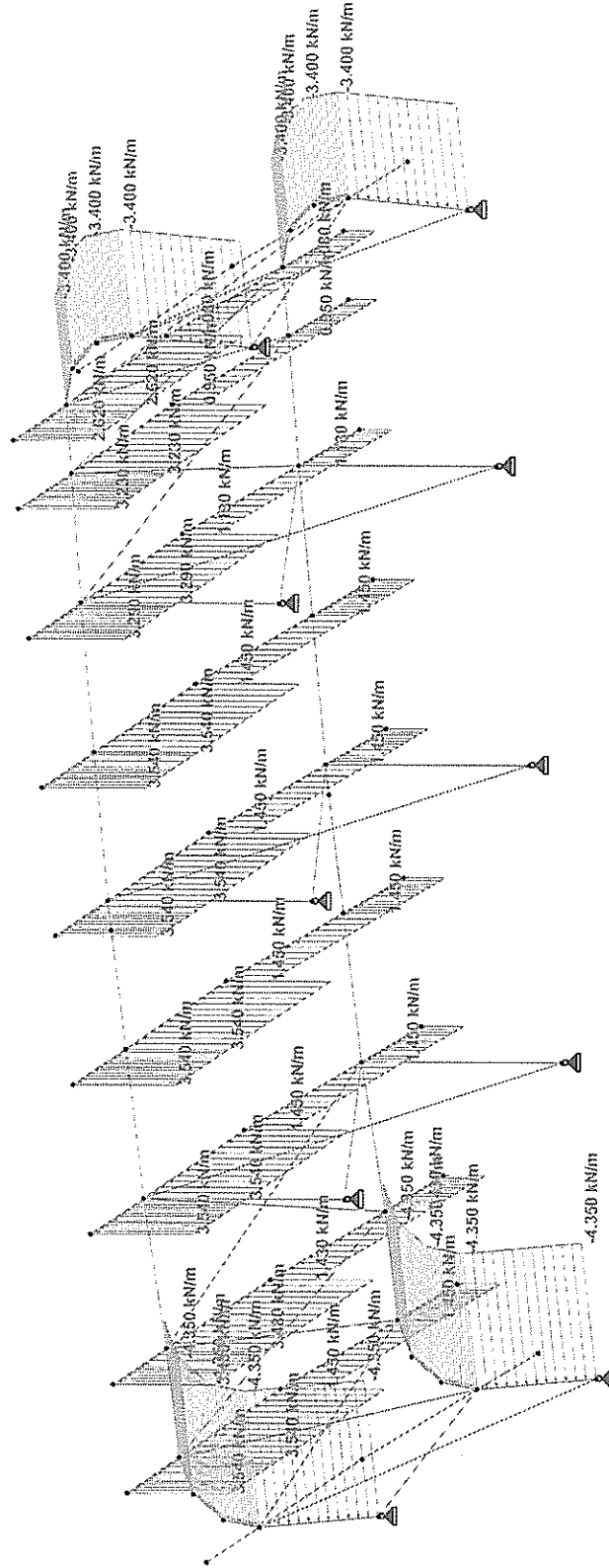
Client

Job No

Sheet No

1

Rev



Load 16

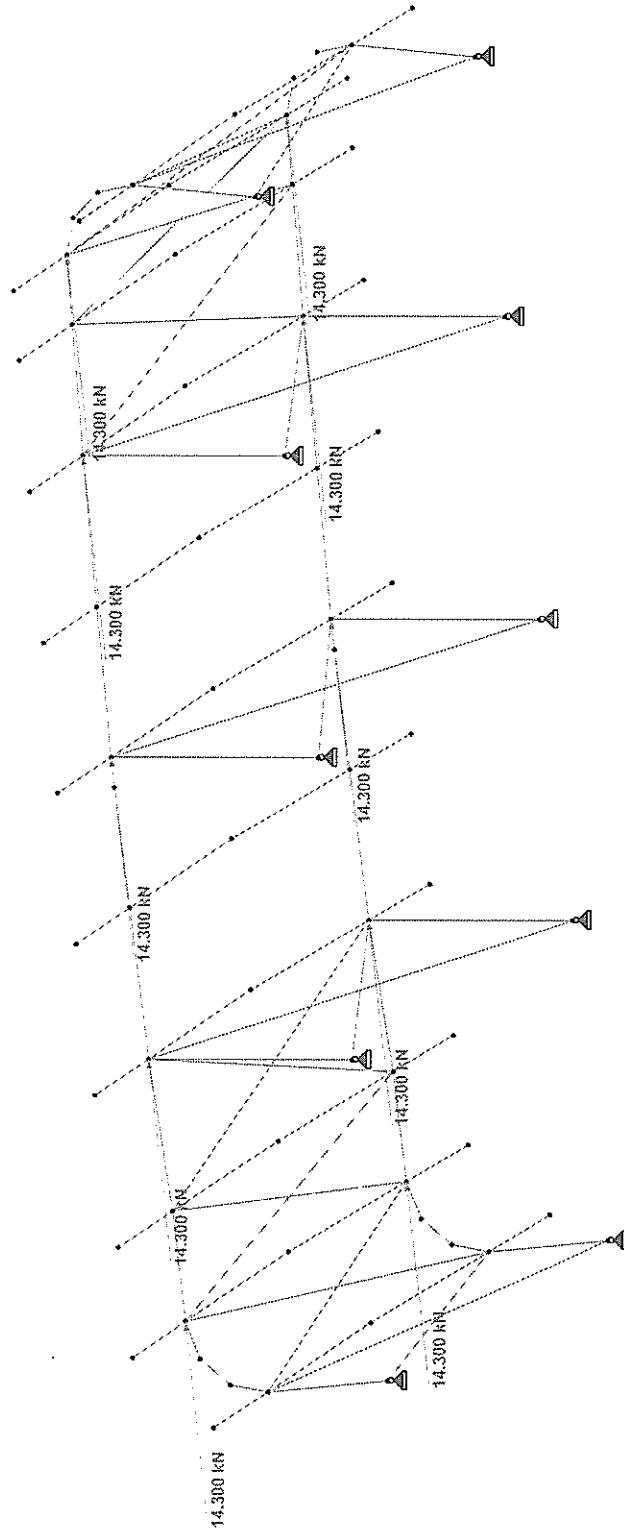


Software licensed to MIPL

Job Title

Client

Job No	Sheet No	Rev
	1	
Part		
Ref		
By	Date	MAY 2007 Chd
File Waste Stock Yard Rev 1- Date/Time 15-May-2007 15:25		





Software licensed to MIPL

Job Title

Client

Job No

Sheet No

Rev

1

Part

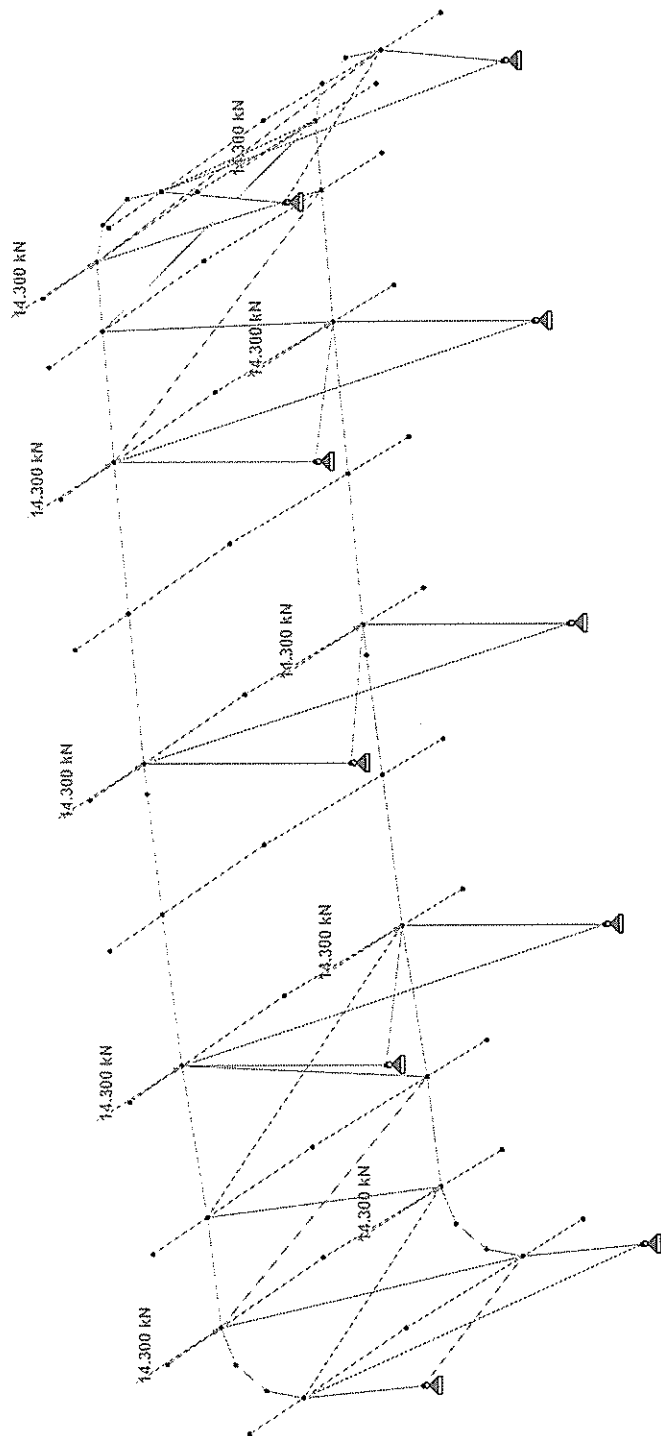
Ref

By

Date MAY 2007

Chd

File Waste Stock Yard Rev 1- Date/Time 15-May-2007 15:25



Load 22

* STAAD.Pro
* Version 2006 Bld 1005.US
* Proprietary Program of
* Research Engineers Intl.
* Date= MAY 15 2007
* Time= 15:25:50
*
* USER ID: MIPL

1. STAAD SPACE

INPUT FILE: Waste Stock Yard Rev 1- Rashidiya.STD

2. START JOB INFORMATION

3. ENGINEER DATE MAY 2007

4. END JOB INFORMATION

5. INPUT WIDTH 79

6. UNIT METER KN

7. JOINT COORDINATES

8. JOINT COORDINATES

9. 102 -11.345 2.484 -7.600; 105 -10.211 3.913 -7.600

10. 108 -3.312 3.951 -7.600; 107 -5.212 4.085 -7.600

11. 111 -3.488 4.128 -7.600; 110 0.588 4.160 -7.600

12. 113 8.897 3.903 -7.600; 112 6.388 4.045 -7.600

13. 117 11.565 2.404 -7.600; 114 10.219 3.913 -7.600

15. 201 -11.335 0.000 -5.000; 202 -11.552 2.481 -5.000

16. 203 -11.416 3.212 -5.000; 204 -11.923 3.770 -5.000

17. 205 -10.214 3.994 -5.000; 206 -9.375 3.770 -5.000

18. 207 -5.212 4.166 -5.000; 208 -7.312 4.016 -5.000

19. 209 0.000 4.244 -5.000; 210 0.588 4.209 -5.000

20. 211 3.488 4.185 -5.000; 212 6.388 4.157 -5.000

21. 215 8.897 4.040 -5.000; 214 10.214 3.954 -5.000

22. 217 10.223 3.770 -5.000; 216 11.416 3.212 -5.000

23. 302 -11.552 2.481 -5.000; 218 11.335 0.000 -5.000

25. 302 -11.552 2.481 -5.000; 218 11.335 0.000 -5.000

26. 305 -10.219 4.629 0.000; 306 -8.112 4.121 0.000

27. 307 -5.212 4.320 0.000; 308 -2.312 4.365 0.000

28. 310 0.588 4.392 0.000; 309 -2.312 4.365 0.000

29. 312 6.388 4.284 0.000; 311 -3.488 4.342 0.000

30. 314 10.219 4.150 0.000; 313 8.897 4.189 0.000

32. 401 -11.335 0.000 5.000; 402 11.565 2.629 0.000

33. 403 -11.416 3.212 5.000; 404 -11.923 3.770 5.000

34. 405 -10.214 3.994 5.000; 406 -9.375 3.770 5.000

35. 407 -5.212 4.166 5.000; 408 -7.312 4.016 5.000

36. 409 0.000 4.244 5.000; 410 0.588 4.209 5.000

37. 411 3.488 4.185 5.000; 412 6.388 4.157 5.000

38. 413 8.897 4.040 5.000; 414 10.214 3.954 5.000

39. 417 10.223 3.770 5.000; 416 11.416 3.212 5.000

40. 417 10.223 3.770 5.000; 416 11.416 3.212 5.000

42. 502 -11.552 2.481 5.000; 418 11.335 0.000 5.000

43. 506 -8.112 3.951 7.600; 505 -10.211 3.913 7.600

44. 508 -2.312 4.128 7.600; 510 0.588 4.160 7.600

* STAAD SPACE

45. 511 3.488 4.103 7.600; 512 6.388 4.045 7.600

46. 513 8.897 3.963 7.600; 514 10.219 3.913 7.600

47. 517 11.565 2.404 7.600; 515 10.219 3.913 7.600

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53. 402 6.388 0.000 -5.000

55. MEMBER INCIDENCES

56. 21 2007 207; 22 2010 210; 23 2012 212

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59. 201 201 202 217; 401 401 402 417

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64. 4001 107 207; 4002 207 307; 4003 307 407; 4004 407 507

65. 5001 108 208; 5002 208 308; 5003 308 408; 5004 408 508

66. 7001 110 210; 7002 210 310; 7003 310 410; 7004 410 510

67. 8001 111 211; 8002 211 311; 8003 311 411; 8004 411 511

68. 9001 112 212; 9002 212 312; 9003 312 412; 9004 412 512

69. 10001 113 213; 10002 213 313; 10003 313 413; 10004 413 513

70. 11001 114 214; 11002 214 314; 11003 314 414; 11004 414 514

71. 12001 117 217; 12002 217 317; 12003 317 417; 12004 417 517

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Waste Stock Yard Rev 1- Rashidiya
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275. 20409 608 44

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162. 9001 9002 UNI GY 3.23
163. 9003 9004 UNI GY 0.95
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164. 10001 10002 UNI GY 2.62
165. 10003 10004 UNI GY 1.96
166. 201 TO 204 UNI GZ 1.97
167. 21 22 UNI GZ 3.94
168. 23 UNI GZ 3.68
169. 214 TO 217 UNI GZ 1.71
170. 401 TO 404 UNI GZ 2.52
171. 41 42 UNI GZ 5.05
172. 43 UNI GZ 4.71
173. 401 TO 417 UNI GZ 2.18
174. 401 TO 417 UNI GZ 2.18
175. 401 TO 417 UNI GZ 2.18
176. MEMBER LOAD
177. 3001 TO 3004 UNI GY -1.80
178. 3001 TO 3004 UNI GY -1.80
179. 4001 TO 4004 UNI GY -1.77
180. 7001 TO 7004 UNI GY -1.80
181. 8001 TO 8004 UNI GY -1.67
182. 9001 TO 9004 UNI GY -1.18
183. 10001 TO 10004 UNI GY -1.33
184. 201 TO 204 UNI GZ -1.80
185. 21 22 UNI GZ -3.60
186. 23 UNI GZ -3.35
187. 214 TO 217 UNI GZ -1.56
188. 401 TO 404 UNI GZ 1.80
189. 41 42 UNI GZ 3.60
190. 43 UNI GZ 3.35
191. 411 TO 417 UNI GZ 1.56
192. 411 TO 417 UNI GZ 1.56
193. LOAD 14 WL4 (INTERNAL PRESSURE, WIND ANGLE 90)
194. MEMBER LOAD
195. 2001 TO 2004 UNI GY 3.25
196. 3001 TO 3004 UNI GY 3.19
197. 4001 TO 4004 UNI GY 3.19
198. 7001 TO 7004 UNI GY 3.25
199. 8001 TO 8004 UNI GY 3.02
200. 9001 TO 9004 UNI GY 2.13
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202. 201 TO 204 UNI GZ -3.60
203. 21 22 UNI GZ 7.19
204. 23 UNI GZ 7.19
205. 214 TO 217 UNI GZ 3.11
206. 401 TO 404 UNI GZ 3.60
207. 41 42 UNI GZ 7.15
208. 43 UNI GZ 6.71
209. 411 TO 417 UNI GZ 3.11
210. 411 TO 417 UNI GZ 3.11
211. LOAD 15 WL5 (INTERNAL SUCTION, WIND ANGLE 0)
212. MEMBER LOAD
213. 2001 TO 2004 UNI GY 1.74
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215. 3001 TO 3004 UNI GY 1.71
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217. 4001 TO 4004 UNI GY -1.77
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219. 8001 TO 8004 UNI GY 1.62
220. 9001 TO 9004 UNI GY 1.67
221. 10001 TO 10004 UNI GY 1.67
222. 9003 9004 UNI GY -1.18
0 STAAD SPACE
--- PAGE NO. 5

223. 10001 10002 UNI GY 1.29
224. 10003 10004 UNI GY -1.33
225. 201 TO 204 UNI GZ -1.25
226. 401 TO 404 UNI GZ -1.25
227. 214 TO 217 UNI GZ -6.5
228. 414 TO 417 UNI GZ -6.5
229. 414 TO 417 UNI GZ -6.5
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232. 3001 TO 3004 UNI GY 3.45
233. 3001 TO 3004 UNI GY 1.43
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243. 10003 10004 UNI GY 1.96
244. 201 TO 204 UNI GZ -4.35
245. 401 TO 404 UNI GZ -4.35
246. 214 TO 217 UNI GZ -3.4

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247. 414 TO 417 UNI GZ -3.4
248. LOAD 21 EQ1 (EARTHQUAKE LOAD)
249. MEMBER LOAD
250. 2001 TO 2004 UNI GY 1.77
251. 3001 TO 3004 UNI GY 1.77
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253. 4001 TO 4004 UNI GY 1.77
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257. 1.4 2 1.4 3 1.6
258. 1.4 2 1.4 3 1.6
259. LOAD COMB 1011 1.2DL + 1.2LL + 1.2WL1
260. 1.2 2 1.2 3 1.2 11 1.2
261. LOAD COMB 1111 1.4DL + 1.4WL1
262. 1.4 2 1.4 11 1.4
263. LOAD COMB 1211 1.0DL + 1.4WL1
264. 1.0 2 1.0 11 1.4
265. LOAD COMB 1012 1.2DL + 1.2LL + 1.2WL2
266. 1.2 2 1.2 3 1.2 12 1.2
267. LOAD COMB 1112 1.4DL + 1.4WL2
268. 1.4 2 1.4 11 1.4
269. LOAD COMB 1212 1.4DL + 1.4WL2
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271. LOAD COMB 1013 1.2DL + 1.2LL + 1.2WL3
272. 1.2 2 1.2 3 1.2 13 1.2
273. LOAD COMB 1113 1.4DL + 1.4WL3
274. 1.4 2 1.4 13 1.4
275. LOAD COMB 1213 1.0DL + 1.4WL3
276. 1.0 2 1.0 13 1.4
277. LOAD COMB 1014 1.2DL + 1.2LL + 1.2WL4
278. 1.2 2 1.2 3 1.2 14 1.2
279. LOAD COMB 1114 1.4DL + 1.4WL4
280. 1.4 2 1.4 14 1.4
281. LOAD COMB 1214 1.0DL + 1.4WL4
282. 1.0 2 1.0 14 1.4
0 STAAD SPACE
283. LOAD COMB 1015 1.2DL + 1.2LL + 1.2WL5
284. 1.2 2 1.2 3 1.2 15 1.2
285. LOAD COMB 1115 1.4DL + 1.4WL5
286. 1.4 2 1.4 15 1.4
287. LOAD COMB 1215 1.0DL + 1.4WL5
288. 1.0 2 1.0 15 1.4
289. LOAD COMB 1016 1.2DL + 1.2LL + 1.2WL6
290. 1.2 2 1.2 3 1.2 16 1.2
291. LOAD COMB 1116 1.4DL + 1.4WL6
292. 1.4 2 1.4 16 1.4
293. LOAD COMB 1216 1.0DL + 1.4WL6
294. 1.0 2 1.0 16 1.4
295. LOAD COMB 1021 1.2DL + 0.5LL + 1.0EQ1
297. 1.2 2 1.2 3 0.5 21 1.0
298. LOAD COMB 1022 1.2DL + 0.5LL + 1.0EQ2
299. 1.2 2 1.2 3 0.5 22 1.0
300. LOAD COMB 1121 1.2DL + 0.5LL + 1.0EQ1
301. 1.2 2 1.2 3 0.5 21 -1.0
302. LOAD COMB 1122 1.2DL + 0.5LL + 1.0EQ2
303. 1.2 2 1.2 3 0.5 22 -1.0
304. LOAD COMB 1221 0.9DL + 1.0EQ1
305. 0.9 2 0.9 21 1.0
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307. 1.0 9 2 0.9 21 1.0
308. LOAD COMB 1321 0.9DL + 1.0EQ1
309. 1.0 9 2 0.9 21 1.0
310. LOAD COMB 1322 0.9DL + 1.0EQ2
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312. *SL5
313. LOAD COMB 2001 1.0DL + 1.0LL
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315. LOAD COMB 2002 1.0DL + 0.5LL
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317. LOAD COMB 2011 DL + LL + WL1
318. 1.1 0 2 1.0 3 1.0 11 1.0
319. LOAD COMB 2012 DL + LL + WL2
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321. LOAD COMB 2013 DL + LL + WL3
322. 1.1 0 2 1.0 3 1.0 13 1.0
323. LOAD COMB 2014 DL + LL + WL4
324. 1.1 0 2 1.0 3 1.0 14 1.0
325. LOAD COMB 2015 DL + LL + WL5
326. 1.1 0 2 1.0 3 1.0 15 1.0
327. LOAD COMB 2016 DL + LL + WL6
328. 1.1 0 2 1.0 3 1.0 16 1.0
330. PERFORM ANALYSIS
0 STAAD SPACE
0

--- PAGE NO. 7

Page 4

PROBLEM STATISTICS
Waste Stock Yard Rev 1- Rashidiya
NUMBER OF JOINTS/MEMBER+ELEMENTS/SUPPORTS = 75/ 106/ 10
ORIGINAL/FINAL BAND-WIDTH= 55/ 10/ 66 DOF
TOTAL MEMBER LOAD CASES = 11, TOTAL DEGREES OF FREEDOM = 420
SIZE OF STIFFNESS MATRIX = 12.5/ 402894.1 MB
REQD/AVAIL. DISK SPACE =

0
**NOTE-Tension/Compression converged after 1 iterations, Case= 1
**NOTE-Tension/Compression converged after 1 iterations, Case= 2
**START ITERATION NO. 2
**NOTE-Tension/Compression converged after 2 iterations, Case= 3
**START ITERATION NO. 2
**START ITERATION NO. 3
**NOTE-Tension/Compression converged after 3 iterations, Case= 11
**START ITERATION NO. 2
**START ITERATION NO. 3
**NOTE-Tension/Compression converged after 4 iterations, Case= 12
**START ITERATION NO. 2
**NOTE-Tension/Compression converged after 2 iterations, Case= 13
**START ITERATION NO. 2
**START ITERATION NO. 3
**NOTE-Tension/Compression converged after 4 iterations, Case= 14
**START ITERATION NO. 2
**START ITERATION NO. 3
**NOTE-Tension/Compression converged after 4 iterations, Case= 15
0 STAAD SPACE
**START ITERATION NO. 2
**START ITERATION NO. 3
**NOTE-Tension/Compression converged after 3 iterations, Case= 16
**START ITERATION NO. 2
**START ITERATION NO. 3
**NOTE-Tension/Compression converged after 3 iterations, Case= 21
**START ITERATION NO. 2
**NOTE-Tension/Compression converged after 2 iterations, Case= 22
331. LOAD LIST 1001 1011 TO 1016 1021 1022 1111 TO 1116 1121 1122 -
Page 5

332. 1211 TO 1216 1221 1222 1321 1322
333. PRINT SUPPORT REACTION ALL
0 SUPPORT REACTION ALL
0 STAAD SPACE
0
-- PAGE NO. 9

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE						
JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y
201	1001	8.63	94.35	-6.12	0.00	0.00
1011	1011	4.61	57.21	-30.41	0.00	0.00
1211	1211	1.75	44.07	-35.17	0.00	0.00
1012	1012	0.47	24.52	-33.52	0.00	0.00
1112	1112	-3.01	34.96	-27.79	0.00	0.00
1212	1212	-3.73	18.11	-32.11	0.00	0.00
1013	1013	10.24	98.70	-30.47	0.00	0.00
1113	1113	8.32	91.01	-3.36	0.00	0.00
1213	1213	7.04	72.46	-2.34	0.00	0.00
1014	1014	1.06	41.41	-2.34	0.00	0.00
1214	1214	-2.39	25.64	-2.41	0.00	0.00
1114	1114	-3.67	6.08	-0.77	0.00	0.00
1015	1015	14.66	70.05	-5.31	0.00	0.00
1115	1115	13.47	59.05	-5.88	0.00	0.00
1215	1215	12.19	39.49	-4.23	0.00	0.00
1016	1016	17.45	46.46	-5.25	0.00	0.00
1116	1116	16.74	31.52	-5.81	0.00	0.00
1216	1216	-14.61	47.87	-4.16	0.00	0.00
1021	1021	5.08	57.65	-29.18	0.00	0.00
1121	1121	24.88	85.68	-14.84	0.00	0.00
1221	1221	5.19	75.86	-3.87	0.00	0.00
1022	1022	-16.86	25.07	-2.92	0.00	0.00
1122	1122	2.82	34.89	-27.83	0.00	0.00
1222	1222	22.62	62.92	-3.50	0.00	0.00
1023	1023	2.94	53.09	20.42	0.00	0.00
1123	1123	-5.89	77.89	-6.22	0.00	0.00
1223	1223	-3.43	49.36	-27.18	0.00	0.00
1024	1024	-1.34	41.21	-31.48	0.00	0.00
1124	1124	0.52	24.30	-29.78	0.00	0.00
1224	1224	2.71	28.21	-28.89	0.00	0.00
1025	1025	3.53	16.57	-33.49	0.00	0.00
1125	1125	-7.06	78.07	-31.78	0.00	0.00
1225	1225	-5.58	74.71	-7.37	0.00	0.00
1026	1026	-4.77	57.80	-5.62	0.00	0.00
1126	1126	-0.27	36.19	-7.02	0.00	0.00
1226	1226	2.35	25.85	-6.27	0.00	0.00
1027	1027	3.16	8.94	-5.36	0.00	0.00
1127	1127	18.65	48.18	-6.04	0.00	0.00
1227	1227	24.42	39.83	-4.33	0.00	0.00
1028	1028	25.24	22.52	-5.42	0.00	0.00
1128	1128	33.74	26.39	-6.10	0.00	0.00
1228	1228	21.82	14.42	-4.40	0.00	0.00
1029	1029	-25.21	82.05	-3.19	0.00	0.00
1129	1129	-3.33	47.90	-28.56	0.00	0.00
1229	1229	-3.33	47.90	-28.56	0.00	0.00
0	STAAD SPACE					
-- PAGE NO. 10						

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE						
JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y
1121	1121	18.41	31.10	-5.19	0.00	0.00
1221	1221	-1.47	65.75	18.00	0.00	0.00
1022	1022	-23.64	63.52	-23.03	0.00	0.00
1122	1122	-1.76	29.37	-3.83	0.00	0.00
1222	1222	19.97	12.57	-1.83	0.00	0.00
1023	1023	-1.91	46.72	19.35	0.00	0.00
1123	1123	8.63	94.35	-6.12	0.00	0.00
1223	1223	9.36	99.62	4.54	0.00	0.00
1024	1024	7.29	93.55	4.98	0.00	0.00
1124	1124	6.01	73.99	3.33	0.00	0.00
1224	1224	0.37	64.54	2.02	0.00	0.00
1025	1025	-0.91	32.62	0.39	0.00	0.00
1125	1125	-0.91	33.07	0.39	0.00	0.00
1225	1225	-0.91	33.07	0.39	0.00	0.00
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Waste Stock Yard Rev 1- Rashidiya									
JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM-Z	Waste Stock	Yard Rev 1- Rashidiya
1013	10.20	98.53	3.20	0.00	0.00	0.00	0.00	1011	-0.37
1113	8.28	92.28	3.42	0.00	0.00	0.00	0.00	1111	-0.20
1213	7.00	77.73	1.78	0.00	0.00	0.00	0.00	1211	-0.14
1014	0.83	41.27	2.24	0.00	0.00	0.00	0.00	1012	-0.19
1114	-2.66	25.47	2.30	0.00	0.00	0.00	0.00	1112	-0.01
1214	3.32	101.32	0.65	0.00	0.00	0.00	0.00	1212	-0.43
1015	20.14	101.32	2.23	0.00	0.00	0.00	0.00	1013	-0.47
1115	19.84	91.96	4.19	0.00	0.00	0.00	0.00	1113	-0.32
1215	18.56	72.43	3.50	0.00	0.00	0.00	0.00	1213	-0.32
1016	20.60	64.68	5.17	0.00	0.00	0.00	0.00	1014	-0.32
1116	20.40	52.78	4.07	0.00	0.00	0.00	0.00	1114	-0.32
1216	19.12	33.23	4.71	0.00	0.00	0.00	0.00	1214	-0.32
1021	-14.61	47.83	5.26	0.00	0.00	0.00	0.00	1021	-0.32
1121	24.88	85.68	5.35	0.00	0.00	0.00	0.00	1121	-0.32
1221	15.28	57.83	4.76	0.00	0.00	0.00	0.00	1221	-0.32
1022	-15.28	25.07	3.92	0.00	0.00	0.00	0.00	1022	-0.32
1122	15.28	25.07	3.92	0.00	0.00	0.00	0.00	1122	-0.32
1222	15.28	25.07	3.92	0.00	0.00	0.00	0.00	1222	-0.32
1023	-15.28	25.07	3.92	0.00	0.00	0.00	0.00	1023	-0.32
1123	15.28	25.07	3.92	0.00	0.00	0.00	0.00	1123	-0.32
1223	15.28	25.07	3.92	0.00	0.00	0.00	0.00	1223	-0.32
1024	-15.28	25.07	3.92	0.00	0.00	0.00	0.00	1024	-0.32
1124	15.28	25.07	3.92	0.00	0.00	0.00	0.00	1124	-0.32
1224	15.28	25.07	3.92	0.00	0.00	0.00	0.00	1224	-0.32
1025	-15.28	25.07	3.92	0.00	0.00	0.00	0.00	1025	-0.32
1125	15.28	25.07	3.92	0.00	0.00	0.00	0.00	1125	-0.32
1225	15.28	25.07	3.92	0.00	0.00	0.00	0.00	1225	-0.32
1026	-15.28	25.07	3.92	0.00	0.00	0.00	0.00	1026	-0.32
1126	15.28	25.07	3.92	0.00	0.00	0.00	0.00	1126	-0.32
1226	15.28	25.07	3.92	0.00	0.00	0.00	0.00	1226	-0.32
1027	-15.28	25.07	3.92	0.00	0.00	0.00	0.00	1027	-0.32
1127	15.28	25.07	3.92	0.00	0.00	0.00	0.00	1127	-0.32
1227	15.28	25.07	3.92	0.00	0.00	0.00	0.00	1227	-0.32
1028	-15.28	25.07	3.92	0.00	0.00	0.00	0.00	1028	-0.32
1128	15.28	25.07	3.92	0.00	0.00	0.00	0.00	1128	-0.32
1228	15.28	25.07	3.92	0.00	0.00	0.00	0.00	1228	-0.32
1029	-15.28	25.07	3.92	0.00	0.00	0.00	0.00	1029	-0.32
1129	15.28	25.07	3.92	0.00	0.00	0.00	0.00	1129	-0.32
1229	15.28	25.07	3.92	0.00	0.00	0.00	0.00	1229	-0.32
1030	-15.28	25.07	3.92	0.00	0.00	0.00	0.00	1030	-0.32
1130	15.28	25.07	3.92	0.00	0.00	0.00	0.00	1130	-0.32
1230	15.28	25.07	3.92	0.00	0.00	0.00	0.00	1230	-0.32
1031	-15.28	25.07	3.92	0.00	0.00	0.00	0.00	1031	-0.32
1131	15.28	25.07	3.92	0.00	0.00	0.00	0.00	1131	-0.32
1231	15.28	25.07	3.92	0.00	0.00	0.00	0.00	1231	-0.32
1032	-15.28	25.07	3.92	0.00	0.00	0.00	0.00	1032	-0.32
1132	15.28	25.07	3.92	0.00	0.00	0.00	0.00	1132	-0.32
1232	15.28	25.07	3.92	0.00	0.00	0.00	0.00	1232	-0.32
1033	-15.28	25.07	3.92	0.00	0.00	0.00	0.00	1033	-0.32
1133	15.28	25.07	3.92	0.00	0.00	0.00	0.00	1133	-0.32
1233	15.28	25.07	3.92	0.00	0.00	0.00	0.00	1233	-0.32
1034	-15.28	25.07	3.92	0.00	0.00	0.00	0.00	1034	-0.32
1134	15.28	25.07	3.92	0.00	0.00	0.00	0.00	1134	-0.32
1234	15.28	25.07	3.92	0.00	0.00	0.00	0.00	1234	-0.32
1035	-15.28	25.07	3.92	0.00	0.00	0.00	0.00	1035	-0.32
1135	15.28	25.07	3.92	0.00	0.00	0.00	0.00	1135	-0.32
1235	15.28	25.07	3.92	0.00	0.00	0.00	0.00	1235	-0.32
1036	-15.28	25.07	3.92	0.00	0.00	0.00	0.00	1036	-0.32
1136	15.28	25.07	3.92	0.00	0.00	0.00	0.00	1136	-0.32
1236	15.28	25.07	3.92	0.00	0.00	0.00	0.00	1236	-0.32
1037	-15.28	25.07	3.92	0.00	0.00	0.00	0.00	1037	-0.32
1137	15.28	25.07	3.92	0.00	0.00	0.00	0.00	1137	-0.32
1237	15.28	25.07	3.92	0.00	0.00	0.00	0.00	1237	-0.32
1038	-15.28	25.07	3.92	0.00	0.00	0.00	0.00	1038	-0.32
1138	15.28	25.07	3.92	0.00	0.00	0.00	0.00	1138	-0.32
1238	15.28	25.07	3.92	0.00	0.00	0.00	0.00	1238	-0.32
1039	-15.28	25.07	3.92	0.00	0.00	0.00	0.00	1039	-0.32
1139	15.28	25.07	3.92	0.00	0.00	0.00	0.00	1139	-0.32
1239	15.28	25.07	3.92	0.00	0.00	0.00	0.00	1239	-0.32
1040	-15.28	25.07	3.92	0.00	0.00	0.00	0.00	1040	-0.32
1140	15.28	25.07	3.92	0.00	0.00	0.00	0.00	1140	-0.32
1240	15.28	25.07	3.92	0.00	0.00	0.00	0.00	1240	-0.32
1041	-15.28	25.07	3.92	0.00	0.00	0.00	0.00	1041	-0.32
1141	15.28	25.07	3.92	0.00	0.00	0.00	0.00	1141	-0.32
1241	15.28	25.07	3.92	0.00	0.00	0.00	0.00	1241	-0.32
1042	-15.28	25.07	3.92	0.00	0.00	0.00	0.00	1042	-0.32
1142	15.28	25.07	3.92	0.00	0.00	0.00	0.00	1142	-0.32
1242	15.28	25.07	3.92	0.00	0.00	0.00	0.00	1242	-0.32
1043	-15.28	25.07	3.92	0.00	0.00	0.00	0.00	1043	-0.32
1143	15.28	25.07	3.92	0.00	0.00	0.00	0.00	1143	-0.32
1243	15.28	25.07	3.92	0.00	0.00	0.00	0.00	1243	-0.32
1044	-15.28	25.07	3.92	0.00	0.00	0.00	0.00	1044	-0.32
1144	15.28	25.07	3.92	0.00	0.00	0.00	0.00	1144	-0.32
1244	15.28	25.07	3.92	0.00	0.00	0.00	0.00	1244	-0.32
1045	-15.28	25.07	3.92	0.00	0.00	0.00	0.00	1045	-0.32
1145	15.28	25.07	3.92	0.00	0.00	0.00	0.00	1145	-0.32
1245	15.28	25.07	3.92	0.00	0.00	0.00	0.00	1245	-0.32
1046	-15.28	25.07	3.92	0.00	0.00	0.00	0.00	1046	-0.32
1146	15.28	25.07	3.92	0.00	0.00	0.00	0.00	1146	-0.32
1246	15.28	25.07	3.92	0.00	0.00	0.00	0.00	1246	-0.32
1047	-15.28	25.07	3.92	0.00	0.00	0.00	0.00	1047	-0.32
1147	15.28	25.07	3.92	0.00	0.00	0.00	0.00	1147	-0.32
1247	15.28	25.07	3.92	0.00	0.00	0.00	0.00	1247	-0.32
1048	-15.28	25.07	3.92	0.00	0.00	0.00	0.00	1048	-0.32
1148	15.28	25.07	3.92	0.00	0.00	0.00	0.00	1148	-0.32
1248	15.28	25.07	3.92	0.00	0.00	0.00	0.00	1248	-0.32
1049	-15.28	25.07	3.92	0.00	0.00	0.00	0.00	1049	-0.32
1149	15.28	25.07	3.92	0.00	0.00	0.00	0.00	1149	-0.32
1249	15.28	25.07	3.92	0.00	0.00	0.00	0.00	1249	-0.32
1050	-15.28	25.07	3.92	0.00	0.00	0.00	0.00	1050	-0.32
1150	15.28	25.07	3.92	0.00	0.00	0.00	0.00	1150	-0.32
1250	15.28	25.07	3.92	0.00	0.00	0.00	0.00	1250	-0.32
1051	-15.28	25.07	3.92	0.00	0.00	0.00	0.00	1051	-0.32
1151	15.28	25.07	3.92	0.00	0.00	0.00	0.00	1151	-0.32
1251	15.28	25.07	3.92	0.00	0.00	0.00	0.00	1251	-0.32
1052	-15.28	25.07	3.92	0.00	0.00	0.00	0.00	1052	-0.32
1152	15.28	25.07	3.92	0.00	0.00	0.00	0.00	1152	-0.32
1252	15.28	25.07	3.92	0.00	0.00	0.00	0.00	1252	-0.32
1053	-15.28	25.07	3.92	0.00	0.00	0.00	0.00	1053	-0.32
1153	15.28	25.07	3.92	0.00	0.00	0.00	0.00	1153	-0.32
1253	15.28	25.07	3.92	0.00	0.00	0.00	0.00	1253	-0.32
1054	-15.28	25.07	3.92	0.00	0.00	0.00	0.00	1054	-0.32
1154	15.28	25.07	3.92	0.00	0.00	0.00	0.00	1154	-0.32
1254	15.28	25.07	3.92	0.00	0.00	0.00	0.00	1254	-0.32
1055	-15.28	25.07	3.92	0.00	0.00	0.00	0.00	1055	-0.32
1155	15.28	25.07	3.92	0.00	0.00	0.00	0.00	1155	-0.32
1255	15.28	25.07	3.92	0.00	0.00	0.00	0.00	1255	-0.32
1056	-15.28	25.07	3.92	0.00	0.00	0.00	0.00	1056	-0.32
1156	15.28	25.07	3.92	0.00	0.00	0.00	0.00	1156	-0.32
1256	15.28	25.07	3.92	0.00	0.00	0.00	0.00	1256	-0.32
1057	-15.28	25.07	3.92	0.00	0.00	0.00	0.00	1057	-0.32
1157	15.28	25.07	3.92	0.00	0.00	0.00	0.00	1157	-0.32
1257	15.28	25.07	3.92	0.00	0.00	0.00	0.00	1257	-0.32
1058	-15.28	25.07	3.92	0.00	0.00	0.00	0.00	1058	-0.32
1158	15.28	25.07	3.92	0.00	0.00	0.00	0.00	1158	-0.32
1258	15.28	25.07	3.92	0.00	0.00	0.00	0.00	1258	-0.32
1059	-15.28	25.07	3.92	0.00	0.00	0.00	0.00	1059	-0.32
1159	15.28	25.07	3.92	0.00	0.00	0.00	0.00	1159	-0

MEMBER FORCE ENVELOPE

ALL UNITS ARE KN METE

MEMB	DISTANCE	FY	LD	MZ	LD	FZ	LD	MY	LD
21	0.00	MAX	19.79	1114	0.00	1322	9.82	1321	0.00
		MIN	-18.31	1211	0.00	1322	-11.76	1021	0.00
	0.83	MAX	11.58	1013	11.59	1211	9.82	1321	8.18
		MIN	-9.52	1211	-12.58	1114	-9.82	1321	-8.18
	1.67	MAX	7.98	1013	15.86	1211	9.82	1321	9.80
		MIN	-2.36	1212	-19.30	1013	-11.76	1021	-16.36
	2.50	MAX	9.57	1011	12.79	1211	9.82	1321	24.50
		MIN	-6.83	1214	-24.45	1013	-11.76	1021	-29.40
	3.33	MAX	17.82	1111	7.87	1212	9.82	1321	32.73
		MIN	-15.21	1214	-26.61	1013	-11.76	1021	-39.20
	4.17	MAX	26.62	1111	10.97	1214	9.82	1321	40.91
		MIN	-23.60	1214	-25.76	1013	-11.76	1021	-49.00

MAX/MIN FORCE VALUES FOR MEMB										21, AMONGST ALL SECT LOCATIONS									
		FY/		DIST		LD		MZ/		DIST		LD		FY		DIST		LD	
		LD		LD		LD		LD		LD		LD		LD		LD		LD	

		Waste Stock Yard Rev 1 - Rashidiya											
		MIN		-7.00 1212		-3.89 1212		-0.63 1214					
1.74	MAX	33.80	1013			1.43	1116		1.48	1001	-0.05 1212		
	MIN	37.74	1212			-1.05	1001		-0.25	1214	1.19 1013		
2.32	MAX	35.46	1013			-0.82	1212		-0.48	1001	-0.25 1214		
	MIN	35.46	1212			-20.51	1112		-0.86	1001	2.03 1014		
2.90	MAX	33.22	1013			8.92	1212		2.86	1001	-0.86 1004		
	MIN	-7.72	1212			-39.59	1013		-0.63	1214	-0.99 1214		

MAX/MIN FORCE VALUES FOR MEMB		207 - AMONGST ALL SECT LOCATIONS	
EX/	DIST.		

MAX.	34.67	0.00	1013	58.85	0.00	1001	10.92	C	0.00	1015
	1.48	0.00	1001	2.89	2.90	1013				
MIN.	-7.72	2.90	1212	-39.59	2.90	1013	8.32	T	2.90	1216
	-0.03	2.90	1214	-1.48	0.00	1011				

208	0.00	MAX	8.86	1216	-8.91	1212	1.01	1214	2.83	1011
	0.46	MIN	-2.03	1013	-39.59	1013	-1.81	1011	-0.82	1214

[illegible]

	FZ	DIST	LD	MY	DIST	LD	FX	DIST	LD
MAX.	8.86	0.00	1216	28.82	2.31	1021			
	1.01	0.00	1216	2.82					
	-28.18	2.31	1013	-39.59	0.00	1011	10.44	C	0.00 1015
MIN.	-1.81	2.31	1011	-1.56	2.31	1011	8.10	T	2.31 1216

		209, AMONGST ALL SECT LOCATIONS	
MAX/MIN FORCE VALUES FOR MEMB EV ₅₀			
0.24	MIN	-28.09	10113
	MAX	77.65	12116
0.35	MIN	-28.15	10113
	MAX	77.60	12116
0.47	MIN	-28.21	10113
	MAX	76.55	12116
0.59	MIN	-27.50	12113
	MAX	-28.32	10113

[illegible]

	MAX	-26.17	1013	15-75	1121	1-58	1001	0.32	1214
1.10	MIN	-11.21	1214	-11.21	1221	-2.54	1214	-0.56	1011
	MAX	-27.88	1013	6.45	1321	1-58	1001	0.39	1022
1.74	MIN	-6.37	1214	-46.31	1021	-2.54	1214	-1.15	1214
	MAX	-27.60	1013	8.72	1216	1-58	1001	1-30	1001
2.32	MIN	-6.61	1214	-26.57	1013	-2.54	1214	-2.62	1214
	MAX	-27.31	1013	11.18	1216	1-58	1001	2.21	1001

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[illegible]

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	MAX/MIN FORCE VALUES FOR MEMB				403, AMONGST ALL SECT LOCATIONS							
	FY	FZ	DIST	LD	MY	MZ	DIST	LD	FX	FZ	DIST	LD
MAX.	40.87	0.00	1013		90.70	0.00	1121					
	1.77	0.74	1214		0.79	0.00	1013		70.92	c	0.00	1015
MIN.	-7.24	0.74	1214		-60.52	0.74	1221					
	-3.34	0.00	1013		-1.09	0.74	1013		11.54	w	0.74	1214

MAX.	69.75	0.00	1015	74.83	0.00	1121
	5.51	0.74	1214	2.73	0.74	1214
MIN.	-12.44	0.74	1214	-78.02	0.74	1021
	-2.32	0.74	1001	-2.71	0.74	1001

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	MAX/MIN FORCE VALUES FOR MEMB			405, AMONGST ALL SECT LOCATIONS			FX	FY	DIST
	Fx/ Fz	DIST LD	LD	NY/ MZ	DIST LD	LD			
MAX.	27.11	0.00	1121	45.29	0.00	1321			
	1.45	0.00	1001	2.37	0.00	1214	18.54	C	0.00 1015
MIN.	-14.58	2.10	1121	-77.99	0.00	1021			
	-1.37	2.10	1314	-1.68	0.00	1001	6.98		2.10 1216

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MIN	-44.89	1013	-38.19	1013	-0.48	1113	-0.45	1214
MAX	8.72	1214	2.14	1214	0.18	1222	1.32	1013
MIN	-45.18	1013	-12.90	1015	-0.48	1113	-0.59	1013
MAX	8.48	1214	22.36	1021	0.18	1222	1.24	1001
MIN	-45.47	1013	-9.95	1321	-0.48	1113	-0.74	1214
MAX	8.74	1214	48.32	1021	0.18	1222	1.23	1011
MIN	-45.74	1013	-11.61	1321	-0.48	1113	-0.88	1214
MAX	8.70	1214	71.46	1021	0.18	1222	1.27	1011
MIN	-46.05	1013	-13.15	1321	-0.48	1113	-1.03	1214

MAX/MIN FORCE VALUES FOR MEMB								
FY/	DIST	LD	MZ/	DIST	LD	FX	DIST	LD
MAX.	9.28	0.00	74.46	2.90	1021	16.09	C	0.00
MIN.	-46.05	2.90	-64.18	0.00	1013	7.06	T	2.90
	-0.48	2.90	-1.03	2.90	1216			

407	0.00	MAX	34.67	1013	58.86	1013	0.85	1214	1.64	1011
		MIN	-6.12	1214	-11.01	1013	-1.54	1011	-0.75	1011
	0.58	MAX	34.38	1013	38.84	1015	0.85	1214	0.52	1014
		MIN	-6.36	1214	-7.39	1214	-1.54	1011	-0.52	1014
	1.16	MAX	34.09	1013	19.16	1015	0.85	1214	0.06	1212
		MIN	-6.60	1214	-3.64	1214	-1.54	1011	-0.42	1013
	1.74	MAX	33.80	1013	1.39	1116	0.85	1214	0.46	1214
		MIN	-6.84	1214	-1.05	1001	-1.54	1011	-0.48	1001
	2.32	MAX	33.51	1013	4.30	1214	0.85	1214	0.95	1214
		MIN	-7.08	1214	-20.23	1013	-1.54	1011	-2.03	1001
	2.90	MAX	32.72	1013	8.47	1214	0.85	1214	1.44	1214
		MIN	-7.32	1214	-39.59	1013	-1.54	1011	-2.89	1001

STAND SPACE

MAX/MIN FORCE VALUES FOR MEMB								
FY/	DIST	LD	MZ/	DIST	LD	FX	DIST	LD
MAX.	34.67	0.00	58.86	0.00	1013	15.81	C	0.00
MIN.	-6.12	2.90	-64.18	0.00	1011	5.10	T	2.90
	-1.34	2.90	-1.89	2.90	1001			

408	0.00	MAX	6.48	1214	8.48	1214	1.80	1001	1.59	1214
		MIN	-27.03	1013	-39.59	1013	-0.79	1214	-2.79	1001
	0.46	MAX	6.29	1214	5.53	1214	1.80	1001	1.23	1214
		MIN	-27.26	1013	-27.95	1015	-0.79	1214	-1.96	1001
	0.92	MAX	6.10	1214	4.79	1221	1.80	1001	0.86	1214
		MIN	-27.49	1013	-17.16	1221	-0.79	1214	-1.13	1001
	1.39	MAX	5.90	1214	10.86	1221	1.80	1001	0.49	1214
		MIN	-27.72	1013	-12.23	1221	-0.79	1214	-0.30	1001
	1.85	MAX	5.71	1214	19.53	1021	1.80	1001	0.72	1013
		MIN	-27.95	1013	-9.71	1321	-0.79	1214	-0.07	1216
	2.31	MAX	5.61	1214	28.82	1021	1.80	1001	1.51	1013
		MIN	-28.18	1013	-7.62	1321	-0.79	1214	-0.24	1214

MAX/MIN FORCE VALUES FOR MEMB								
FY/	DIST	LD	MZ/	DIST	LD	FX	DIST	LD
MAX.	6.48	0.00	28.82	2.31	1021	14.92	C	0.00
MIN.	-28.18	2.31	-39.59	0.00	1013	5.18	T	2.31
	-0.79	2.31	-2.79	0.00	1001			

409	0.00	MAX	5.45	1214	28.82	1021	1.80	1001	1.51	1013
		MIN	-28.04	1013	-7.62	1321	-0.79	1214	-0.24	1214
	0.12	MAX	5.40	1214	31.19	1021	1.80	1001	1.71	1013
		MIN	-28.10	1013	-7.09	1321	-0.79	1214	-0.33	1214
	0.24	MAX	5.35	1214	33.57	1021	1.80	1001	1.91	1013
		MIN	-28.16	1013	-6.66	1214	-0.79	1214	-0.43	1214
	0.35	MAX	5.31	1214	35.96	1021	1.80	1001	2.11	1013
		MIN	-28.21	1013	-7.29	1214	-0.79	1214	-0.52	1214
	0.47	MAX	5.26	1214	38.36	1021	1.80	1001	2.31	1013

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Waste Stock Yard Rev 1- Rashidiya	-28.27	1013	-7.91	1214	-0.79	1214	-0.61	1214
MIN	0.59	MAX	5.21	1214	40.82	1013	1.80	1001
MAX	-28.33	1013	-0.79	1214	-0.79	1214	-0.71	1214
STAND SPACE								

STAND SPACE

MAX/MIN FORCE VALUES FOR MEMB								
FY/	DIST	LD	MZ/	DIST	LD	FX	DIST	LD
MAX.	5.45	0.00	40.82	0.59	1013	15.36	C	0.59
MIN.	-28.33	0.59	-8.53	0.59	1214	5.18	T	0.00
	-0.79	0.59	-0.71	0.59	1214			

410	0.00	MAX	29.10	1015	44.05	1015	0.10	1216	2.90	1013
		MIN	-5.72	1214	-8.37	1214	-1.98	1013	-0.75	1013
	0.58	MAX	28.82	1015	27.46	1121	0.10	1216	0.10	1216
		MIN	-5.96	1214	-8.22	1221	-1.98	1013	-0.06	1216
	1.16	MAX	28.53	1015	15.75	1121	0.10	1216	0.60	1013
		MIN	-6.21	1214	-11.21	1221	-1.98	1013	-0.01	1322
	1.74	MAX	28.24	1015	16.45	1121	0.10	1216	0.05	1216
		MIN	-6.45	1214	-16.31	1021	-1.98	1013	-0.56	1113
	2.32	MAX	27.95	1015	26.03	1214	0.10	1216	0.11	1216
		MIN	-6.69	1214	-9.98	1013	-1.98	1013	-1.70	1013
	2.90	MAX	27.66	1015	42.48	1013	0.10	1216	0.17	1216
		MIN	-6.93	1214	-42.48	1013	-1.98	1013	-2.85	1013

MAX/MIN FORCE VALUES FOR MEMB								
FY/	DIST	LD	MZ/	DIST	LD	FX	DIST	LD
MAX.	29.10	0.00	44.05	0.00	1015	18.02	C	2.90
MIN.	-6.93	2.90	-42.48	2.90	1013	3.77	T	0.00
	-1.98	2.90	-2.85	2.90	1013			

411	0.00	MAX	6.87	1214	10.01	1214	1.33	1001	0.54	1214
		MIN	-29.96	1013	-47.47	1013	-5.03	1011	-2.86	1214
	0.58	MAX	6.63	1214	6.00	1214	1.33	1001	0.13	1016
		MIN	-30.25	1013	-25.00	1013	-2.50	1214	-2.70	1013
	1.16	MAX	6.39	1214	2.32	1214	1.63	1013	-0.03	1216
		MIN	-30.54	1013	-8.11	1015	-0.27	1216	-2.48	1114
	1.74	MAX	6.15	1214	10.43	1013	2.81	1014	0.14	1222
		MIN	-30.83	1013	-1.32	1214	-0.27	1216	-1.61	1214
	2.32	MAX	5.90	1214	28.39	1013	5.30	1011	1.08	1011
		MIN	-31.12	1013	-4.81	1214	-0.27	1216	-0.34	1216
	2.90	MAX	5.66	1214	46.53	1013	7.82	1114	4.55	1114
		MIN	-31.40	1013	-8.17	1214	-0.27	1216	-0.49	1216

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MAX/MIN FORCE VALUES FOR MEMB								
FY/	DIST	LD	MZ/	DIST	LD	FX	DIST	LD
MAX.	6.87	0.00	46.53	2.90	1013	18.89	C	2.90
MIN.	-31.40	2.90	-42.47	0.00	1013	4.18	T	0.00
	-5.02	0.00	-2.70	0.58	1013			

412	0.00	MAX	44.89	1121	63.62	1121	0.40	1216	4.58	1114
		MIN	-13.00	1221	-22.14	1221	-7.57	1114	-0.44	1216
	0.50	MAX	44.64	1121	41.14	1121	0.40	1216	1.66	1013
		MIN	-13.18	1221	-15.56	1221	-5.39	1114	-0.24	1216
	1.00	MAX	44.39	1121	18.79	1121	0.40	1216	0.76	1001
		MIN	-13.37	1221	-8.90	1221	-3.20	1114	-0.95	1214
	1.51	MAX	44.14	1121	-0.19	1216	0.40	1216	0.44	1001
		MIN	-13.56	1221	-5.27	1013	-1.04	1011	-1.95	1214
	2.01	MAX	43.79	1121	25.52	1121	0.40	1216	0.41	1016
		MIN	-13.79	1221	-11.67	1121	-0.77	1022	-1.86	1214
	2.51	MAX	43.63	1121	-47.52	1121	-0.77	1022	-0.72	1114

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MAX/MIN FORCE VALUES FOR MEMB										412, AMONGST ALL SECT LOCATIONS									
		FZ		DIST		LD		MZ		MY		DIST		LD		FX		DIST	
MAX.		44.89		0.00		1121		63.62		0.00		1121		24.72		C		2.51 1115	
MIN.		-13.93		2.51		1121		-47.52		2.51		1121		5.08		T		0.00 1321	
		-7.57		0.00		1114		-1.95		1.51		1214							
413										413, AMONGST ALL SECT LOCATIONS									
		FZ		DIST		LD		MZ		MY		DIST		LD		FX		DIST	
MAX.		22.48		1321		11.67		1321		0.33		1322		0.67		1216			
MIN.		-47.52		1121		-1.76		1121		-0.93		1113		-0.71		1114			
		25.18		1321		19.04		1321		0.93		1113		0.36		1114			
		-52.95		1121		-1.21		1022		2.10		1214		0.76		1216			
		26.43		1321		-58.34		1121		-1.21		1022		-0.84		1111			
		-29.99		1021		3.25		1214		-1.21		1022		0.94		1214			
		-30.12		1021		-63.70		1121		-1.21		1022		-1.34		1214			
		22.09		1321		41.30		1214		-1.21		1022		-1.26		1022			
		-69.03		1121		-1.21		1022		-1.21		1022		-1.58		1022			
		-74.32		1121		-1.21		1022		-1.21		1022		-1.58		1022			
STAND SPACE										STAND SPACE									

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MAX/MIN FORCE VALUES FOR MEMB										413, AMONGST ALL SECT LOCATIONS									
		FZ		DIST		LD		MZ		MY		DIST		LD		FX		DIST	
MAX.	22.48	0.00		1321		48.77		1.32		1221		24.38		C		1.32		1115	
	3.54	1.32		1214		2.65		1.32		1214									
MIN.	-30.38	1.32		1021		-74.32		1.32		1214		5.20		T		0.00		1321	
	-1.70	0.00		1113		-1.58		1.32		1022									

414																			
0.00	MAX	6.42		1214		48.74		1221		1.86		1001		2.13		1214			
	MTN	-51.09		1013		-74.38		1121		-4.56		1214		-2.15		1001			
0.15	MAX	6.36		1214		54.10		1221		1.86		1001		1.50		1214			
	MIN	-51.17		1013		-72.58		1121		-3.92		1214		-0.87		1001			
0.30	MAX	6.30		1214		59.47		1221		1.86		1001		0.97		1214			
	MIN	-51.24		1013		70.76		1121		-3.27		1214		-1.59		1001			
0.45	MAX	6.24		1214		64.84		1221		1.86		1001		-1.32		1001			
	MTN	-51.31		1013		-68.94		1121		-2.62		1214		-1.32		1001			
0.59	MAX	6.18		1214		71.07		1021		1.86		1001		0.19		1214			
	MTN	-51.38		1013		-67.94		1321		-1.97		1214		-1.04		1001			
0.74	MAX	6.12		1214		78.19		1021		1.86		1001		-0.06		1214			
	MTN	-51.45		1013		-67.82		1321		-1.33		1214		-0.83		1013			

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MAX/MIN FORCE VALUES FOR MEMB										414, AMONGST ALL SECT LOCATIONS									
		FZ		DIST		LD		MZ		MY		DIST		LD		FX		DIST	
MAX.		6.42		0.00		1214		78.19		0.74		1021		43.71		C		0.74	
MIN.		-51.45		0.74		1013		-74.38		0.00		1121		19.28		T		0.00	
		-4.56		0.00		1214		-2.15		0.00		1001							

15		0.00		MAX		3.67		1316		78.19		1021		1.86		1001		-0.04	
		MIN		-30.67		1013		-67.85		1021		-1.86		1001		-0.83		1013	
		0.15		MAX		3.57		1316		86.78		1021		-1.33		1214		0.62	
		MIN		-30.49		1013		-68.24		1021		-1.68		1214		-0.52		1019	
		0.30		MAX		3.52		1214		83.73		1031		1.88		1001		0.04	
		MIN		-30.54		1013		-64.65		1321		-0.03		1214		-0.37		1013	
		0.45		MAX		3.48		1214		86.50		1021		2.11		1013		0.11	
		MIN		-30.59		1013		-63.05		1321		0.30		1216		-0.20		1214	
		0.60		MAX		3.44		1214		89.29		1021		2.39		1013		0.35	
		MIN		-30.64		1013		-61.45		1321		0.30		1216		-0.06		1214	
		0.74		MAX		3.40		1214		97.08		1021		2.69		1113		0.64	
		MIN		-30.68		1013		-59.85		1321		0.30		1216		0.18		1216	
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1002	0.00	MAX	0.01	0.00	1212	1.23	1111	-9.39	1322	24.46	1114
	MIN	0.00	-0.12	0.00	1212	-0.26	1013	-16.25	1114	13.00	1222
	MAX	1.00	0.00	0.00	1212	1.26	1111	7.22	1222	10.14	1112
	MIN	0.00	-0.12	0.00	1212	-0.17	1213	-12.87	1114	4.69	1222
	MAX	2.00	0.01	0.00	1212	1.30	1111	5.95	1222	1.13	1322
	MIN	0.00	-0.12	0.00	1212	-0.12	1213	-9.48	1114	-1.74	1112
	MAX	3.00	0.01	0.00	1212	1.34	1111	-2.87	1222	-5.00	1221
	MIN	0.00	-0.12	0.00	1212	-0.06	1213	-6.10	1114	-9.08	1114
	MAX	4.00	0.01	0.00	1212	1.38	1111	-0.70	1222	-7.19	1222
	MIN	0.00	-0.12	0.00	1212	-0.12	1213	-2.95	1122	-13.50	1114
	MAX	5.00	0.01	0.00	1212	1.42	1111	1.71	1022	-6.81	1222
	MIN	0.00	-0.12	0.00	1212	-0.05	1213	-0.29	1322	-14.53	1114

0 STAAD SPACE

MAX/MIN FORCE VALUES FOR MEMB 1002, AMONGST ALL SECT LOCATIONS

MAX.	0.01	0.00	1212	1.42	5.00	1111	51.86	C	0.00	1111
MIN.	-16.25	0.00	1022	-0.25	0.00	1013	2.76	T	5.00	1322

1003	0.00	MAX	0.23	1111	1.42	1111	-0.13	1322	-6.81	1222
	MIN	0.00	-0.01	1322	0.05	1213	-1.37	1111	-14.53	1114
	MAX	1.00	0.00	1111	1.19	1111	2.73	1114	-6.78	1222
	MIN	0.00	-0.01	1322	0.00	1213	1.12	1222	-13.49	1114
	MAX	2.00	0.00	1111	1.06	1116	6.11	1114	4.57	1222
	MIN	0.00	-0.01	1322	-0.06	1213	3.29	1222	-9.06	1114
	MAX	3.00	0.00	1111	0.97	1116	9.49	1114	-0.19	1222
	MIN	0.00	-0.01	1322	-0.11	1213	5.47	1222	-1.32	1122
	MAX	4.00	0.00	1111	0.88	1116	12.88	1114	9.98	1113
	MIN	0.00	-0.01	1322	-0.17	1213	7.64	1222	6.03	1221
	MAX	5.00	0.00	1111	0.79	1116	16.26	1114	24.51	1114
	MIN	0.00	-0.01	1322	-0.24	1013	9.81	1222	14.96	1221

MAX/MIN FORCE VALUES FOR MEMB 1003, AMONGST ALL SECT LOCATIONS

MAX.	0.23	0.00	1111	1.42	0.00	1111	51.87	C	5.00	1111
MIN.	-1.37	0.00	1322	-0.24	5.00	1013	2.77	T	0.00	1322

1004	0.00	MAX	0.00	1322	0.00	1322	-5.55	1322	11.43	1116
	MIN	0.00	-0.01	1322	0.00	1322	-8.42	1322	7.32	1116
	MAX	0.52	0.00	1322	0.00	1322	-7.03	1322	4.76	1322
	MIN	0.00	-0.01	1322	0.00	1322	-3.39	1322	4.12	1116
	MAX	1.04	0.00	1322	0.00	1322	-5.27	1116	2.65	1322
	MIN	0.00	-0.01	1322	0.00	1322	-2.26	1322	1.83	1116
	MAX	1.56	0.00	1322	0.00	1322	-3.52	1116	1.18	1322
	MIN	0.00	-0.01	1322	0.00	1322	-1.13	1322	0.29	1322
	MAX	2.08	0.00	1322	0.00	1322	-1.76	1116	0.00	1322
	MIN	0.00	-0.01	1322	0.00	1322	0.00	1322	0.00	1322
	MAX	2.60	0.00	1322	0.00	1322	0.00	1322	0.00	1322
	MIN	0.00	-0.01	1322	0.00	1322	0.00	1322	0.00	1322

0 STAAD SPACE

MAX/MIN FORCE VALUES FOR MEMB 1004, AMONGST ALL SECT LOCATIONS

MAX.	0.00	0.00	1001	0.00	0.00	1001	0.00	2.60	1001
	0.00	2.60	1001	11.43	0.00	1001	0.00		
MIN.	-8.79	0.00	1116	0.00	2.60	1322	0.26	T	0.00
									1116

2001	0.00	MAX	0.00	1322	0.00	1322	0.00	1322	0.00	1322
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0.52	0.00	1322	0.00	1322	0.00	1322	0.00	1322	0.00	1322
MIN	0.00	-0.12	0.00	1213	-0.00	1322	0.00	1322	0.00	1322
MAX	1.04	0.00	0.00	1216	4.07	1013	0.00	1322	0.00	1322
MIN	0.00	-0.12	0.00	1216	-1.20	1216	0.00	1322	0.00	1322
MAX	1.56	0.00	0.00	1216	9.17	1013	0.00	1322	0.00	1322
MIN	0.00	-0.12	0.00	1216	-2.70	1216	0.00	1322	0.00	1322
MAX	2.08	0.00	0.00	1216	16.30	1013	0.00	1322	0.00	1322
MIN	0.00	-0.12	0.00	1216	-4.81	1216	0.00	1322	0.00	1322
MAX	2.60	0.00	0.00	1216	25.46	1013	0.00	1322	0.00	1322
MIN	0.00	-0.12	0.00	1216	-7.51	1216	0.00	1322	0.00	1322

MAX/MIN FORCE VALUES FOR MEMB 2001, AMONGST ALL SECT LOCATIONS

MAX.	5.77	2.60	1212	25.46	2.60	1013	0.18	C	2.60	1212
	0.00	0.00	1001	0.00	0.00	1001				
MIN.	-19.58	2.60	1013	-7.51	2.60	1216	0.61	T	2.60	1013
	0.00	2.60	1322	0.00	2.60	1322				

2002	0.00	MAX	35.71	1013	26.28	1013	0.03	1322	1.12	1011
	MIN	0.00	-9.65	1214	-7.53	1216	-0.13	1322	-0.13	1214
	MAX	1.00	0.00	1001	1.73	1214	0.02	1322	0.02	1214
	MIN	0.00	-0.01	1322	-8.68	1214	0.03	1322	0.03	1322
	MAX	2.00	0.00	1013	30.10	1013	-0.15	1112	-0.44	1214
	MIN	0.00	-0.01	1322	-13.78	1214	0.03	1322	0.03	1322
	MAX	3.00	0.00	1013	46.99	1013	-0.15	1112	-0.43	1214
	MIN	0.00	-0.01	1322	-17.08	1214	0.03	1322	0.03	1322
	MAX	4.00	0.00	1013	56.35	1013	-0.15	1112	-0.43	1214
	MIN	0.00	-0.01	1322	-18.57	1214	0.03	1322	0.03	1322
	MAX	5.00	0.00	1013	58.17	1013	-0.15	1112	-0.42	1214
	MIN	0.00	-0.01	1322	-1.94	1013	0.03	1322	0.03	1322

0 STAAD SPACE

MAX/MIN FORCE VALUES FOR MEMB 2002, AMONGST ALL SECT LOCATIONS

MAX.	35.71	0.00	1013	26.28	0.00	1013	63.45	C	0.00	1013
	0.03	0.00	1322	1.12	5.00	1011				
MIN.	-9.65	0.00	1214	-58.17	5.00	1013	11.26	C	5.00	1322
	-0.15	5.00	1112	-0.45	0.00	1214				

2003	0.00	MAX	5.99	1111	18.57	1214	0.07	1016	0.53	1001
	MIN	0.00	-0.01	1322	-17.09	1214	-0.07	1016	-0.42	1214
	MAX	1.00	0.00	1013	56.35	1013	-0.08	1212	-0.38	1214
	MIN	0.00	-0.01	1322	-13.80	1214	0.07	1016	0.65	1001
	MAX	2.00	0.00	1013	46.99	1013	-0.08	1212	-0.34	1214
	MIN	0.00	-0.01	1322	-8.69	1214	0.07	1016	0.71	1001
	MAX	3.00	0.00	1013	30.10	1013	-0.08	1212	-0.38	1212
	MIN	0.00	-0.01	1322	-5.68	1216	0.07	1016	0.76	1001
	MAX	4.00	0.00	1013	26.34	1015	-0.08	1212	-0.46	1212
	MIN	0.00	-0.01	1322	-6.98	1214	0.07	1016	0.82	1001
	MAX	5.00	0.00	1013	6.98	1214	-0.08	1212	-0.54	1212

MAX/MIN FORCE VALUES FOR MEMB 2003, AMONGST ALL SECT LOCATIONS

MAX.	9.64	5.00	1214	26.34	5.00	1015			
	0.07	0.00	1016	0.82	5.00	1001	63.45	C	5.00 1013
MIN.	-35.71	5.00	1013	-58.17	0.00	1013			
	-0.08	5.00	1212	-0.54	5.00	1212	11.26	C	0.00 1322

2004	0.00	MAX	19.58	1015	25.46	1015	0.00	1322	0.00	1322
		MIN	-4.72	1214	-6.34	1214	0.00	1322	0.00	1322
	0.52	MAX	13.46	1015	16.30	1015	0.00	1322	0.00	1322
		MIN	-3.77	1214	-3.93	1214	0.00	1322	0.00	1322
	1.04	MAX	13.77	1015	3.93	1214	0.00	1322	0.00	1322
		MIN	-7.83	1015	-2.21	1214	0.00	1322	0.00	1322
	1.56	MAX	7.83	1015	4.07	1015	0.00	1322	0.00	1322
		MIN	-7.83	1015	-4.07	1015	0.00	1322	0.00	1322

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Waste Stock Yard Rev 1- Rashidiya 4002, AMONGST ALL SECT LOCATIONS																									
MAX/MIN FORCE VALUES FOR MEMB		FY/ DIST		LD		MZ/ DIST		LD		FX		DIST													
		FY/ DIST		LD		MZ/ DIST		LD		FX		DIST													
MAX.		36.79		0.00 1013		51.14		0.00 1013		43.91 C		0.00 1011													
MIN.		-8.72		0.00 1216		-38.73		5.00 1013		12.50 T		0.00 1214													
		-0.04		5.00 1222		-0.20		0.00 1212																	

4003		0.00		MAX		6.70 1111		5.56 1212		0.04 1322		0.30 1013													
		MIN		-0.37 1214		-38.73 1013		-0.04 1222		-0.03 1214		0.03 1011													
		1.00		MAX		3.51 1212		3.16 1214		0.04 1322		0.29 1013													
		MIN		-6.66 1013		-35.83 1013		-0.04 1222		-0.02 1214		0.02 1011													
		2.00		MAX		3.26 1214		0.80 1214		0.04 1322		0.28 1013													
		MIN		-14.19 1013		-25.39 1013		-0.04 1222		-0.01 1214		0.01 1011													
		3.00		MAX		5.07 1214		0.37 1222		0.04 1322		0.31 1011													
		MIN		-21.72 1013		-7.43 1013		-0.04 1222		-0.05 1222		0.05 1222													
		4.00		MAX		6.89 1214		18.08 1013		0.04 1322		0.34 1011													
		MIN		-29.25 1013		-9.35 1214		-0.04 1222		-0.09 1222		0.09 1222													
		5.00		MAX		8.70 1214		51.11 1013		0.04 1322		0.37 1011													
		MIN		-36.78 1013		-17.15 1214		-0.04 1222		-0.14 1222		0.14 1222													

MAX/MIN FORCE VALUES FOR MEMB																									
		FY/ DIST		LD		MZ/ DIST		LD		FX		DIST													
		FY/ DIST		LD		MZ/ DIST		LD		FX		DIST													
MAX.		8.70		5.00 1214		51.11		5.00 1013		44.29 C		5.00 1011													
MIN.		-36.78		5.00 1013		-38.73		0.00 1013		12.50 T		5.00 1214													
		-0.04		5.00 1222		-0.14		5.00 1222																	

4004		0.00		MAX		19.58 1015		25.46 1015		0.00 1322		0.00 1322													
		MIN		-4.72 1214		-6.14 1214		0.00 1322		0.00 1322		0.00 1322													
		0.52		MAX		15.66 1015		16.30 1015		0.00 1322		0.00 1322													
		MIN		-3.77 1214		-3.93 1214		0.00 1322		0.00 1322		0.00 1322													
		1.04		MAX		11.75 1015		9.17 1015		0.00 1322		0.00 1322													
		MIN		-2.83 1214		-2.21 1214		0.00 1322		0.00 1322		0.00 1322													
		1.56		MAX		7.83 1015		4.07 1015		0.00 1322		0.00 1322													
		MIN		-1.89 1214		-0.98 1214		0.00 1322		0.00 1322		0.00 1322													
		2.08		MAX		3.92 1015		1.02 1015		0.00 1322		0.00 1322													
		MIN		-0.94 1214		-0.25 1214		0.00 1322		0.00 1322		0.00 1322													
		2.60		MAX		0.00 1322		0.00 1322		0.00 1322		0.00 1322													
		MIN		0.00 1322		0.00 1322		0.00 1322		0.00 1322		0.00 1322													

MAX/MIN FORCE VALUES FOR MEMB																									
		FY/ DIST		LD		MZ/ DIST		LD		FX		DIST													
		FY/ DIST		LD		MZ/ DIST		LD		FX		DIST													
MAX.		19.58		0.00 1011		25.46		0.00 1011		0.15 C		0.00 1214													
MIN.		-8.72		0.00 1214		-6.14		0.00 1214		0.61 T		0.00 1015													
		0.00		2.60 1322		0.00		2.60 1322																	

5001		0.00		MAX		0.00 1322		0.00 1322		0.00 1322		0.00 1322													
		MIN		0.00 1322		0.00 1322		0.00 1322		0.00 1322		0.00 1322													
		0.52		MAX		1.15 1212		1.02 1013		0.00 1322		0.00 1322													
		MIN		-3.92 1013		-0.30 1216		0.00 1322		0.00 1322		0.00 1322													
		1.04		MAX		2.31 1216		4.07 1013		0.00 1322		0.00 1322													
		MIN		-7.83 1013		-1.20 1216		0.00 1322		0.00 1322		0.00 1322													
		1.56		MAX		3.46 1216		9.17 1013		0.00 1322		0.00 1322													
		MIN		-11.75 1013		12.70 1216		0.00 1322		0.00 1322		0.00 1322													
		2.08		MAX		15.66 1013		16.30 1013		0.00 1322		0.00 1322													
		MIN		-15.66 1013		-4.81 1216		0.00 1322		0.00 1322		0.00 1322													
		2.60		MAX		5.77 1216		25.46 1013		0.00 1322		0.00 1322													
		MIN		-19.58 1013		-7.51 1216		0.00 1322		0.00 1322		0.00 1322													

MAX/MIN FORCE VALUES FOR MEMB																									
		FY/ DIST		LD		MZ/ DIST		LD		FX		DIST													
		FY/ DIST		LD		MZ/ DIST		LD		FX		DIST													
MAX.		5.77		2.60 1212		25.46		2.60 1013		0.18 C		2.60 1212													
MIN.		0.00		0.00 1001		0.00		0.00 1001																	

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MIN.		-19.58		2.60 1013		-7.51		2.60 1216		0.61 T		2.60 1013	
0.00		2.60 1322		0.00		2.60 1322		0.00		2.60 1322		0.00	
5002													
0.00		MAX		40.56 1013		25.53 1013		0.04 1211		0.12 1013		0.12 1013	
MIN		MIN		-9.02 1214		-7.52 1216		-0.01 1222		-0.22 1212		-0.22 1212	
1.00		MAX		33.03 1013		1.96 1214		0.04 1211		0.12 1013		0.12 1013	
MIN		MIN		-7.21 1214		-11.58 1013		-0.01 1222		-0.18 1212		-0.18 1212	
2.00		MAX		25.50 1013		8.72 1214		0.04 1211		0.11 1013		0.11 1013	
MIN		MIN		-15.39 1214		-9.56 1216		-0.01 1222		-0.11 1013		-0.11 1013	
3.00		MAX		3.58 1214		62.10 1013		-0.04 1222		-0.09 1212		-0.09 1212	
MIN		MIN		-10.51 1001		35.43 1214		0.04 1211		0.11 1013		0.11 1013	
4.00		MAX		16.76 1214		-76.51 1013		-0.01 1222		-0.05 1212		-0.05 1212	
MIN		MIN		-6.79 1015		16.28 1214		0.04 1211		0.11 1013		0.11 1013	
5.00		MAX		-0.05 1113		-83.19 1013		-0.01 1222		-0.01 1214		-0.01 1214	
MIN		MIN		-0.05 1113		-83.19 1013		-0.01 1222		-0.01 1214		-0.01 1214	
STAAD SPACE													
5002, AMONGST ALL SECT LOCATIONS													
MAX/MIN FORCE VALUES FOR MEMB		FY/ DIST		LD		MZ/ DIST		LD		FX		DIST	
		FY/ DIST		LD		MZ/ DIST		LD		FX		DIST	
		FY/ DIST		LD		MZ/ DIST		LD		FX		DIST	
MAX.		40.56		0.00 1013		25.53		0.00 1013		4.43 C		0.00 1001	
MIN.		-9.02		0.00 1111		-83.19		5.00 1013		1.92 T		0.00 1214	
		-0.01		5.00 1222		-0.22		0.00 1212					
5003													
0.00		MAX		4.59 1111		16.28 1214		0.04 1211		0.11 1013		0.11 1013	
MIN		MIN		-3.90 1001		-83.19 1013		-0.01 1222		-0.01 1214		-0.01 1214	
1.00		MAX		1.96 1212		35.43 1214		0.04 1211		0.12 1011		0.12 1011	
MIN		MIN		-10.51 1001		-76.51 1013		-0.01 1222		0.02 1222		0.02 1222	
2.00		MAX		15.39 1214		25.50 1013		0.04 1211		0.11 1013		0.11 1013	
MIN		MIN		-5.39 1213		8.72 1214		-0.04 1222		0.00 1322		0.00 1322	
3.00		MAX		25.50 1013		-40.56 1013		-0.01 1222		0.19 1011		0.19 1011	
MIN		MIN		-7.21 1214		1.96 1214		-0.01 1222		-0.23 1011		-0.23 1011	
4.00		MAX		-33.03 1013		-11.28 1013		-0.01 1222		-0.02 1322		-0.02 1322	
MIN		MIN		9.02 1214		25.53 1013		0.04 1211		0.27 1011		0.27 1011	
5.00		MAX		-40.56 1013		-6.16 1214		-0.01 1222		-0.03 1322		-0.03 1322	
MIN		MIN		-40.56 1013		-6.16 1214		-0.01 1222		-0.03 1322		-0.03 1322	
5004													
0.00		MAX		19.58 1015		25.46 1015		0.00 1322		0.00 1322		0.00 1322	
MIN		MIN		-4.72 1214		-6.14 1214		0.00 1322		0.00 1322		0.00 1322	
0.52		MAX		15.66 1015		16.30 1015		0.00 1322		0.00 1322		0.00 1322	
MIN		MIN		-3.77 1214		-3.93 1214		0.00 1322		0.00 1322		0.00 1322	
1.04		MAX		11.75 1015		9.17 1015		0.00 1322		0.00 1322		0.00 1322	
MIN		MIN		-7.83 1214		-2.07 1215		0.00 1322		0.00 1322		0.00 1322	
1.56		MAX		1.89 1214		-0.98 1214		0.00 1322		0.00 1322		0.00 1322	
MIN		MIN		-3.92 1015		1.02 1015		0.00 1322		0.00 1322		0.00 1322	
2.08		MAX		-0.94 1214		-0.25 1214		0.00 1322		0.00 1322		0.00 1322	
MIN		MIN		0.00 1322		0.00 1322		0.00 1322		0.00 1322		0.00 1322	
2.60		MAX		0.00 1322		0.00 1322		0.00 1322		0.00 1322		0.00 1322	
MIN		MIN		0.00 1322		0.00 1322		0.00 1322		0.00 1322		0.00 1322	
STAAD SPACE													
5004, AMONGST ALL SECT LOCATIONS													
MAX/MIN FORCE VALUES FOR MEMB		FY/ DIST		LD		MZ/ DIST		LD		FX		DIST	
		FY/ DIST		LD		MZ/ DIST		LD		FX		DIST	
		FY/ DIST		LD		MZ/ DIST		LD		FX		DIST	
MAX.		19.58		0.00 1011		25.46		0.00 1011		0.15 C		0.00 1214	
MIN.		-4.72		0.00 1001		0.00		0.00 1001		0.61 T		0.00 1015	
		-0.00		0.00 1214		-6.14		0.00 1214					
		0.00		2.60 1322		0.00		2.60 1322					
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[illegible]

MAX/MIN FORCE VALUES FOR MEMB 6001, AMONGST ALL SECT LOCATIONS

	FZ	DIST	LD	MZ	DIST	LD	FX	DIST	LD
MAX.	5.77	2.60	1212	25.46	2.60	1013			
	0.00	0.00	1001	0.00	0.00	1001			
MIN.	-19.58	2.60	1013	-7.51	2.60	1216	0.18	C	2.60 1212
	0.00	2.60	1322	0.00	2.60	1322	0.62	T	2.60 1011
5002	0.00	MAX	37.52 1013	52.39 1013			0.23 1214		
	MIN	-8.56 1214	-17.18 1214	-17.18 1214			0.01 1322		0.06 1322
1.00	MAX	29.99 1013	18.61 1214	18.61 1214			0.23 1314		-0.05 1322
	MIN	-6.75 1214	-9.52 1013	-9.52 1013			0.01 1322		-0.94 1214
2.00	MAX	22.46 1013	5.36 1211	5.36 1211			0.23 1214		0.04 1322
	MIN	-4.93 1214	-7.63 1013	-7.63 1013			0.01 1322		-0.71 1214
3.00	MAX	34.72 1013	6.78 1212	6.78 1212			0.23 1314		0.03 1322
	MIN	-7.03 1213	-27.03 1213	-27.03 1213			0.01 1322		-0.48 1214
4.00	MAX	7.40 1013	37.51 1013	37.51 1013			0.01 1322		-0.21 1322
	MIN	-1.30 1214	-37.51 1013	-37.51 1013			0.23 1214		0.01 1111
5.00	MAX	4.64 1211	5.06 1211	5.06 1211			0.01 1322		-0.06 1014
	MIN	-0.96 1022	-41.15 1013	-41.15 1013			-0.01 1322		-0.06 1014
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MIN	-0.96	1022	-41.15	1013	-0.01	1322	-0.05
STAAD SPACE						--	PAGE NO.

[illegible]

	FY/ FZ	DIST LD	MZ/ WV	DIST LD	FX	DIST LD
MAX.	37.52	0.00 1013	57.39	0.00 1013		
	0.23	0.00 1012	6.38	0.00 1013		
MIN.	-8.56	0.00 1214	-41.15	5.00 1013	29.42 C	0.00 1111
	-0.01	5.00 1322	-1.15	0.00 1214	17.90 T	0.00 1214

MAX/MIN FORCE VALUES FOR MEMB 6003, AMONGST ALL SECT LOCATIONS

	FV/ FZ	LD DIST	LD DIST	MZ/ MY	DIST LD	LD DIST	FX	DIST	LD
MAX.	8.48	5.00	1214	52.45	5.00	1013			
	0.22	5.00	1114	5.00	1114				
	-37.54	5.00	1013	-4.11	5.00	1114	29.86	C	5.00 1111
MIN.	-0.01	5.00	1322	-0.07	5.00	1122	17.89	T	5.00 1214
004	0.00	MAX	19.58	1015	25.46	1015	0.00	1322	0.00 1322
		MIN	-4.72	1214	-6.14	1214	0.00	1322	0.00 1322
	0.52	MAX	15.66	1015	16.30	1015	0.00	1322	0.00 1322
		MIN	-3.77	1214	-3.93	1214	0.00	1322	0.00 1322
	1.04	MAX	11.75	1015	9.17	1015	0.00	1322	0.00 1322

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STAAD SPACE	MIN	0.00	1332	0.00	1332	0.00	1332
2.60	MAX	0.00	1332	0.00	1332	0.00	1332
-0.94	MIN	-0.94	1214	-0.25	1214	0.00	1332
3.92	MAX	1.89	1215	1.92	1015	0.00	1332
1.56	MIN	-1.89	1214	-0.98	1214	0.00	1332
MAX	7.83	1214	7.83	1214	4.67	1015	0.00
MIN	-2.83	1214	-2.83	1214	0.00	1332	0.00

STAAD SPACE

2 0.00
PAGE NO.

MAX/MIN FORCE VALUES FOR MEMB 6004, AMONGST ALL SECT LOCATIONS

	Fz	DIST	LD	MZ	DIST	LD	FX	DIST	LD
MAX.	29.58	0.00	1011	25.46	0.00	1011			
	0.00	0.00	1001	0.00	0.00	1011			
	-4.72	0.00	1214	-6.74	0.00	1214	0.15	C	0.00 1214
MIN.	0.00	2.60	1322	0.00	2.60	1322	0.62	T	0.00 1015

MAX/MIN FORCE VALUES FOR MEMB 7001, AMONGST ALL SECT LOCATIONS

	FV/ FZ	DIST LD	MZ/ MY	DIST LD	PX	DIST LD
MAX.	5.77	2.60 1212	25.46	2.60 1013		
	0.00	0.00 1001	0.00	0.00 1216		
MIN.	-19.58	2.60 1013	-7.51	2.60 1216	0.18 C	2.60 1212
	0.00	2.60 1322	0.00	2.60 1322	0.62 T	2.60 1013
002	0.00 MAX	37.61 1013	25.53 1013	0.16 1214	0.12 1322	
	NIN	6.92 1214	-1.87 1216	-0.16 1314	-0.11 1314	
1.00	MAX	30.08 1013	-8.33 1214	0.03 1322	-0.55 1114	
	NIN	-7.11 1214	0.16 1214	0.16 1114	0.05 1322	
2.00	MAX	22.55 1013	8.07 1214	-0.03 1322	-0.39 1114	
	NIN	5.29 1214	-34.65 1013	0.16 1214	0.02 1322	
3.00	MAX	15.02 1013	12.46 1214	-0.03 1322	-0.24 1114	
	NIN	-3.48 1214	-53.44 1013	0.09 1114	0.00 1216	
4.00	MAX	7.66 1013	6.57 1214	0.16 1214	-0.16 1314	
	NIN	-1.66 1214	15.79 1214	0.16 1114	0.02 1212	
5.00	MAX	4.48 1215	-68.42 1013	-0.03 1322	-0.12 1001	
	NIN	-0.09 1001			-- PAGE NO. 48	
STAAD SPACE						

STAAD SPACE

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MAX/MIN FORCE VALUES FOR MEMB 7002, AMONGST ALL SECT LOCATIONS

	FY/ FZ	DIST	LD	MZ/ MY	DIST	LD	FX	DIST	LD
MAX.	37.61	0.00	1013	25.53	0.00	1013			
	0.16	0.00	1114	0.12	0.00	1322	3.95	C	0.00 1001
MIN.	-8.92	0.00	1214	-68.42	5.00	1013			
	-0.03	5.00	1322	-0.71	0.00	1114	5.04	T	0.00 1214
0003	0.00	MAX	4.58	1111	15.79	1214	0.16	1214	0.10 1214
		MIN	-0.15	1214	-68.42	1013	-0.03	1322	-0.12 1001
1.00	MAX	1.95	1216	0.02	0.02	1214	0.16	1312	0.16 1214
	MIN	-7.48	1013	-64.70	1013	0.03	0.03	1322	-0.12 1001
2.00	MAX	3.48	1214	12.45	1214	0.16	0.16	1214	0.42 1214
	MIN	-15.01	1013	-53.45	1013	0.03	0.03	1322	-0.14 1122
3.00	MAX	5.30	1214	8.05	1214	0.16	0.16	1214	0.58 1214
	MIN	-22.54	1013	-34.66	1013	0.03	0.03	1322	-0.17 1122
4.00	MAX	7.11	1214	1.86	1216	0.16	0.16	1214	0.74 1214
	MIN	-11.11	1013	-48.42	1013	0.03	0.03	1322	-0.12 1001

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		8002, AMONGST ALL SECT LOCATIONS			
MAX/MIN	FORCE VALUES FOR MEMB	FY/	DIST	LD	
		FZ	DIST	LD	
MAX.	34.35	0.00	1013	47.66	0.00 1013
MIN.	-8.27	0.00	1014	-36.18	5.00 1013
	-0.06	5.00	1022	-0.39	0.00 1022

8003	0.00	MAX	6.19	1111	4.52	1212	0.06	1322	-0.03	1216
	1.00	MIN	-0.47	1214	-36.18	1013	-0.06	1322	-0.20	1013
	2.00	MAX	3.13	1212	3.35	1214	0.06	1322	0.01	1222
	3.00	MIN	-0.83	1214	-31.37	1013	-0.06	1322	-0.23	1013
	4.00	MAX	13.56	1013	-23.52	1013	0.06	1322	-0.30	1013
	5.00	MIN	-4.49	1214	-0.41	1222	0.06	1322	0.13	1222
	6.00	MAX	-20.40	1013	-6.63	1013	-0.06	1322	-0.34	1013
	7.00	MIN	6.14	1214	17.30	1013	0.06	1322	0.19	1222
	8.00	MAX	-27.44	1013	-7.65	1214	-0.06	1322	-0.39	1113
	9.00	MIN	7.79	1214	48.27	1013	0.06	1322	0.25	1222
	10.00	MAX	-34.48	1013	-14.62	1214	-0.06	1322	-0.45	1113

		8003, AMONGST ALL SECT LOCATIONS			
MAX/MIN	FORCE VALUES FOR MEMB	FY/	DIST	LD	
		FZ	DIST	LD	
MAX.	7.79	5.00	1214	48.27	5.00 1013
MIN.	-34.48	5.00	1013	-36.18	0.00 1013
	-0.06	5.00	1322	-0.45	5.00 1113

8004	0.00	MAX	18.30	1015	23.80	1015	0.00	1322	0.00	1322
	0.52	MIN	-4.30	1214	-5.59	1214	0.00	1322	0.00	1322
	1.04	MAX	14.04	1015	13.23	1015	0.00	1322	0.00	1322
	1.56	MIN	-0.58	1214	-8.59	1214	0.00	1322	0.00	1322
	2.08	MAX	7.32	1015	3.81	1015	0.00	1322	0.00	1322
	2.60	MIN	-1.72	1214	-0.89	1214	0.00	1322	0.00	1322
	3.12	MAX	3.66	1015	0.95	1015	0.00	1322	0.00	1322
	3.68	MIN	-0.86	1214	-0.22	1214	0.00	1322	0.00	1322
	4.20	MAX	0.00	1322	0.00	1322	0.00	1322	0.00	1322
	4.72	MIN	0.00	1322	0.00	1322	0.00	1322	0.00	1322

0 STAAD SPACE

		8004, AMONGST ALL SECT LOCATIONS			
MAX/MIN	FORCE VALUES FOR MEMB	FY/	DIST	LD	
		FZ	DIST	LD	
MAX.	18.30	0.00	1011	23.80	0.00 1011
MIN.	-4.30	0.00	1001	0.00	0.00 1001
	0.00	2.60	1322	-5.59	0.00 1214
				0.00	2.60 1322

9001	0.00	MAX	0.00	1322	0.00	1322	0.00	1322	0.00	1322
	0.52	MIN	-0.00	1322	-0.00	1322	0.00	1322	0.00	1322
	1.04	MAX	2.89	1216	-0.99	1216	0.00	1322	0.00	1322
	1.56	MIN	-5.31	1013	-2.78	1013	0.00	1322	0.00	1322
	2.08	MAX	4.04	1216	-1.40	1216	0.00	1322	0.00	1322
	2.60	MIN	-7.97	1013	-3.15	1216	0.00	1322	0.00	1322
	3.12	MAX	5.38	1216	11.05	1013	0.00	1322	0.00	1322
	3.68	MIN	-10.62	1013	-5.60	1216	0.00	1322	0.00	1322
	4.20	MAX	6.73	1216	17.27	1013	0.00	1322	0.00	1322
	4.72	MIN	-13.28	1013	-8.75	1216	0.00	1322	0.00	1322

		9001, AMONGST ALL SECT LOCATIONS			
MAX/MIN	FORCE VALUES FOR MEMB	FY/	DIST	LD	
		FZ	DIST	LD	
MAX.	0.00	0.00	1322	0.00	1322
MIN.	-0.00	0.00	1322	-0.00	1322
	0.00	2.60	1322	-5.59	0.00 1214
				0.00	2.60 1322

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5.00	MIN	-30.07	1013	-0.20	1122
	MAX	26.34	1013	-0.03	1322
	MIN	-9.93	1013	0.16	1214
	MIN	-37.60	1013	-0.03	1322

		7003, AMONGST ALL SECT LOCATIONS			
MAX/MIN	FORCE VALUES FOR MEMB	FY/	DIST	LD	
		FZ	DIST	LD	
MAX.	8.93	5.00	1214	25.52	5.00 1013
MIN.	-37.60	0.00	1013	-68.42	0.00 1013
	-0.03	5.00	1322	-0.23	5.00 1122

7004	0.00	MAX	19.58	1015	25.46	1015	0.00	1322	0.00	1322
	0.52	MIN	-4.72	1214	-6.14	1214	0.00	1322	0.00	1322
	1.04	MAX	15.66	1015	16.30	1015	0.00	1322	0.00	1322
	1.56	MIN	-3.77	1214	-3.93	1214	0.00	1322	0.00	1322
	2.08	MAX	11.75	1015	9.17	1015	0.00	1322	0.00	1322
	2.60	MIN	-2.83	1214	-2.21	1214	0.00	1322	0.00	1322
	3.12	MAX	7.83	1015	4.07	1015	0.00	1322	0.00	1322
	3.68	MIN	-1.89	1214	-0.98	1214	0.00	1322	0.00	1322
	4.20	MAX	3.92	1015	1.02	1015	0.00	1322	0.00	1322
	4.72	MIN	-0.84	1214	-0.25	1214	0.00	1322	0.00	1322
	5.24	MAX	0.00	1322	0.00	1322	0.00	1322	0.00	1322
	5.76	MIN	0.00	1322	0.00	1322	0.00	1322	0.00	1322

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		7004, AMONGST ALL SECT LOCATIONS			
MAX/MIN	FORCE VALUES FOR MEMB	FY/	DIST	LD	
		FZ	DIST	LD	
MAX.	19.58	0.00	1011	25.46	0.00 1011
MIN.	-4.72	0.00	1001	-6.14	0.00 1011
	0.00	2.60	1322	0.00	2.60 1322

8001	0.00	MAX	0.00	1322	0.00	1322	0.00	1322	0.00	1322
	0.52	MIN	1.06	1216	0.95	1013	0.00	1322	0.00	1322
	1.04	MAX	-3.66	1013	-0.27	1216	0.00	1322	0.00	1322
	1.56	MIN	2.11	1216	3.81	1013	0.00	1322	0.00	1322
	2.08	MAX	-7.32	1013	-1.10	1216	0.00	1322	0.00	1322
	2.60	MIN	3.17	1216	8.57	1013	0.00	1322	0.00	1322
	3.12	MAX	-10.99	1013	-2.47	1216	0.00	1322	0.00	1322
	3.68	MIN	4.24	1216	13.23	1013	0.00	1322	0.00	1322
	4.20	MAX	-14.58	1013	-23.80	1013	0.00	1322	0.00	1322
	4.72	MIN	18.30	1013	-6.87	1216	0.00	1322	0.00	1322

		8001, AMONGST ALL SECT LOCATIONS			
MAX/MIN	FORCE VALUES FOR MEMB	FY/	DIST	LD	
		FZ	DIST	LD	
MAX.	5.28	2.60	1212	23.80	2.60 1013
MIN.	-18.30	0.00	1001	0.00	0.00 1001
	0.00	2.60	1013	-6.87	2.60 1216
				0.00	2.60 1322

8002	0.00	MAX	34.35	1013	47.66	1013	0.06	1322	0.24	1322
	1.00	MIN	-8.27	1214	-17.02	1214	-0.06	1322	-0.39	1022
	2.00	MAX	27.32	1013	16.81	1013	0.06	1322	0.18	1322
	3.00	MIN	-6.62	1214	-9.57	1214	-0.06	1322	-0.33	1022
	4.00	MAX	20.28	1013	4.48	1211	0.06	1322	0.12	1322
	5.00	MIN	-4.97	1214	-7.00	1013	-0.06	1322	-0.27	1022
	6.00	MAX	13.24	1013	5.80	1212	0.06	1322	0.06	1322
	7.00	MIN	-3.31	1214	-23.77	1013	-0.06	1322	-0.21	1022
	8.00	MAX	6.20	1013	6.18	1212	0.06	1322	0.00	1322
	9.00	MIN	-1.66	1214	-33.49	1013	-0.06	1322	-0.16	1013
	10.00	MAX	3.52	1211	4.52	1212	0.06	1322	-0.03	1216
	11.00	MIN	-1.36	1022	-36.18	1013	-0.06	1322	-0.20	1013

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MAX.	6.73	2.60	1212	17.27	2.60	1013	0.20	C	2.60	1212
MIN.	-13.28	2.60	1013	-8.75	2.60	1216	0.39	T	2.60	1013

9002	0.00	MAX	23.97	1013	17.28	1013	0.11	1222	0.51	1322
	0.00	MIN	-10.55	1216	-8.76	1216	-0.11	1322	-0.57	1022
	1.00	MAX	19.86	1013	1.62	1214	0.11	1322	0.40	1322
	2.00	MAX	13.75	1013	7.15	1013	-0.11	1322	-0.47	1022
	3.00	MAX	-5.38	1216	-20.46	1013	-0.11	1322	0.29	1322
	4.00	MAX	8.65	1013	11.27	1216	-0.11	1322	-0.16	1322
	5.00	MAX	-2.79	1216	-11.67	1013	-0.11	1322	-0.25	1022
	6.00	MAX	3.75	1011	12.78	1216	-0.11	1322	0.08	1322
	7.00	MAX	-1.51	1214	-37.76	1013	-0.11	1322	-0.15	1022
	8.00	MAX	2.45	1212	11.83	1214	-0.11	1322	0.04	1216
	9.00	MAX	-1.56	1013	-38.75	1013	-0.11	1322	-0.09	1113

STAAD SPACE
MAX/MIN FORCE VALUES FOR MEMB 9002, AMONGST ALL SECT LOCATIONS
FY/ DIST LD MZ/ DIST LD FX DIST LD
MAX. 23.97 0.00 1013 17.28 0.00 1013 53.24 C 0.00 1013
MIN. -10.55 0.00 1216 -8.75 5.00 1013 11.85 C 5.00 1322
0.11 5.00 1322

9003	0.00	MAX	3.80	1115	11.83	1214	0.11	1222	0.04	1216
	1.00	MIN	-0.35	1322	-38.75	1013	-0.11	1322	-0.09	1113
	2.00	MAX	2.84	1216	30.95	1214	0.11	1222	0.08	1322
	3.00	MAX	-3.75	1214	-6.89	1013	-0.11	1322	-0.14	1122
	4.00	MAX	-8.65	1013	-31.67	1013	-0.11	1322	0.18	1222
	5.00	MAX	3.60	1214	-5.75	1214	-0.11	1322	-0.25	1322
	6.00	MAX	-13.75	1013	-20.46	1013	-0.11	1322	-0.35	1122
	7.00	MAX	4.64	1214	2.75	1216	-0.11	1322	0.39	1222
	8.00	MAX	-18.86	1013	-4.14	1013	-0.11	1322	-0.46	1122
	9.00	MAX	5.69	1214	17.28	1013	0.11	1222	0.50	1222
	10.00	MAX	-23.97	1013	-3.54	1214	-0.11	1322	-0.56	1122

STAAD SPACE
MAX/MIN FORCE VALUES FOR MEMB 9003, AMONGST ALL SECT LOCATIONS
FY/ DIST LD MZ/ DIST LD FX DIST LD
MAX. 5.69 5.00 1214 17.28 5.00 1013 53.24 C 5.00 1013
MIN. -23.97 5.00 1013 -38.75 0.00 1013 11.85 C 0.00 1322
-0.11 5.00 1322

9004	0.00	MAX	13.28	1015	17.27	1015	0.00	1322	0.00	1322
	0.52	MIN	-2.72	1214	-3.54	1214	0.00	1322	0.00	1322
	1.04	MAX	10.62	1015	11.05	1015	0.00	1322	0.00	1322
	1.56	MAX	-2.46	1214	-2.27	1214	0.00	1322	0.00	1322
	2.08	MAX	-7.59	1015	-6.22	1015	0.00	1322	0.00	1322
	2.60	MAX	1.31	1015	1.77	1214	0.00	1322	0.00	1322
	3.12	MAX	-1.09	1214	-0.59	1215	0.00	1322	0.00	1322
	3.64	MAX	1.06	1015	0.69	1015	0.00	1322	0.00	1322
	4.16	MAX	-0.34	1214	-0.14	1214	0.00	1322	0.00	1322
	4.68	MAX	0.00	1322	0.00	1322	0.00	1322	0.00	1322

STAAD SPACE
MAX/MIN FORCE VALUES FOR MEMB 9004, AMONGST ALL SECT LOCATIONS
FY/ DIST LD MZ/ DIST LD FX DIST LD
MAX. 13.28 0.00 1011 17.27 0.00 1011 0.08 C 0.00 1214
MIN. -2.72 0.00 1214 -3.54 0.00 1214 0.39 T 0.00 1015
0.00 2.60 1322

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10001	0.00	MAX	0.00	1322	0.00	1322	0.00	1322	0.00	1322
	0.52	MIN	-0.80	1216	-0.77	1013	0.00	1322	0.00	1322
	1.04	MAX	-2.97	1013	-0.21	1216	0.00	1322	0.00	1322
	1.56	MAX	1.59	1216	3.09	1013	0.00	1322	0.00	1322
	2.08	MAX	-3.39	1216	-0.83	1216	0.00	1322	0.00	1322
	2.60	MAX	-8.90	1013	6.95	1013	0.00	1322	0.00	1322
	3.12	MAX	-3.19	1216	-1.87	1216	0.00	1322	0.00	1322
	3.64	MAX	3.19	1216	12.35	1013	0.00	1322	0.00	1322
	4.16	MAX	-11.87	1013	-3.32	1216	0.00	1322	0.00	1322
	4.68	MAX	3.99	1216	19.30	1013	0.00	1322	0.00	1322
	5.20	MAX	-14.84	1013	-5.18	1216	0.00	1322	0.00	1322

STAAD SPACE
MAX/MIN FORCE VALUES FOR MEMB 10001, AMONGST ALL SECT LOCATIONS
FY/ DIST LD MZ/ DIST LD FX DIST LD
MAX. 3.99 2.60 1212 19.30 2.60 1013 0.12 C 2.60 1212
MIN. -14.84 0.00 1001 -5.18 2.60 1216 0.46 T 2.60 1013
0.00 2.60 1322

10002	0.00	MAX	26.84	1013	19.88	1013	0.13	1022	1.07	1214
	1.00	MIN	-26.80	1214	-1.13	1216	-0.13	1214	-1.03	1011
	2.00	MAX	-5.58	1214	-4.72	1013	-0.15	1214	-0.92	1214
	3.00	MAX	15.53	1013	6.22	1214	-0.13	1022	-0.72	1011
	4.00	MAX	-4.32	1214	-22.61	1013	-0.15	1214	-0.81	1011
	5.00	MAX	9.82	1013	9.92	1214	-0.13	1022	0.62	1214
	6.00	MAX	-30.88	1214	-35.29	1013	-0.15	1214	-0.70	1011
	7.00	MAX	4.12	1013	12.38	1214	-0.13	1022	0.47	1214
	8.00	MAX	-1.84	1214	-42.26	1013	-0.15	1214	-0.59	1011
	9.00	MAX	2.49	1211	13.60	1214	-0.13	1022	0.32	1214
	10.00	MAX	-1.59	1013	-43.53	1013	-0.15	1214	-0.53	1013

STAAD SPACE
MAX/MIN FORCE VALUES FOR MEMB 10002, AMONGST ALL SECT LOCATIONS
FY/ DIST LD MZ/ DIST LD FX DIST LD
MAX. 26.84 0.00 1013 19.88 0.00 1013 51.84 C 0.00 1013
MIN. 0.13 0.00 1022 -43.53 5.00 1013 8.95 C 5.00 1322
-6.80 0.00 1214 -1.03 0.00 1011

10003	0.00	MAX	4.63	1111	13.60	1214	0.05	1222	0.32	1214
	1.00	MIN	-0.31	1322	-43.53	1013	-0.05	1222	-0.13	1214
	2.00	MAX	2.35	1212	12.41	1213	-0.05	1222	-0.60	1013
	3.00	MAX	-4.12	1013	-42.26	1013	-0.20	1114	-0.13	1216
	4.00	MAX	3.05	1214	9.97	1214	0.05	1222	-0.67	1013
	5.00	MAX	-9.82	1013	-35.29	1013	-0.20	1114	-0.11	1216
	6.00	MAX	4.29	1214	6.30	1214	-0.05	1222	-0.74	1013
	7.00	MAX	-15.53	1013	-22.60	1013	-0.20	1114	-0.09	1216
	8.00	MAX	5.53	1214	2.06	1216	0.05	1222	-0.81	1013
	9.00	MAX	-21.24	1013	-4.21	1013	-0.20	1114	-0.07	1216
	10.00	MAX	6.77	1214	19.95	1013	0.05	1222	-0.90	1122
	11.00	MAX	-26.94	1013	-4.76	1214	-0.20	1114	-0.90	1122

STAAD SPACE
MAX/MIN FORCE VALUES FOR MEMB 10003, AMONGST ALL SECT LOCATIONS
FY/ DIST LD MZ/ DIST LD FX DIST LD
MAX. 6.77 5.00 1214 19.95 5.00 1015 51.84 C 5.00 1013
MIN. 0.05 0.00 1322 0.32 0.00 1214 8.95 C 0.00 1322
-26.94 5.00 1013 -43.53 5.00 1122

10004	0.00	MAX	14.84	1015	19.30	1015	0.00	1322	0.00	1322
	0.52	MIN	-3.22	1214	-4.19	1214	0.00	1322	0.00	1322
	1.04	MAX	11.87	1015	12.33	1015	0.00	1322	0.00	1322

Waste Stock Yard Rev 1- Rashidiya

MAX/MIN FORCE VALUES FOR MEMB 20002, AMONGST ALL SECT LOCATIONS											
	FY/	DIST	LD	MZ/	DIST	LD	FX	DIST	LD		
MAX.	0.57	0.00	1001	0.00	0.00	1001					
MIN.	-0.57	0.00	10.31	0.00	0.00	10.31	6.67 T	10.31	1221		
	0.00	10.31	1322	0.00	10.31	1322	12.98 T	0.00	1114		
20003											
MAX.	0.00	MAX	0.58	0.00	1322	0.00	1322	0.00	1322	0.00	1322
MIN.	2.04	MIN	0.37	0.00	1322	0.00	1322	0.00	1322	0.00	1322
	4.08	MAX	0.00	0.00	1322	0.00	1322	0.00	1322	0.00	1322
	6.12	MIN	0.00	0.00	1322	0.00	1322	0.00	1322	0.00	1322
	8.16	MAX	0.00	0.00	1322	0.00	1322	0.00	1322	0.00	1322
	10.20	MIN	0.00	0.00	1322	0.00	1322	0.00	1322	0.00	1322
		STAAD SPACE	-0.58	0.00	1116	0.00	1322	0.00	1322	0.00	1322

MAX/MIN FORCE VALUES FOR MEMB 20003, AMONGST ALL SECT LOCATIONS											
	FY/	DIST	LD	MZ/	DIST	LD	FX	DIST	LD		
MAX.	0.58	0.00	1001	0.00	0.00	1001					
MIN.	-0.58	0.00	10.20	0.00	10.20	1322	11.77 C	0.00	1322		
	0.00	10.20	1322	0.00	10.20	1322	41.26 T	10.20	1022		
20004											
MAX.	0.00	MAX	0.58	0.00	1116	0.00	1322	0.00	1322	0.00	1322
MIN.	2.04	MIN	0.37	0.00	1322	0.00	1322	0.00	1322	0.00	1322
	4.08	MAX	0.00	0.00	1322	0.00	1322	0.00	1322	0.00	1322
	6.12	MIN	0.00	0.00	1322	0.00	1322	0.00	1322	0.00	1322
	8.16	MAX	0.00	0.00	1322	0.00	1322	0.00	1322	0.00	1322
	10.20	MIN	0.00	0.00	1322	0.00	1322	0.00	1322	0.00	1322
		STAAD SPACE	-0.58	0.00	1116	0.00	1322	0.00	1322	0.00	1322

MAX/MIN FORCE VALUES FOR MEMB 20004, AMONGST ALL SECT LOCATIONS											
	FY/	DIST	LD	MZ/	DIST	LD	FX	DIST	LD		
MAX.	0.58	0.00	1001	0.00	0.00	1001					
MIN.	-0.58	0.00	10.20	0.00	10.20	1322	2.58 T	10.20	1321		
	0.00	10.20	1322	0.00	10.20	1322	26.90 T	0.00	1021		
20005											
MAX.	0.00	MAX	0.58	0.00	1116	0.00	1322	0.00	1322	0.00	1322
MIN.	2.04	MIN	0.37	0.00	1322	0.00	1322	0.00	1322	0.00	1322
	4.09	MAX	0.00	0.00	1322	0.00	1322	0.00	1322	0.00	1322
	6.13	MIN	0.00	0.00	1322	0.00	1322	0.00	1322	0.00	1322
	8.17	MAX	0.00	0.00	1322	0.00	1322	0.00	1322	0.00	1322
	10.22	MIN	0.00	0.00	1322	0.00	1322	0.00	1322	0.00	1322
		STAAD SPACE	-0.58	0.00	1116	0.00	1322	0.00	1322	0.00	1322

MAX/MIN FORCE VALUES FOR MEMB 20005, AMONGST ALL SECT LOCATIONS											
	FY/	DIST	LD	MZ/	DIST	LD	FX	DIST	LD		
MAX.	0.59	0.00	1001	0.00	0.00	1001					
MIN.	-0.59	0.00	10.41	0.00	10.41	1322	7.93 T	0.00	1321		
	0.00	10.41	1322	0.00	10.41	1322	26.03 T	10.41	1013		
20007											
MAX.	0.00	MAX	0.59	0.00	1116	0.00	1322	0.00	1322	0.00	1322
MIN.	2.08	MIN	0.38	0.00	1322	0.00	1322	0.00	1322	0.00	1322
	4.17	MAX	0.00	0.00	1322	0.00	1322	0.00	1322	0.00	1322
	6.25	MIN	0.00	0.00	1322	0.00	1322	0.00	1322	0.00	1322
	8.33	MAX	0.00	0.00	1322	0.00	1322	0.00	1322	0.00	1322
	10.41	MIN	0.00	0.00	1322	0.00	1322	0.00	1322	0.00	1322
		STAAD SPACE	-0.59	0.00	1116	0.00	1322	0.00	1322	0.00	1322

Waste Stock Yard Rev 1- Rashidiya											
	FZ	DIST	LD	MY	DIST	LD	FX	DIST	LD		
MAX.	0.58	0.00	1001	0.00	0.00	1001					
MIN.	-0.58	0.00	10.22	0.00	10.22	1322	11.40 T	0.00	1221		
	0.00	10.22	1322	0.00	10.22	1322	36.86 T	10.22	1013		
20006											
MAX.	0.00	MAX	0.58	0.00	1322	0.00	1322	0.00	1322	0.00	1322
MIN.	2.04	MIN	0.37	0.00	1322	0.00	1322	0.00	1322	0.00	1322
	4.09	MAX	0.00	0.00	1322	0.00	1322	0.00	1322	0.00	1322
	6.13	MIN	0.00	0.00	1322	0.00	1322	0.00	1322	0.00	1322
	8.17	MAX	0.00	0.00	1322	0.00	1322	0.00	1322	0.00	1322
	10.22	MIN	0.00	0.00	1322	0.00	1322	0.00	1322	0.00	1322
		STAAD SPACE	-0.58	0.00	1116	0.00	1322	0.00	1322	0.00	1322

MAX/MIN FORCE VALUES FOR MEMB 20006, AMONGST ALL SECT LOCATIONS											
	FY/	DIST	LD	MZ/	DIST	LD	FX	DIST	LD		
MAX.	0.58	0.00	1001	0.00	0.00	1001					
MIN.	-0.58	0.00	10.22	0.00	10.22	1322	7.40 T	10.22	1322		
	0.00	10.22	1322	0.00	10.22	1322	36.49 T	0.00	1013		
20007											
MAX.	0.00	MAX	0.59	0.00	1116	0.00	1322	0.00	1322	0.00	1322
MIN.	2.08	MIN	0.38	0.00	1322	0.00	1322	0.00	1322	0.00	1322
	4.17	MAX	0.00	0.00	1322	0.00	1322	0.00	1322	0.00	1322
	6.25	MIN	0.00	0.00	1322	0.00	1322	0.00	1322	0.00	1322
	8.33	MAX	0.00	0.00	1322	0.00	1322	0.00	1322	0.00	1322
	10.41	MIN	0.00	0.00	1322	0.00	1322	0.00	1322	0.00	1322
		STAAD SPACE	-0.59	0.00	1116	0.00	1322	0.00	1322	0.00	1322

MAX/MIN FORCE VALUES FOR MEMB 20007, AMONGST ALL SECT LOCATIONS											
	FY/	DIST	LD	MZ/	DIST	LD	FX	DIST	LD		
MAX.	0.59	0.00	1001	0.00	0.00	1001					
MIN.	-0.59	0.00	10.41	0.00	10.41	1322	7.93 T	0.00	1321		
	0.00	10.41	1322	0.00	10.41	1322	26.03 T	10.41	1013		
20008											
MAX.	0.00	MAX	0.59	0.00	1116	0.00	1322	0.00	1322	0.00	1322
MIN.	2.08	MIN	0.38	0.00	1322	0.00	1322	0.00	1322	0.00	1322
	4.17	MAX	0.00	0.00	1322	0.00	1322	0.00	1322	0.00	1322
	6.25	MIN	0.00	0.00	1322	0.00	1322	0.00	1322	0.00	1322
	8.33	MAX	0.00	0.00	1322	0.00	1322	0.00	1322	0.00	1322
	10.41	MIN	0.00	0.00	1322	0.00	1322	0.00	1322	0.00	1322
		STAAD SPACE	-0.59	0.00	1116	0.00	1322	0.00	1322	0.00	1322

MAX/MIN FORCE VALUES FOR MEMB 20008, AMONGST ALL SECT LOCATIONS											
	FY/	DIST	LD	MZ/	DIST	LD	FX	DIST	LD		
MAX.	0.59	0.00	1001	0.00	0.00	1001					
MIN.	-0.59	0.00	10.41	0.00	10.41	1322	3.26 T	10.41	1322		
	0.00	10.41	1322	0.00	10.41	1322					
20009											
MAX.	0.00	MAX	0.59	0.00	1116	0.00	1322	0.00	1322	0.00	1322
MIN.	2.08	MIN	0.38	0.00	1322	0.00	1322	0.00	1322	0.00	1322
	4.17	MAX	0.00	0.00	1322	0.00	1322	0.00	1322	0.00	1322
	6.25	MIN	0.00	0.00	1322	0.00	1322	0.00	1322	0.00	1322
	8.33	MAX	0.00	0.00	1322	0.00	1322	0.00	1322	0.00	1322
	10.41	MIN	0.00	0.00	1322	0.00	1322	0.00	1322	0.00	1322
		STAAD SPACE	-0.59	0.00	1116	0.00	1322	0.00	1322	0.00	1322

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STAAD SPACE

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20019	0.00	MAX	0.57	1116	0.00	1322	0.00	1322	0.00	1322
		MIN	0.37	1322	0.00	1322	0.00	1322	0.00	1322
	2.02	MAX	0.00	1322	0.00	1322	0.00	1322	0.00	1322
		MIN	0.00	1322	0.00	1322	0.00	1322	0.00	1322
	4.03	MAX	0.00	1322	0.00	1322	0.00	1322	0.00	1322
		MIN	0.00	1322	0.00	1322	0.00	1322	0.00	1322
	6.05	MAX	0.00	1322	0.00	1322	0.00	1322	0.00	1322
		MIN	0.00	1322	0.00	1322	0.00	1322	0.00	1322
	8.07	MAX	0.00	1322	0.00	1322	0.00	1322	0.00	1322
		MIN	0.00	1322	0.00	1322	0.00	1322	0.00	1322
	10.09	MAX	-0.37	1322	0.00	1322	0.00	1322	0.00	1322
		MIN	-0.37	1116	0.00	1322	0.00	1322	0.00	1322

MAX/MIN FORCE VALUES FOR MEMB												20019, AMONGST ALL SECT LOCATIONS					
	FX/	FY/	DIST	LD	DIST	LD	MZ/	DIST	LD	DIST	LD	FX	DIST	LD			
MAX.	0.57	0.00	0.00	1001	0.00	0.00	0.00	0.00	1001	0.00	0.00	1001	6.04	T	10.09	1322	
MIN.	-0.57	0.00	0.00	1001	0.00	0.00	0.00	0.00	1001	0.00	0.00	1001	29.62	T	0.00	1013	

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MAX/MIN FORCE VALUES FOR MEMB												2020Z, AMONGST ALL SECT LOCATIONS							
	FZ	FY	LD	DIST	LD	DIST	LD	LD	DIST	LD	LD	LD	DIST	LD	LD	DIST	LD		
MAX.	0.57	0.00	0.00	1301	0.00	0.00	1301	0.00	0.00	1301	0.00	0.00	1301	9.66	1	0.00	1221		
MIN.	-0.57	10.09	1116	0.00	10.09	1332	0.00	10.09	1332	0.00	10.09	1332	0.00	10.09	1332	31.20	10.09 1013		

20021

20021	0.00	MAX	0.58	1116	0.00	1322	0.00	1322
		MAX	0.37	1322	0.00	1322	0.00	1322
	2.04	MAX	0.00	1322	0.00	1322	0.00	1322
		MIN	0.00	1322	0.00	1322	0.00	1322
	4.08	MAX	0.00	1322	0.00	1322	0.00	1322
		MIN	0.00	1322	0.00	1322	0.00	1322
	6.12	MAX	0.00	1322	0.00	1322	0.00	1322
		MIN	0.00	1322	0.00	1322	0.00	1322
	8.16	MAX	0.00	1322	0.00	1322	0.00	1322
		MIN	0.00	1322	0.00	1322	0.00	1322
	10.20	MAX	-0.37	1322	0.00	1322	0.00	1322
		MIN	-0.58	1116	0.00	1322	0.00	1322

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MAX/MIN FORCE VALUES FOR MEMB										2002.1, AMONGST ALL SECT LOCATIONS									
FZ		DST		LD		MZ		DST		LD		FX		DST		LD			
MAX.		0.58	0.00	1001	0.00	0.00	1001	0.00	0.00	1001	0.00	0.00	10.68	T	10.20	1321			
MIN.		-0.58	0.00	1001	0.00	0.00	1001	0.00	0.00	1001	0.00	0.00	22.00	T	0.00	1115			

20022

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STAAD SPACE

MAX/MIN FORCE VALUES FOR MEMB										2002Z, AMONGST ALL SECT LOCATIONS			
	FZ/	FY/	FX/	LD	DIST	LD	MZ/	DIST	LD	FX	DIST	LD	
MAX.	0.58	0.00	0.00	1001	0.00	0.00	0.00	0.00	1001				
	0.00	0.00	0.00	1001	0.00	0.00	0.00	0.00	1001	11.92	C	0.00 1322	
MIN.	-0.58	0.00	0.00	1016	0.00	10.20	0.00	10.20	1322				
	0.00	0.00	0.00	1020	0.00	10.20	0.00	10.20	1322	38.30		10.20 1022	

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Waste Stock Yard Rev 1- Rashidhaya									
MAX/MIN FORCE VALUES FOR MEMB		20024		20023		AMONGST		ALL SECT LOCATIONS	
FV/	DIST	LD		FV/	DIST	LD	FZ/	DIST	LD
MAX.	0.57	0.00	1001	0.00	0.00	1001			
	0.00	0.00	1001	0.00	0.00	1001			
MIN.	-0.57	10.31	1116	0.00	10.31	1322		6.88	10.31
	0.00	10.31	1322	0.00	10.31	1322		12.61	0.00
									1115
20024	0.00	MAX	0.57	1116	0.00	1322		0.00	1322
	2.06	MIN	0.37	1322	0.00	1322		0.00	1322
	4.12	MAX	0.00	1322	0.00	1322		0.00	1322
		MIN	0.00	1322	0.00	1322		0.00	1322
	6.18	MAX	0.00	1322	0.00	1322		0.00	1322
		MIN	0.00	1322	0.00	1322		0.00	1322
	8.24	MAX	0.00	1322	0.00	1322		0.00	1322
		MIN	0.00	1322	0.00	1322		0.00	1322
	10.31	MAX	-0.37	1322	0.00	1322		0.00	1322
		MIN	-0.57	1116	0.00	1322		0.00	1322
STAAD SPACE									
								--	PAGE NO. 68

MAX/MIN FORCE VALUES FOR MEMB 20024, AMONGST ALL SECT LOCATIONS										
	FX/	FY/	DIST	DIST	DIST	MZ/	DIST	DIST	FX	DIST
	LD	LD	LD	LD	LD	LD	LD	LD		LD
MAX.	0.57	0.00	1001	0.00	0.00	1001				
	0.00	0.00	1001	0.00	0.00	1001				
MIN.	-0.57	10.31	1116	0.00	10.31	1322			16.21	C 0.00 1322
	0.00	10.31	1322	0.00	10.31	1322			36.38	T 10.31 1112

***** END OF FORCE ENVELOPE FROM INTERNAL STORAGE *****

335. LOAD LIST 2001 2002 2011 TO 2016 .

336. PRINT JOINT DISPLACEMENTS ALL
 JOINT DISPLACE ALL
 STAAD SPACE ☐

-- PAGE NO. 69

JOINT DISPLACEMENT (CM)		RADIANS		STRUCTURE TYPE = SPACE			
JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN
102	2001	0.0766	0.0019	-0.0371	-0.0003	-0.0009	-0.0013
	2002	0.1176	0.0202	-0.0348	-0.0002	-0.0009	-0.0011
	2003	0.2615	0.0283	0.0696	-0.0002	-0.0013	-0.0010
	2011	0.2754	0.0885	0.0583	0.0000	-0.0012	-0.0004
	2012	-0.0801	-0.0354	-0.0354	-0.0003	-0.0005	-0.0018
105	2001	0.0702	0.0954	-0.0462	-0.0000	-0.0006	-0.0003
	2002	-1.21313	-0.1083	-0.0337	-0.0003	-0.0012	0.0035
	2003	-1.19355	-0.0732	-0.0374	-0.0001	-0.0012	0.0039
	2011	0.2484	0.10490	-0.1109	0.0059	0.0001	0.0035
	2012	0.1975	0.8219	-0.0878	0.0047	0.0001	-0.0029
106	2001	0.1193	1.3203	0.0520	0.0065	0.0005	0.0026
	2002	0.0221	0.4548	0.1471	0.0024	0.0003	-0.0011
	2003	0.3922	1.5495	-0.1596	0.0085	0.0000	-0.0049
	2011	0.0933	0.1352	-0.0009	0.0011	-0.0002	-0.0011
	2012	-1.7295	1.5397	-0.0904	0.0067	0.0002	-0.0014
107	2001	-1.8601	0.6067	0.0110	0.0024	0.0001	-0.0001
	2002	0.2983	0.9043	-0.0789	0.0065	0.0000	0.0012
	2003	0.7413	0.7155	-0.0638	0.0053	0.0000	0.0010
	2011	0.2356	1.3702	0.0564	0.0075	0.0001	0.0009
	2012	0.3906	1.2758	-0.1742	0.0031	0.0001	0.0004
108	2001	0.3906	1.2758	-0.1742	0.0031	0.0001	0.0004
	2002	0.0205	0.6667	0.0012	0.0013	0.0001	0.0004
	2003	-1.7127	1.5357	-0.0695	0.0075	0.0001	0.0003
	2011	-1.8394	0.6310	-0.0381	0.0030	0.0000	-0.0001
	2012	0.1840	0.1932	-0.0380	0.0007	0.0003	0.0010
109	2001	0.1555	0.1551	-0.0306	0.0005	0.0002	0.0003
	2002	0.1505	0.0372	0.1158	0.0001	0.0003	0.0007
	2003	0.0487	0.0475	0.1333	0.0000	0.0002	0.0003
	2011	0.2543	0.4402	-0.0675	0.0016	0.0005	0.0014
	2012	-0.0153	0.3784	-0.0355	0.0014	0.0002	0.0003

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JOINT DISPLACEMENT (CM) RADIANS) STRUCTURE TYPE = SPACE

JOINT LOAD X-TRANS Y-TRANS Z-TRANS X-ROTAN Y-ROTAN Z-ROTAN

305	2001	0.2631	-3.0880	0.0000	0.0000	0.0000	0.0105
	2002	0.2212	-2.4802	0.0000	0.0000	0.0000	0.0082
	2003	0.3080	-3.0475	0.0000	0.0000	0.0000	0.0090
	2004	0.1534	-1.1652	0.1582	0.0000	0.0000	0.0035
	2013	0.3066	-4.3876	0.0000	0.0000	0.0000	0.0171
	2014	-0.0054	-0.6929	0.0095	0.0001	0.0000	0.0047
	2015	-1.7130	-2.9266	0.0313	0.0006	0.0001	0.0118
	2016	-1.8227	-1.0368	0.0228	0.0004	0.0001	0.0044
306	2001	0.3056	-3.5715	0.0000	0.0000	0.0000	0.0006
	2002	0.2497	-2.8947	0.0000	0.0000	0.0000	0.0006
	2003	0.2921	-3.5979	0.1456	0.0000	0.0001	0.0013
	2012	0.1574	-1.4835	0.1482	0.0005	0.0001	0.0004
	2013	0.4468	-4.9750	0.0007	0.0000	0.0000	0.0006
	2014	0.1126	-0.8493	0.0050	0.0000	0.0001	-0.0006
	2015	-1.16456	-3.4010	0.0195	0.0008	0.0001	-0.0002
	2016	-1.7982	-1.3275	0.0188	0.0005	0.0001	-0.0001
307	2001	0.3523	-1.3383	0.0000	0.0000	0.0001	-0.0025
	2002	0.2781	-1.0616	0.0000	0.0000	0.0000	-0.0017
	2011	0.3518	-1.1342	0.1360	0.0006	0.0001	-0.0024
	2012	0.4363	-0.5428	0.1398	0.0004	0.0001	-0.0017
	2013	0.4911	-2.0224	-0.0033	0.0000	0.0000	-0.0033
	2014	0.1236	-0.6874	-0.0034	0.0004	0.0001	-0.0017
	2015	-1.5965	-1.3252	0.0004	0.0000	0.0000	-0.0025
	2016	-1.7574	-0.5489	0.0175	0.0000	0.0001	-0.0016
308	2001	0.3045	-4.0486	0.0000	0.0000	0.0000	-0.0015
	2002	0.2436	-3.0101	0.0000	0.0000	0.0000	-0.0011
	2011	0.1057	-4.0661	0.1379	0.0005	0.0001	-0.0004
	2012	0.4597	-2.2510	-0.1361	0.0003	0.0001	-0.0009
	2013	0.4260	-5.3472	-0.0095	0.0000	0.0000	-0.0020
	2014	0.0944	-1.7188	0.0000	0.0000	0.0001	-0.0008
	2015	-1.6430	-4.1130	0.0041	0.0005	0.0001	-0.0016
	2016	-1.7826	-2.2884	0.0199	0.0003	0.0001	-0.0011
310	2001	0.2454	-1.3726	0.0000	0.0000	0.0000	0.0000
	2002	0.2061	-1.1053	0.0000	0.0000	0.0000	0.0000
	2011	0.2475	-1.1791	0.1429	0.0006	0.0001	0.0000
	2012	0.1137	-0.5891	0.1553	0.0004	0.0002	0.0000
	2013	0.3490	-2.1113	0.0034	0.0000	0.0001	0.0000
	2014	0.0390	-0.7490	0.0165	0.0000	0.0002	0.0010
	2015	-1.7159	-1.3816	0.0043	0.0004	0.0001	0.0009
311	2001	0.2513	-3.3205	0.0000	0.0000	0.0000	0.0012
	2002	0.2018	-3.3205	0.0000	0.0000	0.0000	0.0019
	2011	0.2513	-3.3205	0.0000	0.0000	0.0000	0.0016
	2012	0.3119	-5.3432	0.0000	0.0005	0.0001	0.0014
	2013	0.1408	-1.4927	0.0000	0.0000	0.0001	-0.0002
	2014	0.2993	-4.6752	0.0577	0.0000	0.0001	0.0018
	2015	0.0864	-0.9329	0.1274	0.0000	0.0002	-0.0005
	2016	-1.7499	-3.3098	0.0116	0.0005	0.0001	0.0015
	2017	-1.8231	-1.4365	0.0200	0.0003	0.0000	0.0015
							--- PAGE NO
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JOINT DISPLACEMENT (CM) RADIANS) STRUCTURE TYPE = SPACE

JOINT LOAD X-TRANS Y-TRANS Z-TRANS X-ROTAN Y-ROTAN Z-ROTAN

312	2001	0.1955	-1.1287	0.0000	0.0000	0.0019
2002	0.1608	-0.9917	0.0000	0.0000	0.0000	0.0014
2003	0.1995	-1.0562	0.1461	0.0005	0.0000	0.0018
2004	0.0807	-0.5347	0.1874	0.0004	0.0000	0.0000
2005	0.2614	-1.8812	0.0255	0.0000	0.0000	0.0000
2006	0.0213	-0.6311	0.0847	0.0000	0.0000	0.0000
2007	-1.7635	-1.2334	0.0166	0.0000	0.0000	0.0000
2008	-1.8644	-0.5487	0.0164	0.0000	0.0000	0.0000
2009	0.2284	-2.1227	0.0000	0.0000	0.0000	0.0000
2010	0.1984	-1.1361	0.0000	0.0000	0.0000	0.0000
2011	0.1397	-0.5334	0.0000	0.0000	0.0000	0.0000
2012	0.3115	-2.9129	0.0250	0.0000	0.0000	0.0000
2013	0.0284	-0.5057	0.1177	0.0000	0.0000	0.0000
2014	-1.7335	-2.2189	0.0132	0.0004	0.0000	0.0000
2015	-1.8086	-0.6518	0.0000	0.0000	0.0000	0.0000
2016	0.2410	-2.1933	0.0000	0.0000	0.0000	0.0000
314	2001	0.1891	-1.7702	0.0000	0.0000	0.0000
2002	0.2105	-0.0000	0.0000	0.0000	0.0000	0.0000
2003	0.1455	-2.1513	0.1482	0.0004	0.0000	0.0000
2004	0.1455	-2.1513	0.1482	0.0004	0.0000	0.0000
2005	0.1455	-2.1513	0.1482	0.0004	0.0000	0.0000
2006	0.1455	-2.1513	0.1482	0.0004	0.0000	0.0000
2007	0.1455	-2.1513	0.1482	0.0004	0.0000	0.0000
2008	0.1455	-2.1513	0.1482	0.0004	0.0000	0.0000
2009	0.1455	-2.1513	0.1482	0.0004	0.0000	0.0000
2010	0.1455	-2.1513	0.1482	0.0004	0.0000	0.0000
2011	0.1455	-2.1513	0.1482	0.0004	0.0000	0.0000
2012	0.1455	-2.1513	0.1482	0.0004	0.0000	0.0000

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Waste Stock Yard Rev 1- Rashidiya

JOINT DISPLACEMENT (CM) RADIANS) STRUCTURE TYPE = SPACE

JOINT LOAD X-TRANS Y-TRANS Z-TRANS X-ROTAN Y-ROTAN Z-ROTAN

317	2001	0.3843	-3.0993	0.0206	0.0000	0.0121
2002	0.1095	-0.5118	0.0846	0.0000	0.0001	-0.0031
2003	-1.7354	-2.2597	0.0146	0.0004	0.0000	-0.0067
2004	-1.8136	-0.9407	0.0112	0.0000	0.0000	-0.0029
2005	0.8515	-1.7437	0.0000	0.0000	0.0000	-0.0271
2006	0.7965	-1.7540	0.0000	0.0000	0.0000	-0.0279
2007	0.6306	-1.6506	0.0000	0.0000	0.0000	-0.0279
2008	0.8791	-1.7486	0.0198	0.0000	0.0000	-0.0209
2009	0.8791	-1.7486	0.0198	0.0000	0.0000	-0.0209
2010	0.4738	-1.7826	0.0158	0.0000	0.0000	-0.0218
2011	-0.7742	-1.5670	-0.0075	0.0000	0.0000	-0.0229
2012	0.8828	-1.5977	0.0000	0.0000	0.0000	-0.0254
2013	0.0000	0.0000	0.0000	0.0000	0.0000	0.0014
2014	0.0000	0.0000	0.0000	0.0000	0.0000	0.0011
2015	0.0000	0.0000	0.0000	0.0000	0.0000	0.0011
2016	0.0000	0.0000	0.0000	0.0000	0.0000	0.0011
401	2001	0.0000	0.0000	0.0000	0.0000	0.0000
2002	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
2003	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
2004	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
2005	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
2006	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
2007	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
2008	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
2009	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
2010	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
2011	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
2012	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
2013	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
2014	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
2015	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
2016	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
402	2001	0.0038	-0.0585	-0.0663	-0.0030	-0.0087
2002	0.0075	-0.0491	-0.0711	-0.0029	-0.0078	-0.0028
2003	-0.0293	-0.0772	0.0490	-0.0026	-0.0036	-0.0036
2004	0.0064	-0.0410	0.0327	-0.0024	-0.0071	-0.0019
2005	0.0225	-0.0781	-0.0480	-0.0031	-0.0107	-0.0039
2006	0.0142	-0.0742	-0.0026	-0.0057	-0.0009	-0.0009
2007	-1.8010	-0.1501	-0.0905	-0.0030	-	
2008	-1.7976	-0.1501	-0.0905	-0.0030	-0.0001	-0.0001
2009	-1.7976	-0.1501	-0.0905	-0.0030	-0.0001	-0.0001
2010	-1.7976	-0.1501	-0.0905	-0.0030	-0.0001	-0.0001
2011	-1.7976	-0.1501	-0.0905	-0.0030	-0.0001	-0.0001
2012	-1.7976	-0.1501	-0.0905	-0.0030	-0.0001	-0.0001
2013	-1.7976	-0.1501	-0.0905	-0.0030	-0.0001	-0.0001
2014	-1.7976	-0.1501	-0.0905	-0.0030	-0.0001	-0.0001
2015	-1.7976	-0.1501	-0.0905	-0.0030	-0.0001	-0.0001
2016	-1.7976	-0.1501	-0.0905	-0.0030	-0.0001	-0.0001
403	2001	0.0038	-0.0585	-0.0663	-0.0030	-0.0087
2002	0.0075	-0.0491	-0.0711	-0.0029	-0.0078	-0.0028
2003	-0.0293	-0.0772	0.0490	-0.0026	-0.0036	-0.0036
2004	0.0064	-0.0410	0.0327	-0.0024	-0.0071	-0.0019
2005	0.0225	-0.0781	-0.0480	-0.0031	-0.0107	-0.0039
2006	0.0142	-0.0742	-0.0026	-0.0057	-0.0009	-0.0009
2007	-1.8010	-0.1501	-0.0905	-0.0030	-	

[illegible][illegible]

	Waste	Stock	Yard	Rev	1- Rashid	Divya
2011	0.3469	0.0196	0.1542	0.9000	0.0002	-0.0002
2012	0.2271	0.0839	-0.0003	0.0002	0.0002	-0.0002
2013	0.4639	0.3978	-0.0014	0.0002	0.0002	-0.0002
2014	0.4325	0.0976	-0.0014	0.0002	0.0002	-0.0002
2015	-1.7556	0.0819	-0.0038	-0.0002	0.0002	0.0000
2016	-1.7556	0.0040	0.0257	0.0000	0.0002	-0.0001
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TOTNT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN
1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9
10	10	10	10	10	10	10	10
11	11	11	11	11	11	11	11
12	12	12	12	12	12	12	12
13	13	13	13	13	13	13	13
14	14	14	14	14	14	14	14
15	15	15	15	15	15	15	15
16	16	16	16	16	16	16	16
17	17	17	17	17	17	17	17
18	18	18	18	18	18	18	18
19	19	19	19	19	19	19	19
20	20	20	20	20	20	20	20
21	21	21	21	21	21	21	21
22	22	22	22	22	22	22	22
23	23	23	23	23	23	23	23
24	24	24	24	24	24	24	24
25	25	25	25	25	25	25	25
26	26	26	26	26	26	26	26
27	27	27	27	27	27	27	27
28	28	28	28	28	28	28	28
29	29	29	29	29	29	29	29
30	30	30	30	30	30	30	30
31	31	31	31	31	31	31	31
32	32	32	32	32	32	32	32
33	33	33	33	33	33	33	33
34	34	34	34	34	34	34	34
35	35	35	35	35	35	35	35
36	36	36	36	36	36	36	36
37	37	37	37	37	37	37	37
38	38	38	38	38	38	38	38
39	39	39	39	39	39	39	39
40	40	40	40	40	40	40	40
41	41	41	41	41	41	41	41
42	42	42	42	42	42	42	42
43	43	43	43	43	43	43	43
44	44	44	44	44	44	44	44
45	45	45	45	45	45	45	45
46	46	46	46	46	46	46	46
47	47	47	47	47	47	47	47
48	48	48	48	48	48	48	48
49	49	49	49	49	49	49	49
50	50	50	50	50	50	50	50
51	51	51	51	51	51	51	51
52	52	52	52	52	52	52	52
53	53	53	53	53	53	53	53
54	54	54	54	54	54	54	54
55	55	55	55	55	55	55	55
56	56	56	56	56	56	56	56
57	57	57	57	57	57	57	57
58	58	58	5				

513	2001	0.2563	0.7703	0.0770	-0.0044	0.0001	-0.0003
	2002	0.2093	0.6228	0.0589	-0.0034	0.0001	-0.0006
	2003	0.2093	0.6228	0.0589	-0.0034	0.0001	-0.0006
	2004	0.2824	0.5614	0.2151	-0.0037	0.0001	-0.0006
	2005	0.1669	-0.1231	0.1236	-0.0004	0.0000	-0.0003
	2006	0.1378	0.1715	0.1236	-0.0037	0.0001	-0.0007
	2007	0.3798	0.0658	0.1239	-0.0008	-0.0002	-0.0002
	2008	0.1449	0.0658	0.1239	-0.0008	-0.0002	-0.0002
	2009	-1.6964	0.4727	0.0787	-0.0037	0.0001	-0.0009
	2010	-1.8164	-0.2532	0.1717	-0.0037	0.0001	-0.0009
514	2001	0.3486	0.7378	0.0847	-0.0004	0.0000	-0.0002
	2002	0.3607	0.7378	0.0847	-0.0004	0.0000	-0.0002
	2003	0.1597	0.6662	0.2197	-0.0036	0.0002	0.0012
	2004	0.1397	0.1321	0.2148	-0.0038	0.0003	0.0019
517	2001	0.4289	1.3470	0.1398	-0.0064	0.0002	0.0010
	2002	0.1345	0.1225	0.0906	-0.0006	0.0002	0.0006
	2003	-1.6244	0.6358	0.0880	-0.0039	0.0002	0.0025
	2004	-1.7805	0.0317	0.0263	-0.0012	0.0001	0.0017
519	2001	0.1466	0.0018	0.0411	-0.0002	-0.0010	-0.0002
	2002	0.0648	0.0194	0.0424	-0.0002	-0.0010	-0.0002
	2003	0.2295	-0.0983	0.1119	0.0006	-0.0010	-0.0002
	2004	0.0351	-0.3246	0.1273	0.0006	-0.0010	-0.0002
	2005	0.3859	-0.0553	0.0850	0.0003	-0.0002	-0.0002
	2006	0.1384	-0.0553	0.0850	0.0003	-0.0002	-0.0002
	2007	-1.7873	0.1303	0.0306	0.0001	-0.0007	0.0041
	2008	-1.4917	0.1762	0.0419	0.0000	-0.0010	0.0043
	2009	0.0000	0.0000	0.0000	-0.0009	0.0003	0.0014
	2010	0.0000	0.0000	0.0000	-0.0007	0.0002	0.0012
2007	2001	0.0000	0.0000	0.0000	-0.0007	0.0002	0.0012
	2002	0.0000	0.0000	0.0000	-0.0008	0.0002	-0.0005
	2003	0.0000	0.0000	0.0000	-0.0008	0.0002	-0.0005
	2004	0.0000	0.0000	0.0000	-0.0017	0.0002	-0.0020
2010	2001	0.0000	0.0000	0.0000	-0.0009	0.0004	-0.0028
	2002	0.0000	0.0000	0.0000	-0.0003	0.0002	-0.0008
	2003	0.0000	0.0000	0.0000	-0.0003	0.0002	-0.0008
	2004	0.0000	0.0000	0.0000	-0.0008	0.0000	-0.0008
2012	2001	0.0000	0.0000	0.0000	-0.0010	-0.0001	-0.0007
	2002	0.0000	0.0000	0.0000	-0.0008	-0.0002	-0.0003
	2003	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
	2004	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
2014	2001	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
	2002	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
	2003	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
	2004	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
2016	2001	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
	2002	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
	2003	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
	2004	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
2018	2001	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
	2002	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
	2003	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
	2004	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
2020	2001	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
	2002	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
	2003	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
	2004	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
2022	2001	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
	2002	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
	2003	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
	2004	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
2024	2001	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
	2002	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
	2003	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
	2004	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
2026	2001	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
	2002	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
	2003	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
	2004	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
2028	2001	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
	2002	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
	2003	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
	2004	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
2030	2001	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
	2002	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
	2003	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
	2004	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
2032	2001	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
	2002	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
	2003	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
	2004	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
2034	2001	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
	2002	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
	2003	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
	2004	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
2036	2001	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
	2002	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
	2003	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
	2004	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
2038	2001	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
	2002	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
	2003	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
	2004	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
2040	2001	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
	2002	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
	2003	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
	2004	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
2042	2001	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
	2002	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
	2003	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
	2004	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
2044	2001	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
	2002	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
	2003	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
	2004	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
2046	2001	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
	2002	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
	2003	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
	2004	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
2048	2001	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
	2002	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
	2003	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
	2004	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
2050	2001	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
	2002	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
	2003	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
	2004	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
2052	2001	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
	2002	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
	2003	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
	2004	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
2054	2001	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
	2002	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
	2003	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
	2004	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
2056	2001	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
	2002	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
	2003	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
	2004	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
2058	2001	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
	2002	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
	2003	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
	2004	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
2060	2001	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
	2002	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
	2003	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
	2004	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
2062	2001	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
	2002	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
	2003	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
	2004	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
2064	2001	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
	2002	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
	2003	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
	2004	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
2066	2001	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
	2002	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
	2003	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
	2004	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
2068	2001	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
	2002	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
	2003	0.0000	0.0000	0.0000	-0.0011	-0.0002	-0.0007
	2004	0.0000	0.0000				

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ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

[illegible]

217	TAPERED	PASS	Waste Stock Yard Rev 1- Rashidiya	0.351	2013	0.00
		65.36 C	BS-4.7 (C)	-28.52	2013	0.00
401	TAPERED	PASS	BS-4.7 (C)	0.352	2013	2.49
		83.04 C	BS-4.7 (C)	55.78	2013	2.49
0	STAAD SPACE					

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ALL UNITS ARE - KN

MEMBER	TABLE	RESULT/	CRITICAL COND/	RATIO/	LOADING/
		FX	NY	MZ	LOCATION

402	TAPERED	PASS	BS-4.7 (C)	0.426	2015	0.00
		67.78 C	BS-4.7 (C)	55.84	2015	0.00
403	TAPERED	PASS	BS-4.7 (C)	0.410	2015	0.00
		59.10 C	BS-4.7 (C)	58.13	2015	0.00
404	TAPERED	PASS	BS-4.7 (C)	0.246	2015	0.00
		33.03 C	BS-4.7 (C)	33.89	2015	0.00
405	TAPERED	PASS	BS-4.7 (C)	0.373	2013	0.00
		14.84 C	BS-4.7 (C)	53.48	2013	0.00
406	TAPERED	PASS	BS-4.7 (C)	0.374	2013	0.00
		12.47 C	BS-4.7 (C)	53.48	2013	0.00
407	TAPERED	PASS	BS-4.7 (C)	0.337	2015	0.00
		13.17 C	BS-4.7 (C)	48.90	2015	0.00
408	TAPERED	PASS	BS-4.7 (C)	0.253	2013	0.00
		6.70 C	BS-4.7 (C)	32.39	2013	0.00
409	TAPERED	PASS	BS-4.7 (C)	0.260	2013	0.00
		6.70 C	BS-4.7 (C)	32.39	2013	0.00
410	TAPERED	PASS	BS-4.7 (C)	0.275	2012	0.59
		5.70 C	BS-4.7 (C)	35.40	2012	0.59
411	TAPERED	PASS	BS-4.7 (C)	0.310	2013	0.90
		6.63 C	BS-4.7 (C)	38.77	2013	0.90
412	TAPERED	PASS	BS-4.7 (C)	0.314	2015	0.00
		20.54 C	BS-4.7 (C)	44.70	2015	0.00
413	TAPERED	PASS	BS-4.7 (C)	0.248	2015	0.00
		20.30 C	BS-4.7 (C)	34.50	2015	1.32
414	TAPERED	PASS	BS-4.7 (C)	0.266	2015	0.00
		30.83 C	BS-4.7 (C)	34.58	2015	0.00
415	TAPERED	PASS	BS-4.7 (C)	0.214	2013	0.74
		41.76 C	BS-4.7 (C)	26.44	2013	0.74
416	TAPERED	PASS	BS-4.7 (C)	0.262	2013	0.74
		48.83 C	BS-4.7 (C)	29.11	2013	0.74
417	TAPERED	PASS	BS-4.7 (C)	0.358	2013	0.00
		66.60 C	BS-4.7 (C)	28.99	2013	0.00
1001	TAPERED	PASS	BS-4.7 (C)	0.167	2001	2.60
		0.19 T	BS-4.7 (C)	0.00	2001	2.60
1002	TAPERED	PASS	BS-4.7 (C)	0.375	2002	0.00
		23.98 C	BS-4.7 (C)	0.29	2002	0.00
1003	TAPERED	PASS	BS-4.7 (C)	0.377	2011	0.00
		39.89 C	BS-4.7 (C)	0.04	2011	0.00
1004	TAPERED	PASS	BS-4.7 (C)	0.167	2001	5.00
		0.19 T	BS-4.7 (C)	0.00	2001	5.00
2001	TAPERED	PASS	BS-4.7 (C)	0.135	2013	0.00
		0.31 T	BS-4.7 (C)	2.52	2013	0.00
2002	TAPERED	PASS	BS-4.7 (C)	0.359	2013	2.60
		51.98 C	BS-4.7 (C)	48.61	2013	2.60
2003	TAPERED	PASS	BS-4.7 (C)	0.359	2013	1.59
		51.98 C	BS-4.7 (C)	48.61	2013	1.59
0	STAAD SPACE					

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ALL UNITS ARE - KN

MEMBER	TABLE	RESULT/	CRITICAL COND/	RATIO/	LOADING/
		FX	NY	MZ	LOCATION

2004	TAPERED	PASS	BS-4.3.6	0.135	2011	0.00
		0.51 T	BS-4.3.6	21.22	2011	0.00
3001	TAPERED	PASS	BS-4.3.6	0.133	2013	0.00
		0.34 T	BS-4.3.6	20.88	2013	0.00
3002	TAPERED	PASS	BS-4.7 (C)	0.365	2013	2.60
		49.40 C	BS-4.7 (C)	51.06	2013	2.60
3003	TAPERED	PASS	BS-4.7 (C)	0.365	2013	5.00
		49.40 C	BS-4.7 (C)	51.06	2013	5.00
3004	TAPERED	PASS	BS-4.3.6	0.133	2011	0.00
		0.34 T	BS-4.3.6	20.88	2011	0.00
4001	TAPERED	PASS	BS-4.3.6	0.135	2013	0.00
		0.51 T	BS-4.3.6	21.22	2013	0.00

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4002	TAPERED	PASS	BS-4.7 (C)	0.294	2013	0.00
		24.06 C	BS-4.7 (C)	42.62	2013	0.00
4003	TAPERED	PASS	BS-4.7 (C)	0.293	2013	0.00
		24.06 C	BS-4.7 (C)	42.59	2013	0.00
4004	TAPERED	PASS	BS-4.3.6	0.135	2011	0.00
		0.51 T	BS-4.3.6	21.22	2011	0.00
5001	TAPERED	PASS	BS-4.3.6	0.135	2013	0.00
		0.51 T	BS-4.3.6	21.22	2013	0.00
5002	TAPERED	PASS	BS-4.3.6	0.135	2013	0.00
		2.50 C	BS-4.3.6	69.32	2013	0.00
5003	TAPERED	PASS	BS-4.7 (C)	0.441	2013	0.00
		2.30 C	BS-4.7 (C)	69.32	2013	0.00
5004	TAPERED	PASS	BS-4.3.6	0.135	2011	0.00
		0.51 T	BS-4.3.6	21.22	2011	0.00
6001	TAPERED	PASS	BS-4.3.6	0.135	2013	0.00
		0.51 T	BS-4.3.6	21.22	2013	0.00
6002	TAPERED	PASS	BS-4.7 (C)	0.287	2013	0.00
		4.18 C	BS-4.7 (C)	43.66	2013	0.00
6003	TAPERED	PASS	BS-4.3.6	0.286	2013	0.00
		4.18 C	BS-4.3.6	43.71	2013	0.00
6004	TAPERED	PASS	BS-4.3.6	0.135	2011	0.00
		0.51 T	BS-4.3.6	21.22	2011	0.00
7001	TAPERED	PASS	BS-4.3.6	0.135	2013	0.00
		0.51 T	BS-4.3.6	21.22	2013	0.00
7002	TAPERED	PASS	BS-4.3.6	0.372	2013	0.60
		1.20 C	BS-4.3.6	57.02	2013	0.60
7003	TAPERED	PASS	BS-4.7 (C)	0.362	2013	0.00
		1.20 C	BS-4.7 (C)	57.02	2013	0.00
7004	TAPERED	PASS	BS-4.3.6	0.135	2011	0.00
		0.51 T	BS-4.3.6	21.22	2011	0.00
8001	TAPERED	PASS	BS-4.3.6	0.126	2013	0.00
		0.48 T	BS-4.3.6	19.83	2013	0.00
8002	TAPERED	PASS	BS-4.7 (C)	0.269	2013	2.60
		21.50 C	BS-4.7 (C)	39.72	2013	2.60
0	STAAD SPACE					

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ALL UNITS ARE - KN

MEMBER	TABLE	RESULT/	CRITICAL COND/	RATIO/	LOADING/
		FX	NY	MZ	LOCATION

8003	TAPERED	PASS	BS-4.7 (C)	0.279	2013	0.00
		21.50 C	BS-4.7 (C)	40.23	2013	0.00
8004	TAPERED	PASS	BS-4.3.6	0.135	2011	0.00
		0.48 T	BS-4.3.6	19.83	2011	0.00
9001	TAPERED	PASS	BS-4.3.6	0.091	2013	0.00
		0.33 T	BS-4.3.6	14.39	2013	0.00
9002	TAPERED	PASS	BS-4.7 (C)	0.243	2013	2.60
		43.78 C	BS-4.7 (C)	32.47	2013	2.60
9003	TAPERED	PASS	BS-4.7 (C)	0.243	2013	4.59
		43.78 C	BS-4.7 (C)	32.47	2013	4.59
9004	TAPERED	PASS	BS-4.3.6	0.091	2011	0.00
		0.33 T	BS-4.3.6	14.39	2011	0.00
10001	TAPERED	PASS	BS-4.3.6	0.102	2013	0.00
		0.39 T	BS-4.3.6	16.08	2013	0.00
10002	TAPERED	PASS	BS-4.7 (C)	0.274	2013	2.60
		42.52 C	BS-4.7 (C)	32.47	2013	2.60
10003	TAPERED	PASS	BS-4.7 (C)	0.274	2013	1.59
		42.52 C	BS-4.7 (C)	32.47	2013	1.59
10004	TAPERED	PASS	BS-4.3.6	0.102	2011	0.42
		0.39 T	BS-4.3.6	16.08	2011	0.42
11001	TAPERED	PASS	BS-4.8.2.2	0.167	2001	0.00
		0.19 T	BS-4.8.2.2	0.00	2001	0.00
11002	TAPERED	PASS	BS-4.7 (C)	0.375	2002	2.60
		22.86 C	BS-4.7 (C)	0.37	2002	2.60
11003	TAPERED	PASS	BS-4.7 (C)	0.377	2011	0.00
		36.52 C	BS-4.7 (C)	0.18	2011	0.00
11004	TAPERED	PASS	BS-4.8.2.2	0.167	2001	5.00
		0.19 T	BS-4.8.2.2	0.00	2001	5.00
20001	ST UA90X90X6	PASS	BS-4.6 (T)	0.060	2012	0.00
		24.78 T	BS-4.6 (T)	0.00	2012	0.00
20002	ST UA90X90X6	PASS	BS-4.6 (T)	0.025	2014	0.31
		9.27 T	BS-4.6 (T)	0.00	2014	0.31
20003	ST UA90X90X6	PASS	BS-4.6 (T)	0.060	2011	0.00
		21.93 T	BS-4.6 (T)	0.00	2011	0.00
20004	ST UA90X90X6	PASS	BS-4.6 (T)	0.058	2013	10.20
		21.30 T	BS-4.6 (T)	0.00	2013	10.20
20005	ST UA90X90X6	PASS	BS-4.6 (T)	0.084	2013	0.00
		30.71 T	BS-4.6 (T)	0.00	2013	0.00
20006	ST UA90X90X6	PASS	BS-4.6 (T)	0.083	2013	10.22
		30.41 T	BS-4.6 (T)	0.00	2013	10.22

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	20007 ST UA90X30X6	PASS	Waste Stock Yard Rev 1- Rashidiya	
		21.69 T	85-4.6 (T)	0.059 2013
	20008 ST UA90X30X6	21.69 T	0.00	0.00 2013
		21.69 T	85-4.6 (T)	0.059 2013
	20009 ST UA90X30X6	21.69 T	0.00	0.00 2013
		6.41 T	85-4.6 (T)	0.018 2013
	STAAD SPACE		0.00	0.00 2013

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ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ M/Z	LOADING/ LOCATION
20010 ST UA90X90X6	PASS	25.17 T	BS-4.6 (T)	0.069	2011
20011 ST UA90X90X6	PASS	8.25 T	BS-4.6 (T)	0.023	2013
20012 ST UA90X90X6	PASS	29.14 T	BS-4.6 (T)	0.080	2012
20013 ST UA90X90X6	PASS	3.45 T	BS-4.6 (T)	0.009	2001
20014 ST UA90X90X6	PASS	33.92 T	BS-4.6 (T)	0.093	2012
20017 ST UA90X90X6	PASS	20.97 T	BS-4.6 (T)	0.057	2013
20018 ST UA90X90X6	PASS	20.68 T	BS-4.6 (T)	0.055	2013
20019 ST UA90X90X6	PASS	24.68 T	BS-4.6 (T)	0.067	2013
20020 ST UA90X90X6	PASS	26.00 T	BS-4.6 (T)	0.060	2013
20021 ST UA90X90X6	PASS	18.31 T	BS-4.6 (T)	0.050	2013
20022 ST UA90X90X6	PASS	22.20 T	BS-4.6 (T)	0.061	2012
20023 ST UA90X90X6	PASS	9.01 T	BS-4.6 (T)	0.025	2015
20024 ST UA90X90X6	PASS	25.99 T	BS-4.6 (T)	0.071	2012
			0.00	0.00	2013

END OF TABULATED RESULT OF DESIGN *****

341. STEEL TAKE OFF ALL
STAAD SPACE

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PROFILE	LENGTH(METRE)	WEIGHT(KN)
ISCT: TAPERED	252.11	104.658
ST UA90X90X6	229.11	18.056
	TOTAL =	122.714

END OF DATA FROM INTERNAL STORAGE

342. FINISH

```
***** END OF THE STAAD.PRO RUN *****
*** DATE= MAY 15,2007   TIME= 15:25:55 ***
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**** DATE= MAY 15, 2007  TIME= 15:25:55  ****

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2 FOUNDATION DESIGN CHECK

Document No.: **DM001-E-MAW-CDP-DR-DCC-371948**

Rev A3

Date: 16 May 2007

Page

Calculation/Sketch

Meinhardt Infrastructure Pte Ltd
Consulting Engineers, Planners & Managers



Site	Main Depot Rashidiya - Building 20	Date	Apr '07	Job No.	068010
	Footing Summary				
		Designed by	Rudy	Sheet No.	

Soil Bearing Pressure	=	150	kN/m ²
Soil Density	=	20	kN/m ³
Concrete Density	=	24	kN/m ³
Footing Depth below slab soffit	=	1	m
Column Stump Area	=	0.25	m ²

Individual Footing

Column	Column Reaction (ULS)	Soil Load (kN)	Footing Load (kN)	Total SLS Load (kN)	ULS Load (kN)	Footing size			Area of Footing (m ²)	Soil Pressure (kN/m ²)	Factor	Remarks
						L (m)	B (m)	H (m)				
201	141	34.2	28.224	180	229	1.400	1.400	0.600	1.96	92	1.63	OK

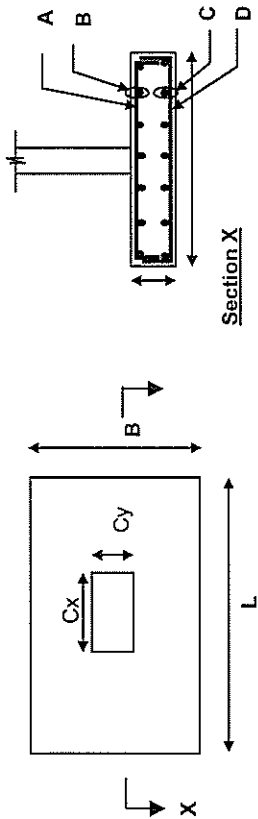
Calculation/Sketch

Meinhardt Infrastructure Pte Ltd
Consulting Engineers, Planners & Managers



Site	Main Depot Rashidiya - Building 20 Footing Detail	Date	Apr '07	Job No.	068010
		Designed by	Rudy	Sheet No.	

Summary



Footing Type	L	B	H	Soil Bearing pressure (kPa)	A	Spacing (mm)	B	Spacing (mm)	C	Spacing (mm)	D	Spacing (mm)
F3	1400	1400	600	150	10T12	143	10T12	143	10T16	143	10T16	143

Calculation/Sketch

Meinhardt Infrastructure Pte Ltd
Consulting Engineers, Planners & Managers



Site	Main Depot Rashidiya - Building 20	Date	Apr '07	Job No.	068010.01
	Footing F3	Designed by	Rudy	Sheet No.	
	For soil bearing pressure = 150kPa				

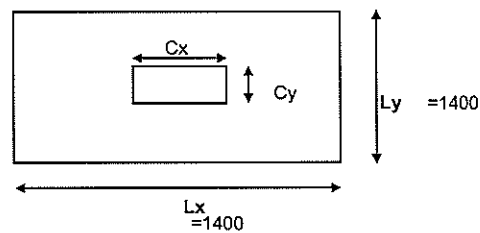
REINFORCED CONCRETE FOOTING DESIGN TO BS8110

1. Material Properties

Concrete grade	f_{cu}	=	35	N/mm ²
Steel grade	f_y	=	460	N/mm ²
Concrete density	γ_c	=	24	KN/m ³
Steel density	γ_s	=	78	KN/m ³

2. Design Data

Depth of Footing	h	=	600	mm
Width of Footing	L_y	=	1400	mm
Length of Footing	L_x	=	1400	mm
Width of Column	C_y	=	450	mm
Length of Column	C_x	=	450	mm
Concrete Ten. Cover	c	=	50	mm
Concrete Comp. Cover	c	=	50	mm
Tension bar dia. (x-dir)	D_x	=	16	mm
Tension bar dia. (y-dir)	D_y	=	16	mm
Comp. bar diameter (x-dir)	D_c	=	12	mm
Comp. bar diameter (y-dir)	D_c	=	12	mm
Effect.depth (x-dir)	d_x	=	542	mm
Effect.depth (y-dir)	d_y	=	526	mm
Area of footing	A	=	1.96	m ²



3. Loadings

Safe bearing pressure	p_b	=	150	KN/m
Load factor	f	=	1.5	
Total axial load (SLS)	N	=	294.00	kN
Eccentricity	e	=	0	mm

4. Analysis

Net pressure (SLS)	p_{max}	=	135.60	KN/m
Net pressure (ULS)	p_{max}	=	203.4	KN/m
Shear force (x-dir)	V_x	=	135.26	kN
Shear force (y-dir)	V_y	=	135.26	kN
Moment (x-dir)	M_x	=	32.12	kNm
Moment (y-dir)	M_y	=	32.12	kNm
Steel required (x-dir)	A_{sreq}	=	1092	mm ²
Steel provided (x-dir)	A_{sprov}	=	2011	mm ²
Steel required (y-dir)	A_{sreq}	=	1092	mm ²
Steel provided (y-dir)	A_{sprov}	=	2011	mm ²
Top bars required (x-dir)	A'_{sreq}	=	1092	mm ²
Top bars provided (x-dir)	A'_{sprov}	=	1131	mm ²
Top bars required (y-dir)	A'_{sreq}	=	1092	mm ²
Top bars provided (y-dir)	A'_{sprov}	=	1131	mm ²

	Nos	ϕ	Spacing	
O.K.	10	16	143	x-dir
O.K.	10	16	143	y-dir
O.K.	10	12	143	x-dir
O.K.	10	12	143	y-dir

Calculation/Sketch

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Site	Main Depot Rashidiya - Building 20	Date	Apr '07	Job No.	068010.01
	Footing F3	Designed by	Rudy	Sheet No.	
	For soil bearing pressure = 150kPa				

5. Shear Check

Checking for punching shear:

Shear force at face $V_f = 395.44$ kN
Shear stress at face $v_f = 0.41$ N/mm² O.K.

Distance from Support (mm)	av (mm)	V (kN)	v (N/mm ²)	v_c (N/mm ²)	Enhanced v_c (N/mm ²)	Remarks
d	542	0.00	0.00	0.45	0.91	No Shear Links
2d	1084	0.00	0.00	0.45	0.45	No Shear Links

Checking for shear (x-dir):

Shear force at face (x-dir) $V_c = 149.63$ kN
Shear stress at face (x-dir) $v_c = 0.20$ N/mm² O.K.

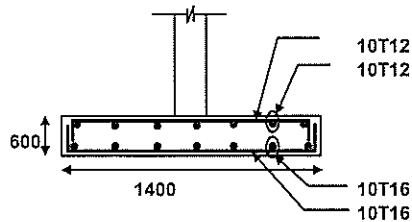
Distance from Support (mm)	av (mm)	V (kN)	v (N/mm ²)	v_c (N/mm ²)	Enhanced v_c (N/mm ²)	Remarks
d	542	0.00	0.00	0.45	0.91	No Shear Links
2d	1084	0.00	0.00	0.45	0.45	No Shear Links

Checking for shear (y-dir):

Shear force at face (y-dir) $V_c = 149.63$ kN
Shear stress at face (y-dir) $v_c = 0.20$ N/mm² O.K.

Distance from Support (mm)	av (mm)	V (kN)	v (N/mm ²)	v_c (N/mm ²)	Enhanced v_c (N/mm ²)	Remarks
d	526	0.00	0.00	0.46	0.92	No Shear Links
2d	1052	0.00	0.00	0.46	0.46	No Shear Links

6. Detailing



3 CONNECTION DESIGN

Document No.: **DM001-E-MAW-CDP-DR-DCC-371948**

Rev A3

Date: 16 May 2007

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Calculation/Sketch

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Site	Dubai Depot Waste Stock Yard (Rashidiya)	Date	May '07	Job No.	068010
		Designed by	Rudy	Sheet No.	

Column Splice Connection Design (Supporting the Beam)

From the STAAD Pro analysis

$F_{xc} =$	143	kN (compression)
$F_{xt} =$	26	kN (tension)
$F_y =$	215	kN (include shear due to beam moment)
$M_z =$	49	kNm

Total Tension

$$T = (n \cdot T_b / d) \cdot [(d-a) + (d-y_1-a) + (d-y_2-a) + \dots]$$

where n = no of row of bolts and T_b is bolt tension capacity

Total Compression

$$= 0.5a \cdot p_c \cdot b$$

where b = stiff bearing length

= web thickness + 2 * plate thickness

Moment Capacity

$$M_c = [n T_b / (d-a)] \cdot [(d-a)^2 + (d-a-y_1)^2 + (d-a-y_2)^2 + \dots]$$

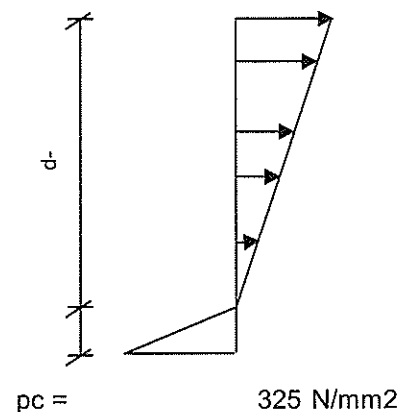
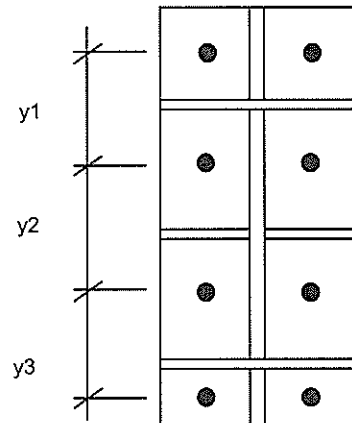
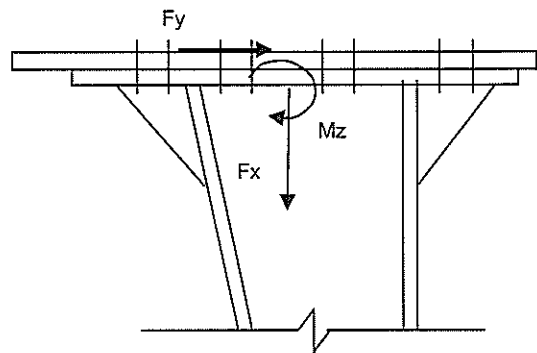
For Gr 8.8 bolt

Bolt size	=	20 mm
T_b	=	110 kN
V_b	=	91.9 kN
n	=	2
n_{total}	=	8
web, t_w	=	8 mm
plate, t_p	=	15 mm
b	=	38 mm

y_1	135	Trial a	=	63 mm
y_2	80	T	=	391.4 kN
y_3	135	C	=	391.4 kN
y_4	0	T-C	=	0.00 kN
y_5	0	M_c	=	84.6 kNm
y_6	0			
Depth, d	<u>350</u>			

$$\begin{aligned} \text{Tension force on the bolt, } T_1 &= (M_z / M_c) \cdot T_b \\ &= 64 \text{ kN} \end{aligned}$$

$$\begin{aligned} [T_1 + (F_{xt} / n_{total})] / T_b + F_y / (n_{total} \cdot V_b) &= 0.90 \\ &< 1.4 \text{ Ok} \end{aligned}$$



Calculation/Sketch

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Site	Dubai Depot Waste Stock Yard (Rashidiya)	Date	May '07	Job No.	068010
		Designed by	Rudy	Sheet No.	

Check the effective weld length on the stiffener plate to resist bolt's tension

Weld size, $s = 6 \text{ mm}$ (up to 15mm thick)
Weld strength, $p_w = 1.05 \text{ kN/mm}$ (based on longitudinal capacity)
Weld length required = $\max(T_1 + 0.5 \cdot F_{xt}, F_y) / p_w$
= 205 mm
= 102 mm per face

Calculation/Sketch

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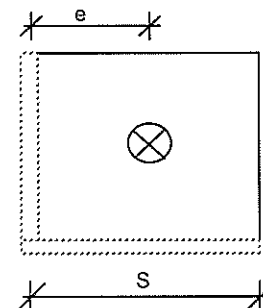


Site	Dubai Depot Waste Stock Yard (Rashidiya)	Date	May '07	Job No.	068010
		Designed by	Rudy	Sheet No.	

From the plate analysis (refer attached) using

Prying Force, $T_o = 10 \text{ kN}$

Plate width, S_o (mm)	Ave. support bending moment, M_o (kNm)
75 312	0.020
100 63.4	0.027
125 6.61	0.035
150 41.6	0.040



Actual - Prying force, T	=	64 kN
Side, S	=	100 mm
Bending Moment, M	=	$(T/T_o) \cdot M_o$
	=	0.172 kNm

Bolt Diameter, ϕ	=	16 mm
Bolt distance, e	=	50 mm
Thickness of plate, t_p	=	15 mm
py allow	=	355 N/mm ²

Effective width of the plate resisting the bending moment, b_p	=	$\min(2e + \phi, S)$
	=	100 mm

Section of modulus of the plate, Z_p	=	$b_p \cdot t_p^2 / 6$
	=	3750 mm ²

$M / Z_p = 46 \text{ N/mm}^2 \quad \text{Ok}$

Calculation/Sketch

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Site	Dubai Depot Waste Stock Yard (Rashidiya)	Date	May '07	Job No.	068010
		Designed by	Rudy	Sheet No.	

For the cantilever end

- Assume shear due to bending is resisted by top and bottom bolt
 - Assume vertical and horizontal shear is resisted by bolt on the web
- Eccentricity of load, e = 60 mm

M_{max} = 25 kNm
 V_{max} = 19 kN

Check shear on the web

- for the secondary beam web

No of bolt provided = 2
Shear per bolt = 9.6 kN
Bolt diameter, ϕ = 16 mm
Shear capacity, V_b = 58.9 kN

- for the main beam web

No of bolt provided = 2
Distance between bolt = 150 mm
Tensile load per bolt = 7.7 kN
Bolt diameter, ϕ = 16 mm
Tensile capacity, V_b = 70.7 kN

Check bolts on the top flange

Beam depth - flange thickness, D_f = 292 mm
 M_{max} bending moment (incl. moment due to ecc), M_e = 26.3 kNm

Shear on the flange, V_f = M_e / D_f
= 89.9 kN

No of bolt provided = 2
Shear per bolt = 45.0 kN
Bolt diameter, ϕ = 16 mm
Shear capacity, V_b = 58.9 kN

Calculation/Sketch

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Site	Dubai Depot Waste Stock Yard (Rashidiya)	Date	May '07	Job No.	068010
		Designed by	Rudy	Sheet No.	

From the analysis for the internal beam

Fy	=	19 kN
Fx	=	1 kN
Mz	=	25 kNm

Eccentricity of load, e = 60 mm

- Assume shear due to bending is resisted by bolt on the flange
- Assume vertical and horizontal shear is resisted by bolt on the web

Check bolts on the flanges

Beam depth - flange thickness, Df	=	292 mm
Mmax bending moment (incl. moment due to ecc), Me	=	26.3 kNm
Shear on the flange, Vf	=	Me / Df
	=	89.9 kN
No of bolt provided	=	2
Shear per bolt	=	45.0 kN
Bolt diameter, ϕ	=	16 mm
Shear capacity, Vb	=	58.9 kN

Check weld length required

Weld size, s =	6 mm	(up to 15mm thick)
Weld strength, pw =	1.05 kN/mm	(based on longitudinal capacity)
Weld length required = Vf / pw		
	= 86 mm	
	= 43 mm	per face

Check shear on the web

- for the secondary beam web

No of bolt provided	=	2
Shear per bolt	=	9.6 kN
Bolt diameter, ϕ	=	16 mm
Shear capacity, Vb	=	58.9 kN

- for the main beam web

No of bolt provided	=	2
Distance between bolt	=	150 mm
Tensile load per bolt	=	7.8 kN
Bolt diameter, ϕ	=	16 mm
Tensile capacity, Vb	=	70.7 kN

Calculation/Sketch

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Site	Dubai Depot Waste Stock Yard (Rashidiya)	Date	May '07	Job No.	068010
		Designed by	Rudy	Sheet No.	

Column Splice Connection Design

From the STAAD Pro analysis

Fxc =	64	kN (compression)
Fxt =	8	kN (tension)
Fy =	18	kN
Mz =	92	kNm

Total Tension

$$T = (n \cdot T_b / d) \cdot [(d-a) + (d-y_1-a) + (d-y_2-a) + \dots]$$

where n = no of row of bolts and T_b is bolt tension capacity

Total Compression

$$= 0.5a \cdot p_c \cdot b$$

where b = stiff bearing length

$$= \text{web thickness} + 2 \cdot \text{plate thickness}$$

Moment Capacity

$$M_c = [n T_b / (d-a)] \cdot [(d-a)^2 + (d-a-y_1)^2 + (d-a-y_2)^2 + \dots]$$

For Gr 8.8 bolt

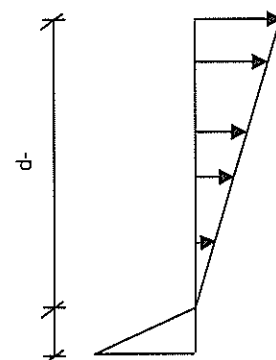
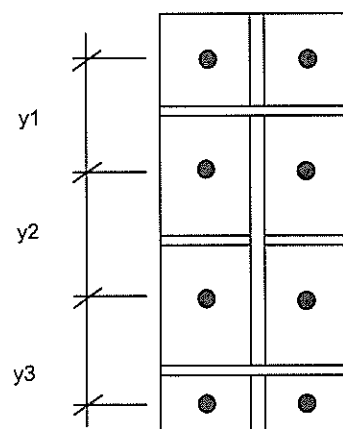
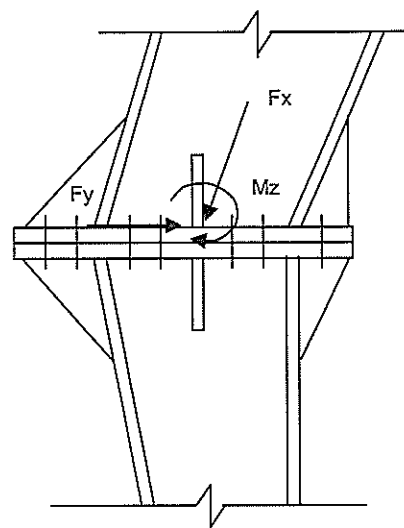
Bolt size =	22 mm
T _b =	136 kN
V _b =	114 kN
n =	2
n total =	12

web, t _w =	8 mm
plate, t _p =	15 mm
b =	38 mm

y1	118	Trial a	=	76 mm
y2	114	T	=	468.7 kN
y3	118	C	=	468.7 kN
y4	0	T-C	=	0.00 kN
y5	0	M _c	=	100.5 kNm
y6	0			
Depth, d	350			

$$\text{Tension force on the bolt, } T_1 = (M_z / M_c) \cdot T_b = 125 \text{ kN}$$

$$[T_1 + (F_{xt} / n_{\text{total}})] / T_b + F_y / (n_{\text{total}} \cdot V_b) = 0.93 < 1.4 \text{ Ok}$$



$$p_c = 325 \text{ N/mm}^2$$

Calculation/Sketch

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Site	Dubai Depot Waste Stock Yard (Rashidiya)	Date	May '07	Job No.	068010
			Designed by	Rudy	Sheet No.

Check the effective weld length on the stiffener plate to resist bolt's tension

Weld size, s = 6 mm (up to 15mm thick)
Weld strength, pw = 1.05 kN/mm (based on longitudinal capacity)
Weld length required = $(T1 + 0.5 \cdot F_{xt}) / pw$
= 122 mm
= 61 mm per face

Calculation/Sketch

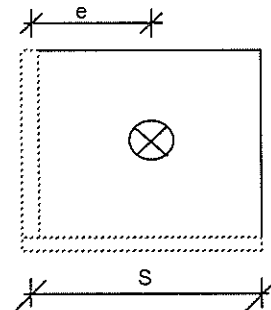
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Site	Dubai Depot Waste Stock Yard (Rashidiya)	Date	May '07	Job No.	068010
			Designed by	Rudy	Sheet No.

From the plate analysis (refer attached) using
Prying Force, $T_o = 10$ kN

Plate width, S_o (mm)	Ave. support bending moment, M_o (kNm)
75	0.020
100	0.027
125	0.035
150	0.040



Actual - Prying force, T	=	124 kN
Side, S	=	75 mm
Bending Moment, M	=	$(T/T_o) * M_o$
	=	0.251 kNm

Bolt Diameter, ϕ	=	24 mm
Bolt distance, e	=	36 mm
Thickness of plate, t_p	=	15 mm
p_y allow	=	355 N/mm ²

Effective width of the plate resisting the bending moment, b_p	=	$\min(2e + \phi, S)$
	=	75 mm

Section of modulus of the plate, Z_p	=	$b_p * t_p^2 / 6$
	=	2812.5 mm ²

M / Z_p	=	89 N/mm ²	Ok
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Calculation/Sketch

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Site Dubai Depot
Waste Stock Yard (Rashidiya)

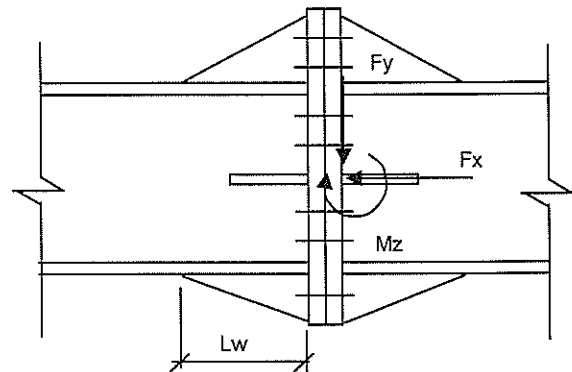
Date May '07 Job No. 068010

Designed by Rudy Sheet No.

Beam Splice Connection Design

From the STAAD Pro analysis

Fxc = 4.43 kN (compression)
Fxt = 1.92 kN (tension)
Fy = 41 kN
Mz = 83 kNm



Total Tension

$$T = (n \cdot T_b / d) \cdot [(d-a) + (d-y_1-a) + (d-y_2-a) + \dots]$$

where n = no of row of bolts and T_b is bolt tension capacity

Total Compression

$$= 0.5a \cdot p_c \cdot b$$

where b = stiff bearing length

= web thickness + 2 * plate thickness

Moment Capacity

$$M_c = [n T_b / (d-a)] \cdot [(d-a)^2 + (d-a-y_1)^2 + (d-a-y_2)^2 + \dots]$$

For Gr 8.8 bolt

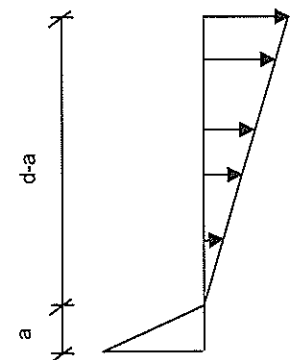
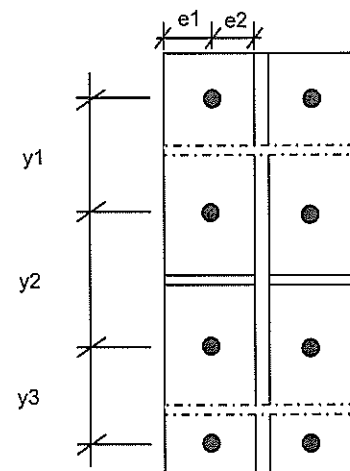
Bolt size = 24 mm
T_b = 159 kN
V_b = 132 kN
n = 2
n total = 8

web, t_w = 6 mm
plate, t_p = 15 mm
b = 36 mm
e₁ = 36 mm
e₂ = 36 mm

y1	118	Trial a	=	90 mm
y2	114	T	=	526.0 kN
y3	118	C	=	526.0 kN
y4	0	T-C	=	0.00 kN
y5	0	M _c	=	108.3 kNm
y6	0			
Depth, d	350			

Tension force on the bolt, T₁ = (M_z/M_c) * T_b
= 122 kN

[T₁ + (F_{xt}/n total)] / T_b + F_y/(n total * V_b) = 0.81
< 1.4 Ok



p_c = 325 N/mm²

Calculation/Sketch

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Site	Dubai Depot Waste Stock Yard (Rashidiya)	Date	May '07	Job No.	068010
			Designed by	Rudy	Sheet No.

Check the effective weld length on the stiffener plate to resist bolt's tension

Weld size, s = 6 mm (up to 15mm thick)
Weld strength, pw = 1.05 kN/mm (based on longitudinal capacity)
Weld length required = $(T1 + 0.5 \cdot F_{xt}) / pw$
= 117 mm
= 59 mm per face

Calculation/Sketch

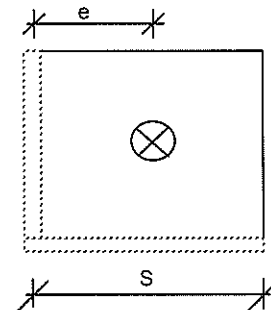
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Site	Dubai Depot Waste Stock Yard (Rashidiya)	Date	May '07	Job No.	068010
			Designed by	Rudy	Sheet No.

From the plate analysis (refer attached) using
Prying Force, $T_o = 10$ kN

Plate width, S_o (mm)	Ave. support bending moment, M_o (kNm)
75	0.020
100	0.027
125	0.035
150	0.040



Actual - Prying force, $T = 122$ kN
Side, $S = 75$ mm
Bending Moment, $M = (T/T_o) * M_o = 0.247$ kNm

Bolt Diameter, $\phi = 24$ mm
Bolt distance, $e = 36$ mm
Thickness of plate, $t_p = 15$ mm
 p_y allow $= 355$ N/mm²

Effective width of the plate resisting the bending moment, $b_p = \min(2e + \phi, S) = 75$ mm

Section of modulus of the plate, $Z_p = b_p * t_p^2 / 6 = 2812.5$ mm²

$M / Z_p = 88$ N/mm² Ok

Calculation/Sketch

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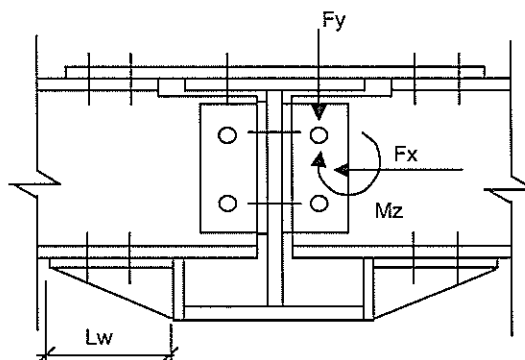
Site	Dubai Depot Waste Stock Yard (Rashidiya)	Date	May '07	Job No.	068010
		Designed by	Rudy	Sheet No.	

Secondary Beam Connection Design

From the analysis for the perimeter beam

F_y	=	46 kN
F_x	=	16 kN
M_z	=	64 kNm

Eccentricity of load, e = 60 mm



- Assume shear due to bending is resisted by bolt on the flange
- Assume vertical and horizontal shear is resisted by bolt on the web

Check bolts on the flanges

Beam depth - flange thickness, D_f	=	242 mm
Mmax bending moment (incl. moment due to ecc), M_e	=	66.9 kNm

Shear on the flange, V_f	=	M_e / D_f
	=	276.6 kN

No of bolt provided	=	4
Shear per bolt	=	69.2 kN
Bolt diameter, ϕ	=	20 mm
Shear capacity, V_b	=	91.9 kN

Check weld length required

Weld size, s =	6 mm	(up to 15mm thick)
Weld strength, p_w =	1.05 kN/mm	(based on longitudinal capacity)
Weld length required = V_f / p_w		
	=	263 mm
	=	132 mm per face

Check shear on the web

- for the secondary beam web

No of bolt provided	=	2
Shear per bolt	=	24.4 kN
Bolt diameter, ϕ	=	16 mm
Shear capacity, V_b	=	58.9 kN

- for the main beam web

No of bolt provided	=	2
Distance between bolt	=	100 mm
Tensile load per bolt	=	25.9 kN
Bolt diameter, ϕ	=	16 mm
Tensile capacity, V_b	=	70.7 kN

Calculation/Sketch

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Site	Dubai Depot Waste Stock Yard (Rashidiya)	Date	May '07	Job No.	068010
		Designed by	Rudy	Sheet No.	

From the analysis for the internal beam

Fy = 41 kN
Fx = 2 kN
Mz = 26 kNm

Eccentricity of load, e = 60 mm

- Assume shear due to bending is resisted by bolt on the flange
- Assume vertical and horizontal shear is resisted by bolt on the web

Check bolts on the flanges

Beam depth - flange thickness, Df = 242 mm
Mmax bending moment (incl. moment due to ecc), Me = 28.0 kNm

Shear on the flange, Vf = Me / Df
= 115.6 kN

No of bolt provided = 2
Shear per bolt = 57.8 kN
Bolt diameter, ϕ = 20 mm
Shear capacity, Vb = 91.9 kN

Check weld length required

Weld size, s = 6 mm (up to 15mm thick)
Weld strength, pw = 1.05 kN/mm (based on longitudinal capacity)
Weld length required = Vf / pw
= 110 mm
= 55 mm per face

Check shear on the web

- for the secondary beam web

No of bolt provided = 2
Shear per bolt = 20.3 kN
Bolt diameter, ϕ = 16 mm
Shear capacity, Vb = 58.9 kN

- for the main beam web

No of bolt provided = 2
Distance between bolt = 100 mm
Tensile load per bolt = 16.7 kN
Bolt diameter, ϕ = 16 mm
Tensile capacity, Vb = 70.7 kN

Calculation/Sketch

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Site	Dubai Depot Waste Stock Yard (Rashidiya)	Date	May '07	Job No.	068010
		Designed by	Rudy	Sheet No.	

For the cantilever end

- Assume shear due to bending is resisted by top and bottom bolt
 - Assume vertical and horizontal shear is resisted by bolt on the web
- Eccentricity of load, e = 60 mm

M_{max} = 25 kNm
 V_{max} = 20 kN

Check shear on the web

- for the secondary beam web

No of bolt provided = 2
Shear per bolt = 9.8 kN
Bolt diameter, ϕ = 16 mm
Shear capacity, V_b = 58.9 kN

- for the main beam web

No of bolt provided = 2
Distance between bolt = 100 mm
Tensile load per bolt = 11.7 kN
Bolt diameter, ϕ = 16 mm
Tensile capacity, V_b = 70.7 kN

Check bolts on the top flange

Beam depth - flange thickness, D_f = 242 mm
 M_{max} bending moment (incl. moment due to ecc), M_e = 26.6 kNm

Shear on the flange, V_f = M_e / D_f
= 110.1 kN

No of bolt provided = 2
Shear per bolt = 55.0 kN
Bolt diameter, ϕ = 20 mm
Shear capacity, V_b = 91.9 kN